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Multi-agent trust management and adaptive authentication for the Internet of Things

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À ma famille,

Avec tout mon amour et ma reconnaissance pour votre soutien constant, votre patience, votre foi en moi, et vos prières constantes. Vous avez été une source constante de force et de réconfort tout au long de ce parcours. C'est grâce à votre amour inconditionnel que ce travail est accompli.

Abstract

This thesis addresses a fundamental challenge in Agent-based Internet of Things (A-IoT) environments: how can autonomous agents make secure decisions when interacting with entities whose reliability and identity remain uncertain? In open Multi-Agent Systems (MAS) operating within IoT contexts, agents operate autonomously in dynamic and uncertain environments. In such settings, agents often find themselves in dependency situations, relying on others for information or services without knowing how those agents will behave. This uncertainty creates the need for trust assessment, allowing agents to estimate the reliability of their potential partners before engaging in collaboration. However, trust can only be meaningful when it is associated with a verified identity. A trust relationship inherently assumes that the interacting entity is indeed who it claims to be. Without authentication, trust becomes fragile and easily exploitable, as malicious agents can impersonate trusted ones and inherit their reputation. At the same time, traditional authentication mechanisms are often too rigid or resource-intensive for heterogeneous IoT environments. This thesis builds trust and authentication as interdependent, co-evolving mechanisms that must adapt to the dynamic, resource-constrained nature of IoT environments. It establishes that authentication without trust consideration becomes resource-inefficient, while trust without authentication remains exploitable by impersonation attacks. By creating a positive feedback loop where authentication reinforces trust reliability and trust dynamically guides authentication decisions, this research enables adaptive security that balances protection with resource efficiency. This thesis makes three interconnected contributions. First, it introduces the concept of opportunistic sensor-based authentication, demonstrating how existing IoT sensor infrastructure can be repurposed to generate diverse, unpredictable authentication factors without additional hardware investments. Second, it develops trust-based adaptive authentication frameworks that dynamically select authentication factors by integrating trust evaluation with multi-objective optimization, balancing security requirements against energy consumption and latency constraints. Third, it extends the notion of trust itself by proposing a bidirectional model of trust as a verifiable asset, that transforms trustees from passive subjects of evaluation into active agents who can demonstrate and manage their trustworthiness through cryptographically verifiable credentials recorded on a blockchain for resource-tolerant IoT environments.

Résumé

Cette thèse aborde un défi fondamental dans les environnements de l'Internet des Objets à base d'agents (A-IoT) : comment des agents autonomes peuvent-ils prendre des décisions sécurisées lorsqu'ils interagissent avec des entités dont la fiabilité et l'identité demeurent incertaines ? Dans les Systèmes Multi-Agents (SMA) ouverts évoluant dans des contextes IoT, les agents opèrent de manière autonome dans des environnements dynamiques et incertains. Dans de telles situations, ils dépendent souvent d'autres agents pour obtenir des informations ou des services, sans savoir comment ces derniers se comporteront. Cette incertitude rend nécessaire l'évaluation de la confiance, permettant d'estimer la fiabilité d'un partenaire potentiel avant toute collaboration. Cependant, la confiance n'a de sens que si elle est associée à une identité vérifiée : sans authentification, elle devient fragile et exploitable, car des agents malveillants peuvent usurper l'identité d'agents réputés et hériter de leur réputation. Les mécanismes d'authentification traditionnels se révèlent souvent trop rigides ou coûteux en ressources pour des environnements IoT à ressources limitées. Cette thèse conçoit la confiance et l'authentification comme deux mécanismes interdépendants et coévolutifs, capables de s'adapter à la nature dynamique et contrainte en ressources des environnements IoT. Elle montre que l'authentification sans confiance devient inefficace, tandis que la confiance sans authentification reste vulnérable aux usurpations. En établissant une boucle de rétroaction où l'authentification renforce la fiabilité de la confiance et où la confiance guide dynamiquement les décisions d'authentification, cette recherche propose une sécurité adaptative conciliant protection et efficacité énergétique. Cette thèse apporte trois contributions principales. Premièrement, elle introduit le concept d'authentification opportuniste fondée sur les capteurs, montrant comment les infrastructures IoT existantes peuvent être réutilisées pour générer des facteurs d'authentification variés sans investissement matériel supplémentaire. Deuxièmement, elle développe des cadres d'authentification adaptative fondés sur la confiance, capables de sélectionner dynamiquement les facteurs d'authentification via une optimisation multi-objectifs équilibrant sécurité, consommation énergétique et latence. Enfin, elle étend la notion de confiance en proposant un modèle bidirectionnel où les entités peuvent démontrer et gérer leur fiabilité grâce à des justificatifs cryptographiquement vérifiables enregistrés sur une blockchain adaptée aux environnements IoT tolérants aux contraintes de ressources.

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1.1 Context and Motivation

The Internet of Things (IoT) represents one of the most significant technological paradigms of the 21st century, fundamentally transforming how we interact with our physical environment through interconnected digital devices. This revolutionary concept extends connectivity to a vast network of heterogeneous physical objects embedded with sensors, actuators, and software, enabling them to connect and exchange data with other devices and systems over the internet, creating an ecosystem where physical and digital worlds converge. IoT's rapid expansion, projected to reach over 30 billion devices by 2030 [1], spans domains such as smart cities, healthcare, agriculture, and industrial automation. Each domain introduces specific constraints and requirements, adding to the complexity of IoT ecosystems. These systems are typically characterized by heterogeneity, resource constraints (e.g., restricted processing power and limited energy), openness, and dynamic topology. IoT environments are composed of a wide variety of devices with different computational capabilities, power availability, communication protocols, and mobility patterns. Moreover, due to their often decentralized and large-scale nature, managing coordination, security, and decision-making in such systems remains a critical challenge [2].

These challenges have led researchers to explore architectural and computational models that can address the unique characteristics of IoT environments. One particularly promising approach is the application of Multi-Agent Systems (MAS) principles to IoT environments [3]. MAS is a mature and well-established paradigm

in distributed artificial intelligence, offering a powerful framework for modeling and implementing systems composed of autonomous, intelligent entities that interact to achieve individual and collective goals. In the context of IoT, MAS provides a natural abstraction that aligns closely with the decentralized, heterogeneous, and dynamic nature of IoT environments [4, 5]. In such settings, each IoT object, or a group of objects, can be modeled as an agent capable of making decisions, exchanging information, and collaborating to accomplish shared goals.

The key characteristics [6] that make MAS particularly suitable for a lot of IoT environments include autonomy, where each agent operates independently without requiring constant external control; social ability, enabling agents to interact and coordinate with others through communication and negotiation protocols; reactivity, allowing agents to respond to environmental changes and unexpected situations; and proactivity, enabling agents to take initiative to achieve their goals rather than merely reacting to external motivators. These characteristics enable MAS-based IoT systems to self-organize, collaborate, and adapt [7]. MAS also enhances scalability, as decentralized agents avoid the bottlenecks of centralized control. Additionally, MAS offers a flexible way to address heterogeneity, since agents can be designed to the capabilities of their devices but still operate within shared protocols and goals. However, the distributed and open nature of MAS and IoT amplifies security vulnerabilities, particularly when agents interact with new or unfamiliar agents [8]. The inherent uncertainty regarding the reliability and intentions of other agents necessitates robust mechanisms to assess trustworthiness and mitigate risks associated with dishonest or malicious behaviors.

In this context, trust emerges as a fundamental requirement in open and dynamic MAS [9], such as MAS deployed in IoT environments. It serves as both a security mechanism and a coordination facilitator, enabling agents to interact and collaborate effectively in dynamic and often unpredictable environments that may involve potentially risky interactions with other agents. A trust relationship involves two roles: a *truster*, the agent who depends on another agent for a service or information, and a *trustee*, the agent providing the service to the *truster*. Trust then generally represents the truster's belief in the trustee's reliability, honesty, and competence, based on past interactions, reputation information, and contextual factors. By prioritizing interactions with trusted peers, agents can detect and isolate those exhibiting malicious behavior, thereby fostering cooperation and ensuring effective decision-making. This notion of trust, when formalized in software agents, creates what is known as a computational trust model: a dynamic and adaptive

approach to managing uncertainty and risk in open environments.

Historically, traditional trust management models have taken a predominantly truster-centric approach, meaning that the truster is solely responsible for determining whether the trustee can actually be trusted. While these models have been widely adopted, they inherently position trustees as passive participants in trust relationships, limiting their role to providing services or information without actively contributing to the trust evaluation process. This passive role means that trustees remain largely unaware of their trust status within the system and cannot leverage their trustworthiness as a strategic advantage for securing better collaboration opportunities or resources. Furthermore, trustees have no visibility into how their trust levels change over time or what specific behaviors contribute to positive or negative trust assessments, preventing them from optimizing their actions to maintain or improve their reputation. Some prior work has explored the other side of trust relationships, recognizing that trust is not only a risk management tool for trusters but also a form of social capital for trustees [10]. Being trusted within a society increases a trustee's likelihood of being selected as an interaction partner and raises the minimum compensation or benefit they can negotiate during interactions. However, models that incorporate this participatory view of trust remain limited in scope and adoption, and the trustee's role is still largely passive in most trust frameworks.

In addition to this modeling issue, the effectiveness of trust management systems fundamentally depends on a critical assumption that is often overlooked: the authenticity of agent identities. Trust relationships are inherently linked to specific identities and are typically established over time through repeated interactions with those identities. For instance, considering two agents, *A* and *B*, when an interaction occurs between these agents, and agent *A* seeks to evaluate the trustworthiness of agent *B*, this assessment is meaningful only if *A* can be certain that it is actually interacting with the genuine agent *B*. The trust relationship is built upon the premise that the agent's claimed identity corresponds to its actual identity, creating a dependency that makes trust vulnerable to identity-based attacks. This dependency reveals a critical vulnerability in trust-only approaches: if a malicious agent can successfully impersonate a trusted identity, it can immediately inherit and exploit all the trust associated with that identity. For example, if a malicious agent *C* successfully impersonates the trusted agent *B*, agent *A* will apply its positive trust assessment of *B* to interactions with *C*, unknowingly depending on the malicious agent *C*. Such impersonation attacks can severely undermine the entire trust model, leading to data manipulation, communication disruption, and reputation damage.

The malicious agent can exploit this inherited trust to propagate false information, provide malicious services, or gain unauthorized access to sensitive resources.

This limitation underscores why trust alone is insufficient for securing MAS environments. While trust mechanisms excel at assessing behavioral reliability and managing uncertainty in agent interactions, they often operate under the assumption that agent identities are verified and authentic. However, without proper authentication mechanisms to establish and verify agent identities, trust becomes a security liability rather than an asset. Therefore, robust authentication is essential for reinforcing the reliability of trust evaluations, ensuring that trust assessments are attributed to the correct agents and that the security benefits of trust-based coordination are fully realized. Building upon the critical need for identity verification, addressing authentication in highly dynamic and resource-constrained IoT environments presents its own unique set of challenges. Traditional authentication mechanisms, often designed for more powerful and less numerous computing entities, are frequently not well suited for the intrinsic characteristics of IoT devices. These include severe resource limitations, immense heterogeneity, the vast scale of deployments, and the dynamic and open nature of IoT environments. Consequently, applying conventional, rigid authentication protocols directly to IoT often leads to inefficiency, vulnerability, or complete impracticality. This underscores the critical necessity for more lightweight and adaptive authentication mechanisms that can dynamically adjust security measures based on contextual factors, effectively balancing security requirements with resource availability.

This thesis is part of the French National Research Agency (ANR) project MaestrIoT (Multi-Agent Trust Decision Process for the Internet of Things) which aims to develop an algorithmic framework for ensuring trust in a multi-agent system handling sensors and actuators of a cyber-physical environment. More specifically, this thesis explores multi-agent trust management and authentication decision-making in IoT through three contributions.

1.2 Research Questions

Building on the motivations presented previously, this thesis is structured around the following research questions:

1. **RQ1:** How can the IoT advantages and the existing IoT infrastructure be leveraged to enhance the identity verification of IoT objects without adding extra burden to the system?

2. **RQ2:** How can MAS effectively integrate trust and authentication mechanisms to enhance security in dynamic and resource-constrained IoT environments, particularly against identity-based attacks?
3. **RQ3:** How can the selection of authentication factors be dynamically optimized based on contextual trust levels and resource constraints, ensuring an adaptive balance between security needs and operational efficiency?
4. **RQ4:** What novel architectural and algorithmic frameworks are required to enable trustees to proactively demonstrate their trustworthiness, moving beyond passive evaluation by trusters?

1.3 Contributions

The contributions of this thesis are organized around two IoT deployment contexts: (i) resource-constrained environments, where adaptive mechanisms are essential due to limited computational power and energy; and (ii) resource-tolerant environments, where higher computational capabilities enable the integration of more sophisticated technologies.

The first contribution of this thesis introduces the concept of opportunistic sensor-based authentication factors in IoT by reimagining the role of embedded sensors originally deployed for non-security purposes. Instead of relying solely on dedicated authentication hardware or protocols, we propose leveraging contextual and behavioral data already available in the environment to build new authentication factors. This approach opens the door to a larger set of possible authentication factors, and thus to enhanced security, while leveraging existing sensor infrastructure.

The second contribution explores the relationship between trust and authentication by proposing a framework for trust-based authentication decision-making. In this model, agents dynamically select the most appropriate authentication factors to be used based on evolving trust levels, contextual factors, and resource constraints, enabling more adaptive and context-aware security policies. The decision-making process of this contribution is then refined through multi-objective optimization. We enhance the adaptive selection of authentication factors by integrating optimization techniques that balance security, energy consumption, and latency before the trust-based decision is applied. This enhancement ensures a more efficient and secure trade-off across the different constraints.

Finally, the third contribution extends the exploration of trust beyond authentication, addressing the limitations of traditional trust management models that treat trustees as passive entities. This contribution shifts the focus to resource-tolerant IoT environments. It introduces a theoretical framework for a bidirectional and verifiable model of trust as an asset in which both trusters and trustees actively participate actively in the trust relationship. In this model, agents can prove and manage their own trustworthiness through verifiable proofs (tokens) recorded on a blockchain.

1.4 Thesis Outline

The remainder of this thesis is organized into five chapters:

- **Chapter 2 - State of the Art:** This chapter presents the theoretical and conceptual foundations of the thesis. It begins by defining the key concepts underlying the Internet of Things, Multi-Agent Systems, and their intersection into Agent-based IoT. It then reviews existing research on trust management and authentication mechanisms in distributed and resource-constrained environments. The chapter identifies open challenges and research gaps that motivate the contributions of this work.
- **Chapter 3 - Trust-based Adaptive Authentication Decision Making:** This chapter introduces the theoretical framework that links trust and authentication in IoT environments. It first presents the concept of opportunistic sensor-based authentication factors, which leverage contextual and behavioral data from existing IoT sensors. It then develops a multi-agent framework for trust-based adaptive authentication, where agents dynamically select authentication factors according to contextual trust and system constraints.
- **Chapter 4 - Experimental Evaluation of the Adaptive Authentication Framework:** This chapter provides the experimental validation of the proposed trust-based adaptive authentication framework. It evaluates both the opportunistic authentication factors and the adaptive decision process through a series of simulations and experiments. The results demonstrate the framework's ability to adapt authentication strategies based on trust evolution and resource availability while maintaining an effective balance between security and efficiency.
- **Chapter 5 - From Passive to Active Trustees: A Bidirectional Model of Trust as an Asset:** This chapter extends the discussion beyond authen-

tication to rethinking trust itself as a verifiable and bilateral process. It introduces a bidirectional trust as an asset model, which enables trustees to actively participate in trust management by proving their trustworthiness through verifiable tokens on a blockchain. The chapter formalizes the model's architecture, defines its operational protocol, and discusses incentive mechanisms that promote cooperative trust management in resource-tolerant IoT environments.

- **Chapter 6 - Conclusions:** The final chapter summarizes the key findings and contributions of the thesis, highlighting how they collectively address the research questions presented in the previous section. It also discusses the limitations of the current work and outlines several directions for future research.

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For over two decades, researchers have extensively investigated various works and challenges associated with trust and authentication in both MAS and IoT contexts. This focus is motivated by the necessity to establish verifiable identities and promote reliable interactions among heterogeneous, resource-constrained devices and autonomous agents. Achieving this under the limitations of constrained computation, memory, and energy resources remains a fundamental challenge. Understanding existing approaches, their motivations, achievements, and their limitations, is essential for identifying open challenges and guiding the development of novel solutions.

As outlined in Section 1.2, this thesis addresses four research questions concerning the enhancement of authentication and trust in IoT and MAS environments. To base these questions, this chapter presents an overview of the current state of the art and establishes the foundational basis for the research presented in this thesis. Formal definitions and challenges are introduced where appropriate to provide a unified and reusable theoretical framework. This chapter is organized in a thematic way. The first part, Section 2.1, introduces the fundamental concepts, definitions, and inherent challenges associated with IoT, MAS, trust management, and authentication. We

explore the interconnections and mutual influences of these concepts, thereby creating the theoretical framework necessary for a shared understanding of the context within which this thesis is situated. Subsequently, the second part of this chapter, Section 2.2, offers a review of the current literature, analyzing existing approaches, methodologies, and solutions that address the intersection of these domains. We examine prior research related to trust management systems in MAS and IoT environments, authentication for resource-constrained devices, and security frameworks for distributed autonomous systems. Through this literature review, we identify key limitations and challenges that motivate the research directions pursued in this thesis.

2.1 Foundational Concepts and Challenges

2.1.1 The Internet of Things (IoT)

The Internet of Things represents a computing paradigm characterized by the interconnection of diverse physical objects embedded with sensors, actuators, processing capabilities, and communication interfaces. These devices form distributed networks capable of data collection, processing, and exchange with minimal human intervention [11].

DEFINITION – INTERNET OF THINGS

The Internet of Things refers to a network of heterogeneous physical objects embedded with sensing, processing, and communication capabilities, enabling interaction and data exchange across physical and digital environments.

This definition encompasses a wide variety of devices, from simple environmental sensors to complex industrial systems. IoT systems are commonly organized into multiple layers that facilitate interactions between physical objects, communication infrastructures, and application services. Although the specific structure may vary depending on the domain [12], a widely adopted reference model includes three layers: perception, network, and application, as shown in Figure 2.1. Other models extend this architecture to five or even seven layers [13], introducing additional levels, such as processing, transport, or business layers, to provide more detailed separation of functions and improve scalability, interoperability, or security management. The perception layer forms the foundation, where sensors capture

environmental data and actuators execute commands. This data is then transmitted through the network layer, responsible for communication and routing, before reaching the application layer, which interfaces with end users through dashboards, alerts, or remote controls, transforming raw data into meaningful and actionable insights. The integration of edge computing paradigms has further evolved IoT architectures, increasing distributed processing capabilities that reduce latency and bandwidth requirements while enhancing system responsiveness [14]. Current IoT deployments span diverse domains, including smart cities, healthcare monitoring, industrial automation, and environmental sensing, creating complex cyber-physical systems that bridge digital and physical environments [15].

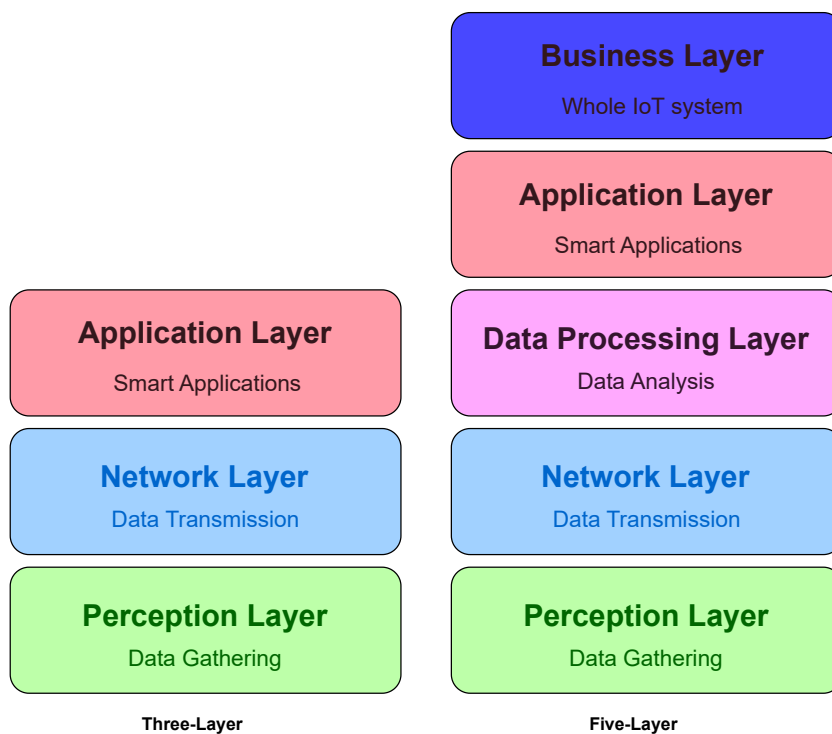


Figure 2.1: IoT General Three and Five-Layer Architecture

IoT systems possess unique characteristics that set them apart from conventional distributed computing environments. Their inherent heterogeneity creates significant interoperability challenges, as devices vary widely in communication protocols, data formats, and computational capabilities. Additionally, the massive scale of IoT deployments introduces unprecedented demands on network management and data processing. Adding to these challenges are the severe resource constraints com-

monly found in IoT environments. Many devices operate under strict limitations in terms of energy consumption, memory capacity, processing power, and communication bandwidth [2]. These constraints are especially critical in battery-powered sensors or devices deployed in remote or inaccessible locations, where maintenance is infrequent or impractical. As a result, all system components, including security mechanisms, must be designed with efficiency and resource optimization as primary considerations. The dynamic nature of IoT environments further complicates system design. Devices can join or leave networks, change geographical locations, modify operational parameters, or experience varying connectivity conditions [16]. This dynamism challenges traditional assumptions about stable network topologies and centralized management structures, necessitating adaptive and self-organizing approaches to system coordination.

From a security perspective, IoT systems face unique vulnerabilities that come from their distributed nature, resource constraints, and often unattended deployment scenarios [17]. The large attack surface resulting from the vast number of interconnected devices, combined with their limited computational resources for implementing robust security measures, creates significant security challenges. Common threats include device impersonation, data injection, unauthorized access, and man-in-the-middle attacks [2]. Ensuring the identity and integrity of communicating entities becomes critical, especially when IoT systems are involved in sensitive operations or critical decision-making processes. Therefore, authentication emerges as a critical security requirement, particularly given the autonomous nature of device interactions and the potential consequences of unauthorized access to IoT systems. However, conventional authentication mechanisms designed for resource-rich computing environments often prove inadequate for IoT contexts due to their computational overhead, reliance on persistent connectivity, and assumption of stable infrastructure. This necessitates the development of lightweight, adaptive authentication solutions specifically adapted to IoT constraints and operational characteristics.

2.1.2 Multi-Agent Systems (MAS)

Understanding Multi-Agent Systems requires first establishing a definition of what constitutes an agent. The concept of an agent in artificial intelligence lacks a universally accepted definition, leading to considerable variation in how different researchers and practitioners characterize these entities. This definitional diversity reflects the multidisciplinary nature of agent-based systems and their application across various domains. Two widely adopted definitions provide a foundational

understanding of agent characteristics. [18] define an agent as "a computer system that is situated in some environment and that is capable of autonomous action in this environment in order to meet its design objectives." This definition emphasizes the autonomous nature of agents and their ability to operate within specific environmental contexts without direct human intervention. Alternatively, [19] characterize an agent as "anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators." This definition focuses on the perceptual and reactive capabilities that enable agents to interact meaningfully with their surroundings, establishing a clear input-output relationship with the environment. Both definitions converge on several key characteristics that distinguish agents from conventional software programs: autonomy in decision-making processes, environmental awareness through perception mechanisms, goal-oriented behavior, and the capability to effect environmental changes through deliberate actions. Additionally, agents typically exhibit reactivity to environmental changes, proactivity in pursuing objectives, and social ability for interaction with other agents or humans [6].

Building on this, when multiple agents are combined within a shared environment, their interactions give rise to emergent behaviors and collective intelligence that exceed the capabilities of individual agents. The complexity of these systems comes from the dynamic relationships between agents, their diverse behavioral patterns, and the unpredictable outcomes that emerge from their collective actions.

DEFINITION – MULTI-AGENT SYSTEM

A Multi-Agent System (MAS) is a system composed of multiple interacting agents that collaborate, compete, or coexist within a shared environment to achieve individual or collective objectives.

MAS exhibit several distinguishing characteristics that differentiate them from single-agent architectures. Distribution represents a fundamental attribute, as agents operate across different computational nodes, geographical locations, or organizational boundaries, eliminating single points of failure and enabling scalable system architectures [20]. Autonomy remains essential, with each agent maintaining independent decision-making capabilities while contributing to system-wide objectives through local actions and interactions. Interaction mechanisms facilitate coordination, negotiation, and information exchange between agents, enabling emergent system behaviors that exceed the capabilities of individual components. These interactions may follow various paradigms, including cooperative proto-

cols where agents collaborate toward shared goals, competitive frameworks where agents pursue conflicting objectives, or mixed scenarios combining elements of both cooperation and competition. Communication in MAS typically occurs through message passing, shared environments, or coordination protocols. More recent studies address the challenges of scalability, decentralization, and dynamic environments by adopting adaptive coordination strategies. These include learning-based approaches that optimize when and how agents communicate, as well as blockchain technologies that provide secure, decentralized agreement among agents [21, 22]. A key driver for these adaptive strategies is the inherent heterogeneity in open MAS, where agents often differ in capabilities, knowledge, behavioral patterns, and interaction protocols, reflecting the diverse requirements and constraints of real-world applications. These open MAS differ from classical MAS in terms of agent heterogeneity, unpredictability, and lack of centralized control. In such systems, agents are often created and managed by different stakeholders with potentially conflicting objectives and without pre-established trust relationships [23].

DEFINITION – OPEN MULTI-AGENT SYSTEM

An Open Multi-Agent System is a multi-agent system in which agents may dynamically join or leave the system, interact with unknown or untrusted peers, and operate under conditions of incomplete information, decentralization, and heterogeneity.

2.1.3 Agent-based IoT

The intersection of IoT and MAS paradigms represents a natural alignment given the complexity and distributed nature of IoT environments. IoT systems, characterized by numerous autonomous devices that interact to achieve individual and collective objectives, align conceptually with MAS principles [3, 24]. Individual IoT devices can be conceptualized as agents possessing autonomy, sensing capabilities, actuation abilities, and communication interfaces. By organizing IoT devices within MAS frameworks, systems can achieve decentralized decision-making, optimized resource allocation, and collaborative task execution among disparate entities, leading to more intelligent and adaptive IoT applications. MAS provide several advantages for IoT environments, including improved scalability through distributed processing, enhanced fault tolerance through redundancy, and increased flexibility through modular agent-based architectures.

Throughout this manuscript, we adopt the term Agent-based IoT (A-IoT) as a unify-

ing designation for systems resulting from the convergence of open MAS and IoT.

DEFINITION – AGENT-BASED IoT (A-IoT)

Agent-based IoT (A-IoT) refers to Internet of Things systems in which devices are modeled as autonomous agents capable of reasoning, local decision-making, interaction, and collaboration, within an open multi-agent system.

A-IoT enables decentralized, adaptive coordination, and dynamic decision-making, addressing key challenges of IoT systems such as centralized bottlenecks, scalability, and flexibility in dynamic environments. By organizing IoT entities as agents within MAS frameworks, A-IoT promotes resilience to single points of failure and support for autonomous behaviors.

A-IoT systems inherit fundamental properties from both IoT and open MAS paradigms [25], distinguishing them from traditional IoT architectures. These inherent characteristics give rise to significant research challenges that must be addressed when developing A-IoT systems:

Properties

Property 1. *Heterogeneity.* *A-IoT environments are inherently heterogeneous across multiple dimensions: hardware (sensors, actuators, edge nodes), software (operating systems, agent platforms), communication protocols, security requirements, and even the design intentions and goals of agents developed by different stakeholders. This diversity is a fundamental characteristic that reflects the distributed and decentralized nature of IoT deployments.*

Property 2. *Openness.* *A-IoT systems operate as open environments where agents can join or leave dynamically and may be unknown at the time of interaction. This openness is a core design principle that enables flexibility and adaptability but introduces uncertainties about agent intentions and behaviors.*

Property 3. *Partial observability.* *Individual agents operate with incomplete views of the global system state due to local sensing limitations, restricted communication ranges, and distributed architecture. This partial observability is an inherent characteristic of distributed systems where no single entity has complete system knowledge.*

Property 4. *Resource constraints.* *Many IoT devices typically operate under strict limitations in energy availability and computational power. Battery-powered*

or energy-harvesting nodes must balance operational needs against limited energy budgets, while low-power processors restrict the complexity of algorithms they can execute. These constraints are fundamental characteristics of IoT hardware.

Challenges

While the properties described above define the fundamental nature of A-IoT systems, they also give rise to a number of open challenges. These challenges capture the practical implications of these characteristics for designers and researchers:

Challenge 1. Interoperability and coordination. *The heterogeneous nature of A-IoT systems creates significant challenges in achieving interoperability. Agents may interpret identical events differently, use incompatible data formats, or employ conflicting decision-making strategies. Developing standardized protocols and translation mechanisms that enable effective coordination despite hardware, software, and protocol diversity remains a key research challenge.*

Challenge 2. Identity and communication security. *The openness of A-IoT systems creates critical security challenges centered around identity verification, message source authentication, and protection against identity attacks. Malicious agents can conduct sophisticated attacks including identity spoofing and message injection. The challenge is compounded by the resource constraints of IoT devices, which limit the computational complexity of cryptographic operations and the frequency of security protocol execution. Traditional security mechanisms designed for resource-rich environments are often unsuitable for battery-powered devices with limited processing capabilities. Developing lightweight, robust, and adaptive security mechanisms that can verify agent identities and authenticate communications without overwhelming resource-constrained devices represents a fundamental research challenge requiring novel approaches that balance security strength with energy efficiency and computational feasibility.*

Challenge 3. Trust-based decision making under uncertainty. *In open and dynamic environments with partial observability, agents frequently encounter dependency situations where they must rely on other agents for critical services, information, or resources to accomplish their objectives. These dependencies create trust relationships where one agent's success depends on another agent's reliability, competence, and benevolence, all under conditions of uncertainty. For example, when an autonomous vehicle agent depends on traffic infrastructure agents for real-time traffic information, or when a healthcare monitoring agent relies on sensor agents for vital signs data, the dependent agent must assess the trustworthiness of its information sources. The challenge is compounded by uncertainty about the other agent's intentions, capabilities,*

current state, and potential conflicts of interest. Incorrect trust assessments can lead to poor decisions, system failures, or security breaches.

Challenge 4. Adaptive behavior under constraints. *The combination of resource constraints, partial observability, and dynamic environments requires agents to continuously adapt their behavior and resource allocation strategies. These constraints force trade-offs across multiple aspects of system functioning, particularly in security, where stronger mechanisms often require more computation and communication. In practice, devices must find a trade-off between efficiency and cost, deciding whether to prioritize performance, responsiveness, or extended operational lifespan. These constraints are fundamental characteristics of IoT hardware.*

Challenge 5. Scalability under resource constraints. *A-IoT systems may involve thousands or millions of agents operating on resource-limited devices. Mechanisms for authentication, trust evaluation, and secure interaction must scale to large populations without relying on centralized control or introducing prohibitive computational and communication overhead. Ensuring that these mechanisms remain lightweight, adaptive, and effective in large-scale deployments represents a significant challenge.*

These properties and challenges are not independent but form a complex web of interdependencies. System openness creates the need for robust trust assessment and identity verification. Resource constraints limit the sophistication of security mechanisms and trust computation algorithms. Partial observability compounds the uncertainty that trust mechanisms must handle, while heterogeneity complicates the establishment of common trust metrics and security protocols.

Understanding and addressing these interdependencies is crucial for developing effective A-IoT solutions. The research presented in this thesis focuses on the intersection of these challenges, developing trust and authentication decision making mechanisms for resource-constrained agents operating in open and dynamic environments.

2.2 Review of Relevant Literature

This second part of the state of the art chapter provides a review of existing literature related to the secure and reliable operation of A-IoT systems, with a focus on trust and authentication mechanisms. Building upon the foundational concepts introduced in section 2.1, this review aims to identify common approaches, analyze their strengths and inherent limitations, and ultimately highlight the research gaps

that necessitate further investigation and form the basis of this dissertation. We will focus on how researchers have attempted to operationalize, manage, and secure agent interactions within these complex environments.

2.2.1 Trust Management Systems

Trust represents a fundamental and inherently complex concept that supports reliable interactions in A-IoT environments. The openness (Property 2) of these environments, where unknown agents can join dynamically, requires trust mechanisms that can assess newcomers without prior interaction history. Additionally, partial observability (Property 3) limits agents' ability to gather complete information about potential interaction partners, forcing trust decisions based on incomplete evidence. The computational interpretation of trust differs significantly from its psychological and sociological counterparts, requiring careful consideration of how trust mechanisms translate from human interactions to distributed computational systems. Understanding trust in these contexts necessitates examining both its theoretical foundations and practical implementation challenges.

Defining Trust in Computational Contexts

Trust fundamentally involves navigating uncertainty about the future behavior of entities within a system (Challenge 3). It enables agents to make informed decisions about interaction partners, facilitate resource sharing, and engage in collaborative activities. [26] provides a foundational computational perspective, defining trust as "a particular level of the subjective probability with which an agent assesses that another agent or group of agents will perform a particular action, both before he can monitor such action and in a context in which it affects his own action." This probabilistic interpretation establishes trust as a measurable quantity that influences decision-making under uncertainty, directly relating to an agent's calculated risk assessment. Complementing this probabilistic view, [27] provide a widely cited relational definition emphasizing vulnerability: "the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the truster, irrespective of the ability to monitor or control that other party." These definitions highlight trust's dual nature as both a quantifiable probability estimate and a behavioral commitment involving vulnerability. In A-IoT contexts, this duality is implemented through computational models that must balance mathematical precision with the practical

uncertainties inherent in distributed agent interactions.

From both definitions, we will adopt the following definition throughout the remainder of this manuscript.

DEFINITION – TRUST

Trust is the degree of confidence an agent (the *truster*) places in the competence, capability, and intention of another agent (the *trustee*) to perform a specific action in a context of uncertainty, where this reliance entails a risk or vulnerability for the truster.

This synthesized definition captures the convergence of probabilistic and relational perspectives, highlighting three essential elements: (i) uncertainty about future behavior, (ii) reliance despite incomplete information or control, and (iii) influence on decision-making processes in environments where complete monitoring or control is impractical or impossible.

Computational Trust Models and Mechanisms

Unlike human trust, which develops through emotional and intuitive processes, computational trust relies on explicit models, metrics, and algorithms to quantify and manage an agent's confidence in another agent's future behavior and to update this estimate as evidence accumulates. This computational approach becomes essential in A-IoT where agents must make decisions about collaboration, resource sharing, and information exchange without direct human oversight. Computational trust extends beyond simple binary classifications of trusted or untrusted agents, encompassing multifaceted and dynamic evaluations of reliability, competence, and behavioral consistency. When agents can evaluate and establish a degree of trust in their peers, they are significantly more likely to engage in cooperative behaviors, share valuable information, and confidently rely on the outputs of other agents. Conversely, a lack of trust can lead to agents adopting defensive strategies, withholding critical information, or refusing to interact, ultimately hindering system performance and potentially leading to system failure [28].

The foundational work of [29] established trust as a measurable entity through mathematical models that quantified relationships using real values in the range $[-1, 1]$ where -1 represents full distrust, 0 is very low trust, and +1 indicates blind trust. This contribution provided the theoretical basis for subsequent research by demon-

strating that trust could inform agent decision-making processes in distributed systems. Building on this foundation, [28] provided comprehensive frameworks for trust and reputation systems that integrated multiple trust dimensions including competence, reliability, and dependence. Their work established taxonomies for trust model classification and identified fundamental design challenges. The dimensional approach to trust assessment recognized that agents might demonstrate varying trustworthiness across different contexts or capabilities, requiring sophisticated trust models that capture these nuanced performance variations.

Trust management systems typically involves two interconnected processes: trust evaluation and trust-based decision-making. Evaluation mechanisms estimate the likelihood an entity will behave as expected, while the decision-making process translate these estimates into actionable policies. The evaluation process draws from multiple information sources, creating a comprehensive picture of agent reliability. Personal interaction histories provide the most reliable foundation, as they reflect actual performance rather than reported behavior. When direct experience is limited, systems aggregate feedback from multiple network participants to establish broader performance patterns. Additionally, explicit recommendations enable trust propagation throughout the network, particularly valuable in dynamic environments where new entities require rapid integration.

Overall, existing research has largely focused on improving the accuracy, robustness, and adaptability of trust evaluation models to address the challenges inherent in open and often resource-constrained environments. By reviewing the multi-agent and IoT trust research literature [30, 31, 32], it can be seen that the different trust models can be categorized according to their approaches, as summarized in Table 2.1, namely:

- **Direct trust models** are based on the agent's own history of interactions, allowing trust to be built through first-hand experience. These models tend to be reliable but are limited in scalability, especially in environments with infrequent or short-lived interactions. Examples of these models are [33, 34].
- **Reputation-based (indirect trust) models** address scalability challenges (Challenge 5) by aggregating third-party feedback and opinions from multiple network agents. While these systems enhance coverage and enable evaluation of previously unknown agents, they remain vulnerable to manipulation (e.g., false reports or collusion attacks), requiring sophisticated filtering mechanisms to ensure information reliability. Examples of these models are [33, 35].

Table 2.1: Comparison of Trust Evaluation Models

Model Type	Information Source	Strengths	Limitations	Typical Use Cases
Direct Trust	Direct personal experience	Accurate and personalized trust estimation	Limited scalability; Cannot leverage trusted peers' opinions	Small-scale networks
Reputation-Based	Third-party feedback or aggregated opinions	Scalable; enables evaluation of unknown agents	Vulnerable to false reports, collusion, and Sybil attacks	Open MAS, online marketplaces
Socio-Cognitive Trust	Beliefs, intentions, goals, and emotional attitudes	Human-like reasoning; supports nuanced, context-aware trust	High computational cost; complex modeling	Social agents (robotics)
Organizational Trust	Roles, rules, norms, and institutional structures	Efficient in structured environments; predefined policies simplify trust	Inflexible; requires prior organizational setup	Industrial IoT, smart cities, regulated MAS

- **Socio-cognitive trust models** represent a more sophisticated approach that incorporates psychological elements such as beliefs, intentions, goals, and emotional attitudes to evaluate trust in ways that mimic human reasoning processes. These models are particularly relevant when the system aims to support more nuanced or context-sensitive interactions, however, they are often more computationally demanding and complex to model. Examples of these models are [36, 37].
- **Organizational trust models** leverage institutional structures, predefined roles, and established norms to simplify trust assessment within structured environments. These models prove especially applicable in regulated MAS, industrial IoT deployments, and smart city infrastructures where entities operate under specific roles and permissions, though they sacrifice flexibility

for computational efficiency and require prior organizational setup. Examples of these models are [38, 39]

However, most existing trust models integrate both direct trust, and indirect trust. This combination allows agents not only to evaluate their own interactions but also to benefit from the wider network's experiences, improving robustness and resilience in dynamic environments. For example, the Beta Reputation System developed by [33] exemplifies the probabilistic approach, leveraging both direct interactions and estimations of the reliability of information obtained from other agents. This probabilistic approach has gained particular prominence in A-IoT environments due to its computational efficiency. The system models trust as a probability updated from interaction outcomes, representing successful and unsuccessful interactions as (α, β) counts, forming an expected trust $\mathbb{E}[T] = \alpha/(\alpha + \beta)$ with uncertainty captured by the variance of $\text{Beta}(\alpha, \beta)$. Time dynamics are introduced via a forgetting factor λ , e.g., $\alpha_t = \lambda\alpha_{t-1} + \text{successes}_t$, $\beta_t = \lambda\beta_{t-1} + \text{failures}_t$ with $\lambda \in (0, 1)$, ensuring recent evidence influences trust assessments more heavily. This model, or modified versions of it, has been applied in various IoT contexts where agents must make trust-based decisions under computational and energy constraints, for example [40] and [41].

Adaptive and Context-Dependent Trust

Sophisticated trust systems recognize that trustworthiness varies significantly across different domains, tasks, and environmental conditions. An agent might demonstrate exceptional reliability in data collection activities while proving less dependable in resource allocation or coordination tasks. This contextual variation demands trust models capable of capturing nuanced performance differences rather than relying on unique trust values that fail to reflect the complexity of agent capabilities and behavioral patterns [42]. Dynamic adaptation represents another critical characteristic of effective trust systems, as trust relationships must evolve based on changing circumstances, modified agent behaviors, and trust-attacks. This capability emphasizes the need for models that accommodate learning, behavioral adaptation, and the natural changes that occur in dynamic distributed environments. Trust assessments should continuously update as new interactions occur and additional information becomes available, ensuring that trust relationships accurately reflect current rather than historical performance patterns. The distributed nature of A-IoT systems where agents represent diverse physical devices and services operating under varying constraints (Challenges 1 and 4), makes centralized trust

management impractical or impossible. In such scenarios, trust mechanisms offer decentralized alternatives for ensuring system coherence and reliability, enabling agents to dynamically assess the credibility and dependability of their peers while adapting interactions based on accumulated experience and system-wide reputation information [43]. Without robust mechanisms for assessing and managing trust, A-IoT are vulnerable to malicious behavior and the breakdown of cooperation caused by uncertainty regarding the intentions or capabilities of interacting agents. Consequently, the development and integration of sophisticated trust models are critical to designing resilient and secure A-IoT capable of functioning effectively in open and dynamic environments.

Trust-Based Decision-Making

Once trust values are computed, they become operationally meaningful only when integrated into decision-making policies that guide agent behavior (Challenge 3). For instance, partner selection algorithms use trust estimates to prioritize highly reliable collaborators, thereby reducing the likelihood of malicious behavior [44]. Similarly, resource allocation strategies exploit trust metrics to direct constrained resources such as energy or computational power toward peers deemed trustworthy. This ensures not only efficiency but also resilience, since unreliable or low-trust entities are excluded from consuming disproportionate resources. For example, trust-based task offloading in mobile edge computing has been shown to improve both throughput and energy efficiency by preferring trustworthy nodes for computational outsourcing [45].

These decision frameworks must carefully balance risk and resource usage, often by setting context-dependent thresholds for collaboration and adaptation. A threshold that is too strict may exclude potentially useful partners, while one that is too indulgent may expose the system to exploitation. Adaptive trust-based decision-making thus represents a trade-off, where operational requirements (e.g., latency, energy, risk) dictate the appropriate calibration of trust thresholds and update policies.

The Truster-Trustee Relationship

Traditional trust and reputation models typically position agents playing the role of truster as active decision-makers, while trustees are evaluated passively through aggregated scores or reputation indices. In this paradigm, trustees serve primarily as subjects of evaluation rather than active participants in trust relationship

formation and maintenance. This positions trustees as essentially reactive entities whose role consists primarily of performing actions that either confirm or violate truster expectations. The passive trustee model simplified computational trust implementations by focusing attention on truster-side trust computation algorithms while treating trustee behavior as externally determined. The passive model fails to capture important dynamics that emerge when trustees become aware of their trust status and actively seek to influence trust relationships for strategic purposes. [10] identified fundamental limitations in passive trustee models that fail to account for trustee agency in trust relationship development. Their theoretical framework identifies several mechanisms through which trust functions as social capital. Trust provides trustees with reputational assets that facilitate access to new collaboration opportunities. High trust levels enable trustees to participate in high-value collaborations that require significant mutual dependency and risk-taking by interaction partners. The social capital perspective introduces dynamic considerations where trustees actively invest in trust relationship development through reputation building activities, consistent performance demonstration, and strategic communication that builds confidence among potential trusters. This perspective suggests that trustees face resource allocation decisions regarding trust capital development similar to other forms of capital investment. Decision-theoretic trust models have explored how trust influences not only initial choice of partners but also the incentive structures within dynamic agent environments, highlighting strategic monitoring, incentives, and delegation protocols built around trust dynamics [44].

2.2.2 Authentication

While trust addresses the willingness to rely on an agent's future behavior, authentication represents the fundamental security process of verifying the claimed identity of an entity within a computational system. While this concept appears straightforward, its implementation in A-IoT environments introduces significant complexity due to distributed architectures, resource constraints, and dynamic operational conditions (Challenges 1, 4, and 5).

Definition and Conceptual Foundations of Authentication

The International Organization for Standardization (ISO/IEC 27000:2018) [46] defines authentication as "provision of assurance that a claimed characteristic of an entity is correct." This definition establishes authentication as a verification process that provides confidence rather than absolute certainty about identity claims. The

emphasis on assurance acknowledges that authentication mechanisms operate under uncertainty and must balance security strength against practical implementation constraints. Complementing on that, the American National Institute of Standards and Technology (NIST SP 800-63-3) [47] characterizes authentication as "the process of establishing confidence in user identities electronically presented to an information system." This definition focuses specifically on digital identity verification and emphasizes the confidence-building aspect of authentication. Both definitions converge on the concept that authentication establishes confidence in identity claims through systematic verification processes. That means that authentication aims to establish a high degree of certainty that an entity is indeed who it claims to be, thereby preventing impersonation and unauthorized access. This confidence-based approach proves particularly relevant in A-IoT systems where absolute identity verification may be impractical or impossible due to network limitations, computational constraints, or operational requirements [48].

The Importance of Authentication in A-IoT

Authentication represents a critical security foundation in A-IoT systems, where the intersection of autonomous agent capabilities and IoT infrastructures creates unique security imperatives that extend far beyond traditional authentication paradigms (Challenges 1, 2, and 4). The autonomous nature of agents fundamentally shifts authentication from human-centered processes to fully automated identity verification systems that must operate reliably across the distributed, heterogeneous environments previously characterized. These characteristics make robust authentication mechanisms particularly critical for:

- **Security and integrity:** In systems where agents exchange sensitive information or control critical physical devices, authenticating agents is vital to prevent malicious attacks, data corruption, and unauthorized control. An unauthenticated agent could inject false data, disrupt operations, or compromise the entire system.
- **Access control:** Authentication forms the basis for effective access control. Once an agent's identity is verified, policies can be enforced to determine what resources it can access, what actions it can perform, and what information it can receive. This is particularly relevant in A-IoT, where agents might have varying levels of permissions.
- **Strengthening trust models:** As previously discussed in the motivation of this thesis, trust is based on the claimed identity during the interaction, and must

rely on consistent and verifiable interactions. Authentication provides the necessary foundation by ensuring that the observed behavior is indeed attributable to the claimed identity, allowing trust assessments to be accurately built and maintained over time.

Authentication in A-IoT extends beyond traditional user-centric models to encompass agent-to-agent identity verification, service authentication, and dynamic credential management across distributed computational environments [49]. The autonomous nature of agents requires authentication mechanisms that operate without human intervention while maintaining security assurance levels appropriate for system requirements. Establishing agent identity is a fundamental challenge, as agents must possess verifiable digital identities that distinguish them from other entities within the system. These identities typically incorporate cryptographic elements such as public-private key pairs, digital certificates, or unique identifiers that enable verification processes [50]. A-IoT environments further amplify authentication complexity due to unique constraints and operational characteristics. Device diversity (Challenge 1) presents significant authentication challenges, as IoT environments typically encompass devices with vastly different computational capabilities, communication protocols, and security implementations. Resource constraints (Challenge 4) represent a fundamental limitation in IoT authentication design. Many IoT devices operate with severe restrictions on processing power, memory capacity, and energy availability. Battery-powered sensors, embedded controllers, and simple actuators cannot support computationally intensive authentication protocols without compromising their primary operational functions or acceptable battery life [51].

Authentication Factors and Approaches

Authentication factors are the technical means used to verify an entity's identity. Authentication factors are commonly categorized into three types depending on the nature of the evidence used to verify identity [52]. A comparative overview of the three categories along with their strengths and limitations is summarized in Table 2.2.

1. Knowledge-based authentication: relies on something only the legitimate entity is supposed to "know".
 - Examples: password, PINs, security questions.

Table 2.2: Comparison of Authentication Factors

Factor Type	Examples	Strengths	Limitations in A-IoT
Knowledge-based	Passwords, PINs, security questions	Simple, widely deployed	Weak against brute-force phishing; not suitable for autonomous agents
Possession-based	Smart cards, tokens, digital certificates, one-time passwords	Stronger security; PKI certificates enable agent-to-agent authentication	Requires storage and computational resources; device loss/duplication risks; overhead for constrained IoT
Inherence-based	Human biometrics (fingerprint); device-specific characteristics (vehicle plate); Device fingerprints (power usage profile, mobility pattern)	Difficult to forge; enables passive and continuous authentication; useful for both human-to-agent and agent-to-agent scenarios	Sensor quality and environment-dependent; may raise privacy concerns

- Applicability in A-IoT: while simple and widely adopted, these methods are vulnerable to brute-force attacks, phishing, and credential leakage. Their direct use for autonomous inter-agent communication is limited due to the absence of a "human" to know the secret. However, they might be relevant for human operators authenticating to an agent system, or for initial bootstrapping of some less critical IoT devices. For resource-constrained IoT devices, securely managing complex knowledge-based credentials can be challenging [53].

2. Possession-based authentication: relies on something only the legitimate entity is supposed to "have".

- Examples: smart cards, security tokens, digital certificates, one-time passwords.
- Applicability in A-IoT: digital certificates, especially those issued by a Public Key Infrastructure (PKI), are widely used in the Internet. These methods provide stronger assurance than knowledge-based approaches. However, the resource-constraints challenges (Challenge 4) led researchers

to build more lightweight alternatives [50, 54, 55]. Each agent can be assigned a unique digital certificate that cryptographically binds its identity to a public key, allowing for secure communication and authentication.

3. Inherence-based authentication: relies on something the legitimate entity "is" (biometric characteristics).
 - Examples: for humans, fingerprints, facial recognition, iris scans, voice recognition, and many other sensor-based factors. For IoT objects and agents, inherence extends to unique physical (e.g., Physically Unclonable Functions (PUFs) [56]) or behavioral (e.g., sensor data patterns [57]) characteristics and patterns.
 - Applicability in A-IoT: while human biometrics are useful when operators interact with IoT systems, device biometrics enable passive and continuous authentication for direct object-to-object communication. These methods depend largely on sensor availability, quality, and environmental stability.

Multi-Factor Authentication

Building on these three categories, Multi-Factor Authentication (MFA) significantly enhances security by requiring an entity to provide two or more different types of authentication factors (e.g., something you know and something you have) [52]. This approach reduces the risk of attacks, as compromising one factor is insufficient to validate the authentication. However, the effectiveness of MFA in distributed systems depends critically on factor independence, ensuring that compromise of one authentication factor does not facilitate attacks against other factors. This independence requirement becomes particularly challenging in IoT environments where devices may share common hardware platforms, communication protocols, or operational environments that could enable correlated attacks across multiple authentication factors [58].

Adaptive and Sensor-Based Authentication

Adaptive MFA represents an evolution of traditional MFA, as the selection and strength of authentication factors are dynamically determined based on real-time risk assessment. Advancements in MFA mechanisms emphasize the integration

of adaptive and context-aware approaches to enhance the security of IoT environments [59, 60, 52]. Adaptive and context-aware authentication refers to dynamically adjusting authentication requirements based on contextual factors such as user behavior (for user authentication), device characteristics, location, and assessed risk levels. For example, an access request from an unfamiliar location or at an unusual time may trigger additional authentication challenges. Unlike static authentication, adaptive approaches aim to balance security and usability by tailoring authentication strength to the perceived threat or sensitivity of an action or device. For example, location-based authentication utilizes the geographical location of the entity, such as GPS coordinates, to verify its identity [61]. This method adds an additional layer of security by ensuring that authentication is tied to a specific location.

Building a signature of entities involves using patterns related to the entities for authentication. These patterns can be based on behavioral or physical factors, which are part of the inherent factors. [62] conducted a survey on behavioral biometrics for continuous user authentication in IoT, highlighting how biometric data such as touchscreen dynamics, eye movements, and other behaviors can strengthen authentication mechanisms. Several studies have also explored the use of sensor data to enhance security in IoT systems. For instance, researchers have investigated the use of accelerometer data to authenticate smartphone users based on their phone usage characteristics [63]. Other studies have examined the potential of using environmental noise and light patterns as supplementary authentication factors [64]. While most studies have primarily focused on user authentication, some have investigated the use of Physically Unclonable Functions (PUFs), which leverage the unique physical properties of hardware components to generate cryptographic keys for device authentication [56, 65]. These findings suggest the potential for extending such methods to other types of sensors.

Given the diversity of authentication factors and the sensitivity of IoT applications, reinforcing and adapting authentication mechanisms is critical. The adaptive selection of authentication factors has been explored as a means to enhance security. For example, [66] proposed an adaptive strategy for selecting authentication factors based on the selection of devices, media, and surrounding contexts. This approach dynamically adjusts the factors used, such as passwords or biometric data, according to performance metrics and contextual information gathered during the authentication process. Such strategies aim to optimize security while accommodating the varying capabilities of devices and the specific requirements of different environments.

However, the adaptive selection of authentication factors inherently involves complex trade-offs between competing objectives: stronger authentication typically requires more computational resources and energy, while lighter mechanisms may compromise security. These conflicting requirements necessitate systematic approaches to balance multiple objectives simultaneously.

Multi-Objective Optimization for Security

Security in A-IoT is then inherently a multi-objective problem, requiring a trade-off between security, efficiency, and usability. Multi-Objective Optimization (MOO) has emerged as a powerful tool for addressing the conflicting requirements of IoT systems, such as balancing security, resource consumption and quality of service. Given the resource constraints in IoT environments, optimizing these trade-offs is crucial for maintaining efficiency and robustness [67]. Unlike single-objective optimization, which yields a single optimal solution, MOO produces a set of Pareto-optimal solutions, each representing a unique trade-off between the objectives. A solution is said to be Pareto-optimal if no objective can be improved without worsening at least one other objective. The set of all Pareto-optimal solutions forms the *Pareto front* that represents the set of non-dominated solutions in the objective space. By examining the Pareto front, decision-makers can select a solution that best aligns with system constraints and priorities.

Traditional methods for solving MOO problems, such as the weighted sum method, aggregate multiple objectives into a single weighted function [68]. While this method is computationally efficient, it requires precise weight tuning and may struggle to capture the full spectrum of trade-offs, especially in dynamic and uncertain IoT environments. This approach also necessitates a predefined trade-off, which may not generalize well across different scenarios.

In contrast, evolutionary algorithms, such as Genetic Algorithms (GA) and Particle Swarm Optimization (PSO), have been widely used to solve multi-objective problems in IoT [69, 70]. These algorithms generate a set of Pareto-optimal solutions without requiring explicit weight assignments, enabling decision-makers to select the most appropriate trade-off. Several comparative studies of different multi-objective optimization methods [71, 72] have demonstrated that the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [73] outperforms other methods by maintaining a well-distributed set of trade-off solutions and achieving better convergence towards the true Pareto front.

Authentication and Trust

With both trust and authentication now defined, it is crucial to differentiate them, as they serve distinct yet complementary roles in securing and enabling agent interactions. Authentication answers the question, "Are you who you say you are?" It is a prerequisite for secure communication and interaction, providing a baseline of identity assurance. Trust, on the other hand, addresses the question, "Will you behave as expected, or can I rely on you?" It is a subjective assessment of an entity's reliability, competence, and benevolence, built upon past interactions and/or reputation. Authentication is very often a binary process (an agent is either authenticated or not), while trust is typically a continuous or probabilistic measure that evolves over time.

However, authentication and trust form an interdependent cycle in A-IoT systems. Authentication serves as a prerequisite for meaningful trust assessment. Trust relationships can only be established with verified identities, as trusting an unverified entity creates vulnerability to impersonation attacks. Conversely, trust assessments can inform adaptive authentication decisions. This creates a positive feedback loop where authentication enables trust formation, and trust history guides authentication intensity, optimizing the security-efficiency trade-off based on relationship context.

2.2.3 Trust Management for Security

A significant body of research has focused on risk-based adaptive authentication, wherein the system dynamically evaluates the risk associated with each access attempt and adjusts authentication requirements accordingly. In addition to contextual factors discussed previously such as device type, role, location, and time, trust management has emerged as a fundamental security mechanism in A-IoT systems, where traditional cryptographic approaches alone prove insufficient to address the dynamic and resource-constrained nature of distributed environments [74]. The intersection of openness (Property 2) and identity and communication security (Challenge 2) creates a security paradigm where agents must make informed decisions about interaction partners without prior knowledge or guaranteed identity verification (Challenge 3). In this context, trust-based security frameworks enable agents to navigate uncertainty by establishing confidence levels in other agents' reliability and intentions. The importance of having a reliable authentication process that relies on a trust management system was highlighted by [75]. This work proposed attaching a measure of trust to an identifier rather than to the agent it is

supposed to represent, circumventing the difficulty of directly assessing an agent's trustworthiness.

The resource constraints challenge (Challenge 4) significantly influences trust-based security design, necessitating lightweight and robust mechanisms that balance security effectiveness with computational and energy limitations. Recent literature demonstrates increasing recognition of trust management as a cornerstone of IoT security architecture. Studies emphasize the critical role of trust in addressing the inherent complexities and vulnerabilities present in IoT networks, particularly highlighting the need to transition from static security models toward more adaptive, context-aware frameworks capable of responding dynamically to evolving threats and environmental conditions [76, 77, 78]. These adaptive trust management systems can adjust security assessments in real-time based on behavioral observations and environmental context, providing enhanced security flexibility and resilience compared to traditional static approaches [79].

The integration of trust mechanisms into authentication and access control directly addresses the intersection of identity and communication security (Challenge 2) and trust-based decision making under uncertainty (Challenge 3) challenges. Research demonstrates how trust and reputation mechanisms can be embedded into authentication schemes, as evidenced by work in the Internet of Vehicles domain where these factors ensure secure and reliable communication [80]. This integration enables authentication systems to make more informed decisions by considering not only the credentials presented but also the trustworthiness of the presenting agent based on historical behavior and network reputation. Another example, [81] propose a trust-based authentication technique for cluster-based vehicular ad hoc networks (VANETs). In their approach, each vehicle maintains a trust degree, computed from direct interactions and indirect recommendations, that determines its eligibility for security-sensitive roles such as cluster head (CH) or verifier. Authentication is then carried out through digital signatures and certificates distributed by a Certification Authority (CA). If a node's trust degree falls below a threshold, it is flagged as malicious and isolated from the network. This design demonstrates the benefits of combining trust and authentication: trust provides a dynamic measure of node reliability, while cryptographic techniques ensure identity verification. However, an important limitation emerges from the ordering of these mechanisms. Trust is associated with a claimed identity and used in decision-making before verifying that the node presenting that identity is indeed the legitimate one. This creates a potential vulnerability: if a malicious node impersonates a trusted identity, it may temporarily inherit that trust score and gain privileged roles (e.g., as a CH

or verifier) before the cryptographic layer detects the fraud. Thus, while effective in resource-constrained and highly dynamic environments such as VANETs, this category of approaches remains susceptible to impersonation and trust attacks.

2.3 Conclusion

Adaptive authentication has been investigated as a way to balance security requirements with the limited capacities of IoT devices. While several works have proposed to link trust with authentication, these schemes typically rely on trust as a filter (granting or denying access) or as a criterion for assigning roles, thereby applying trust to the claimed identity before its verification. To the best of our knowledge, no prior work has used trust as a guiding parameter to adapt the authentication mechanisms themselves, i.e., to dynamically select the most appropriate authentication factors in response to context. Furthermore, much of the literature treats trust primarily as a measure of whether the claimed identity is indeed the real one (identity-oriented trust), whereas trust in our research is defined in terms of the behavior of the entity over time (behavior-oriented trust). The distinction between these two notions of trust remains underexplored in authentication research, which motivated our research questions (RQ2 and RQ3) that would leverage behavioral trust to build adaptive, resource-aware authentication factor selection in A-IoT systems.

We have also seen how different studies have explored the use of sensors to build new authentication factors to enhance security, but most of them were oriented to user authentication. However, there is a significant research gap in applying behavioral or physical signatures for object-to-object authentication in IoT environments. The increasing deployment of IoT devices also presents an opportunity to revisit authentication itself. Many objects are already equipped with sensors that can generate behavioral or physical signatures, yet these remain underexploited for object-to-object verification. This motivates our first research question (RQ1), which asks how existing IoT infrastructure can be leveraged to enhance identity verification without adding extra burden to the system.

Finally, most existing trust models evaluate trust passively, focusing on whether an agent appears trustworthy. This observation motivates our fourth research question (RQ4), which explores the architectural and algorithmic frameworks required to enable agents to demonstrate their trustworthiness proactively and participate in the trust relationship. While earlier works conceptualized trust as relational

capital and discussed potential benefits, actual model implementations usually stop at passive reputation scores or scores stored on chain, without enabling portable, agent-owned credentials that can be mutually exchanged and staked as explicit capital.

In summary, the state of the art shows that although trust and authentication are both well studied, their integration, particularly in resource-constrained A-IoT environments, remains limited. Our four research questions emerge directly from these gaps and frame the contributions of the following chapters.

3

Trust-based Adaptive Authentication Decision-Making

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Building upon the research challenges identified in previous chapters, this chapter introduces the theoretical framework that supports the subsequent development of an adaptive authentication decision-making that dynamically adjusts its security rigor based on the specific context of an interaction, particularly on the trust-worthiness of the agent being authenticated. The proposed approach addresses the fundamental research question of how trust relationships between agents can inform and optimize authentication processes, moving beyond static security protocols toward adaptive authentication mechanisms.

The framework developed here is based on the observation that trust and authentication form a positive feedback loop. More specifically, we argue that: (i) first, authentication strengthens trust by validating the identity of agents and securing the reliability of exchanged information, and (ii) second, trust guides authentication by informing the choice of security measures to apply and allowing agents to optimize resource use when verifying others. The objective then is to exploit this

feedback loop to find a balance between providing robust security against identity attacks and ensuring efficient utilization of the limited resources available to IoT devices.

The chapter introduces three key theoretical contributions that collectively form the foundation of our adaptive authentication framework. First, Section 3.1 presents the concept of opportunistic sensor-based authentication factors, showing with an illustrative scenario how existing IoT infrastructure can be repurposed to provide additional authentication factors without requiring hardware investments. This concept leverages the inherent sensor richness of IoT environments to create authentication factors from data streams originally deployed for other purposes. Second, Section 3.2 develops a decision-making workflow that integrates trust evaluation with authentication factor selection. Third, Section 3.3 presents two developed approaches to authentication factor selection that address different operational contexts and computational constraints. The linear approach (AAT-L) provides a straightforward exhaustive weighted strategy, while the multi-objective optimization approach (AAT-MOO) employs NSGA-II to navigate complex trade-offs between security, energy consumption, and authentication delay.

3.1 Leveraging IoT Existing Systems for Authentication

Despite the constraints and challenges of A-IoT environments, A-IoT also offers advantages and opportunities that can be leveraged to improve security measures. The extensive deployment of sensors across various IoT environments provides a rich source of data. These sensors continuously collect real-time information about their surroundings and the objects they interact with, which, if utilized effectively, can significantly enhance existing security protocols.

We introduce here a novel concept aimed at reinforcing IoT authentication mechanisms by leveraging the existing sensors already deployed in IoT environments. We propose the idea of opportunistic sensor-based authentication factors, which utilizes the data already being collected by these sensors for purposes beyond their primary function. This approach repurposes sensor data to create additional authentication factors, enhancing the security of IoT systems without the need for additional hardware investments.

In this section, we define the concept of opportunistic authentication factors, providing an understanding of their characteristics and potential applications. To clarify the practicality of these factors, we present an illustrative scenario that effectively demonstrates their advantages.

3.1.1 Opportunistic Sensor-Based Authentication Factors

The core idea of opportunistic sensor-based authentication factors is to leverage the data collected by the existing IoT sensors deployed in different environments, which are not typically used for authentication purposes, to reinforce existing authentication mechanisms. This approach repurposes sensor data to serve as supplementary authentication factors, thereby enhancing the security of IoT systems without the need for additional hardware investments. By leveraging existing IoT systems, we can integrate various independent systems to work together collectively, thereby enhancing the overall security infrastructure. This transformation of sensor data into authentication factors enables a new level of authentication, making it more robust and reliable.

OPPORTUNISTIC AUTHENTICATION FACTORS

Opportunistic authentication factors are authentication elements derived from data produced by existing IoT sensors that were originally deployed for non-authentication purposes. Instead of requiring additional dedicated hardware, this approach repurposes sensor outputs as supplementary authentication factors.

It is the responsibility of the authentication protocol designer to examine the set of already deployed sensors in a given environment and determine which of them can be reliably integrated into the authentication process. This design choice ensures that opportunistic factors are both context-appropriate and security-relevant. The opportunistic authentication factors are designed to complement existing authentication systems. They do not replace them, but rather enhance them by providing supplementary verification based on opportunistic data. This integration ensures that even if one authentication factor is compromised, the system remains secure due to the presence of additional factors.

For example, temperature sensors can monitor heat levels, accelerometers can detect motion patterns, and light sensors can measure the intensity of ambient light. To utilize sensor data for authentication, relevant features must be extracted from

the raw data. This involves processing the data to identify unique characteristics that can be used to distinguish between different entities. For example:

- Visual data: shape patterns and object recognition from camera streams.
- Temperature data: temperature fluctuations, heat distribution patterns, and thermal signatures from temperature sensors.
- Motion data: movement patterns, acceleration profiles, and behavioral signatures from accelerometers.

Once the relevant features are extracted, they are used to generate authentication factors. These factors can be integrated into multi-factor authentication schemes, providing an additional layer of security. Moreover, using this kind of information can often be performed without the need for additional communication between objects or with the user. For instance, in a smart home environment, access to a secure area could be authenticated not only by a keypad entry code, but also by opportunistically recognizing the unique walking pattern of an individual detected by floor pressure sensors.

3.1.2 Illustrative Scenario

To better illustrate the concept of opportunistic authentication factors, we now present a simple scenario that demonstrates how existing systems, initially deployed for purposes unrelated to authentication, can be repurposed to enhance security. This example is not intended as a definitive application, but rather as an illustration of the feasibility of the approach and its potential benefits. The scenario involves a private parking area with access control, where only authorized vehicles are allowed to enter. The verification process of a vehicle is performed at the entry to decide whether the vehicle should be allowed access. Three systems have already been deployed in the parking area, each designed for its own functionality. Figure 3.1 (simulated with CARLA [82]) represents this illustrative setup. The following provides a description of the three existing systems.

Description of the Systems

System 1: Barrier management system. Initially deployed to authenticate entering vehicles, the barrier of the parking is managed using a Radio Frequency Identification (RFID) scanner. Each authorized vehicle is equipped with an RFID tag that is scanned at the entry to verify its identity.

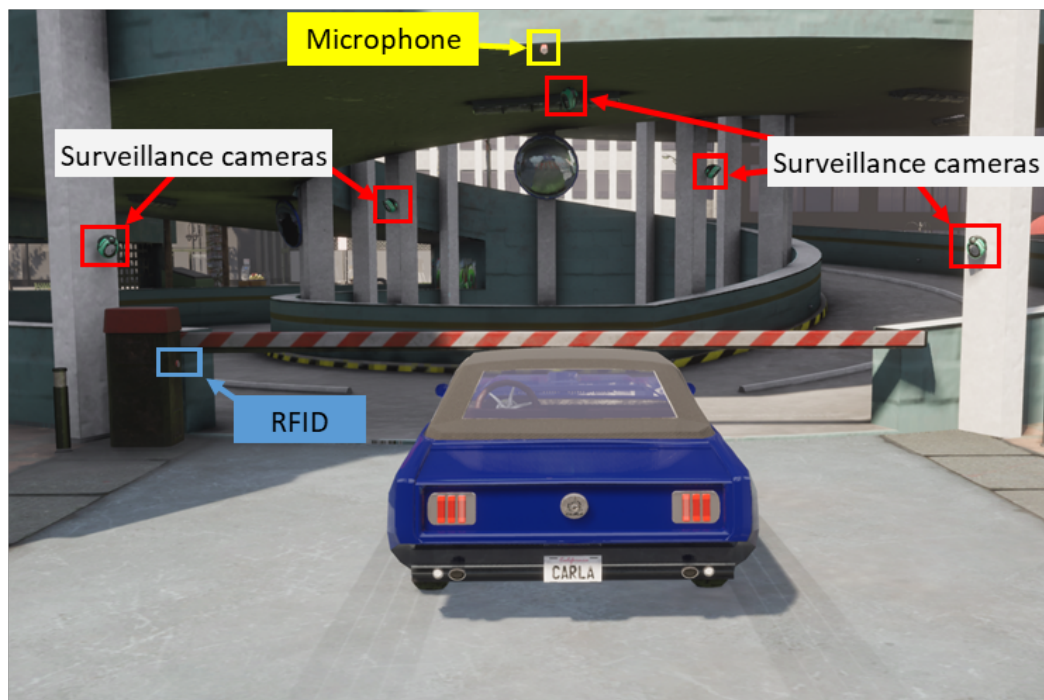


Figure 3.1: Illustrative scenario: parking entry.

System 2: Surveillance system. Deployed to ensure the overall safety of the parking area, including pedestrians and drivers, this system features strategically placed cameras that capture and record images and video footage of different areas. This enables real-time monitoring and enhances security measures.

System 3: Acoustic monitoring system. Deployed for safety reasons, this system detects abnormal sound patterns that may indicate incidents or aggressions. It includes microphones placed in specific locations, and uses signal processing techniques to analyze recorded sounds and identify specific types of sounds or frequencies associated with alarms or other events to ensure security.

In this scenario, the objective is to exploit the existing systems, initially deployed for other purposes than authentication, and the data they collect through their embedded sensors. This exploitation provides new opportunistic authentication factors that could enhance the currently deployed authentication system.

Utilizing Existing Systems

We explore here how existing systems can be repurposed to enhance vehicle authentication in the illustrative scenario. By leveraging data from already deployed acoustic monitoring and surveillance systems, we aim to add new layers of security.

Acoustic monitoring system|Sensors: microphones. While initially designed for safety applications, we propose using this system to strengthen vehicle authentication processes. Microphones placed at parking entrances can capture the engine sounds of vehicles waiting to enter. By comparing these sounds with previously recorded engine sounds of authorized vehicles, we can help in verifying the identity of each vehicle. If the captured sound is sufficiently similar to the known sound of an authorized vehicle, the sound factor is considered validated. This new recording can then be used for future reference. If there is a significant discrepancy between the sounds, it may indicate an unauthorized vehicle or an attempted security breach.

Surveillance system|Sensors: cameras. Surveillance cameras, originally installed for security monitoring, can also enhance vehicle authentication. Various details can be extracted from images of vehicles at the parking entrance. These details can then be compared with information from authorized vehicles to verify identity. If the captured details are sufficiently similar to those of an authorized vehicle, the visual factor is considered validated. Significant discrepancies may suggest an unauthorized vehicle or a potential security threat.

The presence of a diverse set of authentication factors, each with its unique profile (security, energy consumption, etc), creates a complex decision space. This inherent complexity underscores the need for an adaptive selection mechanism, which is the focus of the subsequent sections. The framework must navigate these trade-offs to ensure that authentication decisions are both secure and resource-efficient, responding dynamically to the requirements of the A-IoT environment.

3.2 Trust-Based Adaptive Authentication Framework

Existing studies generally overlook how trust relationships between agents can inform decision-making during authentication processes. Although trust metrics are used to evaluate the reliability of agents, there is a need for frameworks that dynamically adjust authentication requirements based on these trust levels. The

proposed adaptive authentication decision process introduces trust as a two-faceted concept: it serves both as a measure of belief in the reliability of information and as a determinant of the authentication strategy employed. This dual-role of trust enables a more nuanced and context-sensitive approach to security, allowing authentication protocols to be dynamically adjusted based on the trustworthiness of the agents involved. Building upon the fact that IoT offers a large set of possible authentication factors, and leveraging the concept of opportunistic sensor-based authentication factors, the challenge shifts to intelligently selecting the most appropriate combination of these factors for any given authentication instance. As discussed, the diverse characteristics of available authentication factors and the inherent variability of A-IoT environments makes static authentication approaches inefficient and potentially insecure. A fixed set of authentication factors, regardless of its robustness, will inevitably be either overly resource-intensive for simple low-risk interactions or insufficiently secure for critical high-risk scenarios. This highlights the imperative for adaptive authentication to dynamically adjust the stringency and resource expenditure of authentication based on the context and, crucially, the perceived trustworthiness of the communicating agents.

3.2.1 Attack Model

The security model underlying the proposed framework focuses primarily on impersonation attacks, with a particular emphasis on the decision-making process regarding authentication in distributed MAS and IoT environments. An impersonation attack occurs when a malicious agent attempts to assume the identity of another legitimate agent by claiming its unique identifier, for example, in message communications. This type of attack is particularly dangerous in trust-based systems because it compromises the fundamental assumption that trust relationships are established with legitimate agents.

Impersonation attacks, in the context of our framework, specifically target the trust model by claiming identities of highly trusted agents. Once in possession of a trusted identity, attackers can inject false or misleading information into the network, thereby influencing decision-making processes and disrupting the overall functionality of the system. By pretending to be a legitimate agent, the attacker can interfere with or disrupt ongoing communications and transactions. Attackers typically focus on high-trust identities to maximize the impact of their malicious activities. The consequences of such attacks extend beyond the immediate security breach. By compromising previously trusted identities, attackers compromise the credibility of those agents, damage their reputation, and destroy the integrity of

the trust relationships upon which the system relies.

3.2.2 General Architecture

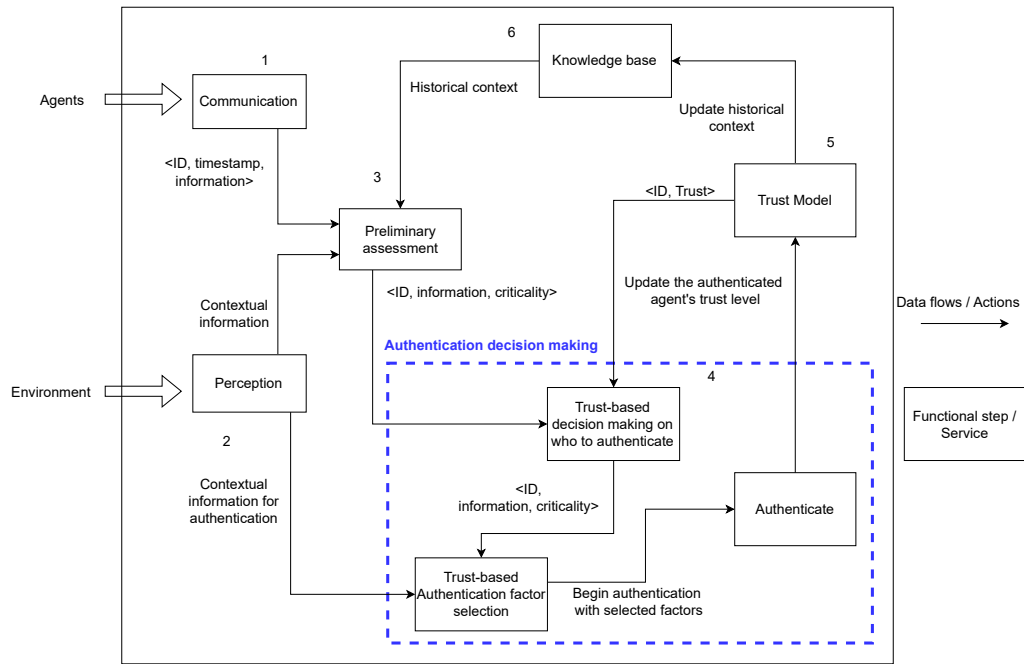


Figure 3.2: Components related to authentication and trust management

The proposed adaptive authentication framework is designed for deployment in A-IoT systems that operate under different challenges, such as resource constraints and dynamic conditions (Challenges 2, 3 and 4). Specific components for trust-based authentication are introduced, designed to be integrated into application agents within a A-IoT system, as shown in Figure 3.2. We consider a heterogeneous set of agents representing IoT objects, ranging from low-power sensors to more complex devices. The agent architecture comprises several integrated components that collectively support trust-based authentication decisions. The communication component (1) serves as the primary interface for agent interactions, allowing agents to communicate between each other. Each agent is assigned a unique identifier (ID) and is capable of sending and receiving messages through this component. For simplicity, we assume that received messages follow the format (claimed_ID, timestamp, content), where:

- Claimed Identity: The declared identifier of the sending agent.
- Timestamp: Indicating when the message was sent.
- Content: The actual information being shared, which varies according to the sender's purpose and the nature of the information being shared.

The perception module (2) enables agents to detect and analyze their immediate environment, providing sensing capabilities that are often limited to adjacent areas in the operational environment. This component can gather contextual information that can be used for authentication, as shown in the illustrative scenario (section 3.1.2). The preliminary assessment service (3) performs initial risk evaluation by analyzing incoming requests and environmental conditions. This component can help in determining the initial appropriate level of authentication rigor required based on the assessed risk level. The authentication component (4) is responsible for verifying whether the claimed ID truly corresponds to the sender's identity, thereby ensuring that trust relationships are only established with legitimate agents. It operates through two trust-based decisions. The first determines whether a given *claimed_ID* should be authenticated at all, while the second enforces multi-factor authentication mechanisms that can be dynamically adapted according to the current trust level.

Agents maintain and update trust values assigned to other agents through a trust model (5), which assesses the trustworthiness of a given ID based on past interactions and the quality of information exchanged. Each agent independently evaluates trust levels, assesses authentication requests, and makes decisions based on its own knowledge and, when available, information from external or third-party sources. Although the choice of trust management system is not the main focus here, it does represent an important decision to be made when implementing the application agent. Additionally, the knowledge base (6) maintains agent profiles along with historical data about the environment. The environment in which agents evolve is characterized by its dynamic nature and the occurrence of unpredictable events. We also focus on the types of environment where agents can be led to simultaneously receive the same type of information from several agents. However, the veracity of this information is sometimes variable and may indicate malicious behavior or an attack attempt by one or more agents.

Overall, the decision-making process enables agents to evaluate the trustworthiness of messages based on the trust assigned to the sender's identity, dynamically

adjusting the authentication rigor accordingly. By considering the risk and resource requirements, the process ensures that agents are authenticated only when necessary, balancing security with the need to optimize energy, computation, and communication resources. This approach is designed to meet the specific constraints of A-IoT environments, aiming to provide a solution that maintains security while minimizing resource consumption.

3.2.3 Authentication Decision-Making Workflow

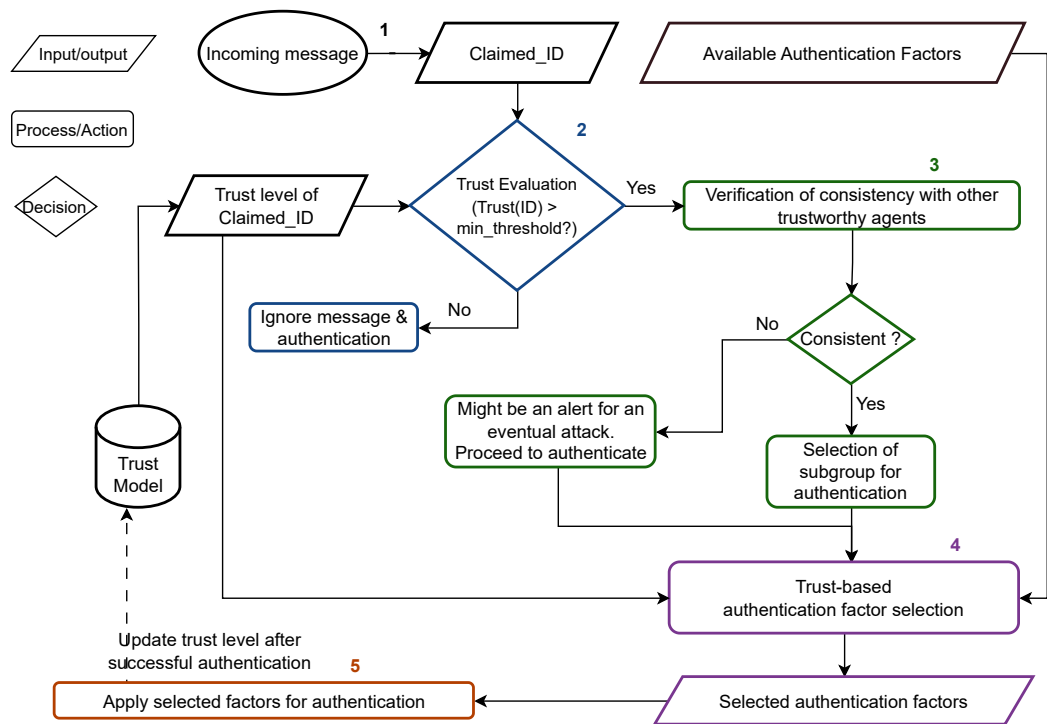


Figure 3.3: Diagram of authentication decision-making process

The authentication decision-making process presented here is performed from the truster's perspective, who must decide whether to authenticate a potential trustee and, if so, which combination of authentication factors to adopt. The model has 3 inputs: (1.) the claimed identity to authenticate, (2.) the trust level associated with the claimed ID from the truster's perspective (retrieved from the trust model), and (3.) the set of available authentication factors that can be used. The output

represents the combination set of authentication factors to be used to authenticate the claimed ID. This process includes five essential steps, each of which may contain several sub-stages, as detailed below. Figure 3.3 illustrates this decision-making process, which comprises the following stages:

1. *Incoming Message*. The decision-making process is triggered when an agent (truster) receives a message from another agent (the potential trustee). The primary objective of authentication is to verify whether the *claimed_ID* accurately represents the sender's true identity. Upon receiving the message, the truster extracts the claimed identity and proceeds to assess whether authentication is necessary in the subsequent step.

2. *Trust Evaluation*. The truster evaluates the trustworthiness of the sender's *claimed_ID* based on the trust model. We define two trust thresholds: a minimum trust threshold, Θ_{min} , at which the *truster* agent accepts to deploy resources for authentication, and a high trust threshold, Θ_{high} , at which we consider the agent to be trustworthy. For each message received, three scenarios are considered based on the level of trust placed in the claimed identity:

- If the claimed identity is that of a trustworthy agent (trust level greater than Θ_{high}), the authentication is initiated to verify the sender's identity and prevent trust exploitation. In this case, a high level of security is applied for authentication, to verify that the claimed identity is indeed the trusted identity. The *truster* agent may however perform a preliminary assessment of the criticality of the information received. It then compares this information, if possible, with other messages received in the same context from other agents claiming trustworthy identities. This comparison of information is intrinsically linked to the specific application in which the agents are deployed, underlining the importance of an adapted methodological approach.
- If the claimed identity is that of an agent the *truster* does not trust (trust level below Θ_{min}), the message is discarded. This is because the *truster* agent would not consider the shared information reliable regardless, due to the insufficient trust in the behavior of the sender's claimed identity.
- If the level of trust attached to the claimed identity is uncertain (trust level between Θ_{min} and Θ_{high}) due to a lack of direct interactions or third-party feedback, a medium security level is applied for identity verification. Alternatively, authentication can be disregarded if other messages on the same

information from trustworthy identities are available.

3. *Consistency check and Context Analysis.* This step is relevant when the *truster* agent receives multiple messages about the same information from agents claiming trustworthy identities, and is particularly relevant to the specific application. For example, consider receiving several messages indicating the temperature at a specific location. If the information is inconsistent, there is a suspicion of a possible attack, leading to the authentication of all agents. If the information is consistent, the *truster* agent selects a subgroup of agents for thorough authentication. We assume that it is highly unlikely that all agents in the initial group are compromised simultaneously while sharing consistent information, thus reducing the number of agents needed for authentication without significantly impacting security.

4. *Trust-based Authentication Factor Selection.* In this step, the truster selects the security level to be applied, which corresponds to the most appropriate authentication factors according to the trust value associated with the `claimed_ID`. Generally, high-trust claimed identities trigger the selection of stricter authentication solutions, prioritizing security to prevent potential trust exploitation through identity spoofing. This precaution is necessary to protect trust-dependent operations, ensuring that the claimed identity corresponds to the true sender. Conversely, lower-trust identities may justify selecting more lightweight, energy-efficient solutions, as the reliance on their information is already limited. The specific details of this selection process are further explained in two different approaches in section 3.3.

5. *Authentication Execution and Evaluation.* Once the authentication factors are selected, the truster assess whether the agent has successfully responded to all chosen factors. If the authentication is successful (the identity is confirmed), the `claimed_ID` is accepted as authentic, and the message content is considered, contributing to future trust evaluations and updates. If the agent fails to respond to the required authentication factors, the identity is considered unverified, and the received information is disregarded. This ensures that only authenticated agents can influence the decision-making process and trust assessments.

3.3 Authentication Factor Selection

We propose here two approaches for the trust-based authentication factor selection (Step 4 of the decision-making workflow). The first, called AAT-L (Adaptive

Authentication based on Trust - Linear approach), adopts an exhaustive linear evaluation strategy: for each authentication request, the system iterates over the entire set of available factors to determine the most suitable subset to apply. While straightforward, this approach incurs a computational cost proportional to the number of factors and must be repeated for every request. To address this limitation, AAT-MOO (Adaptive Authentication based on Trust - Multi-Objective Optimization approach) is presented as an enhancement of AAT-L. It formulates the problem as a multi-objective optimization task. This allows the trust-based selection to be done on the Pareto-optimal solutions only.

3.3.1 Representation of Authentication Factors

Authentication factors are characterized by their ability to provide security while consuming resources. Each authentication factor is defined by three attributes. Let $\mathcal{F}_{\text{avail}} = \{f_1, f_2, \dots, f_n\}$ denote the set of available authentication factors, where each factor f_i is characterized by:

- SL_i (Security level of factor f_i): a normalized measure of robustness against impersonation attacks, on a scale where higher values denote stronger security.
- E_i (Energy consumption of factor f_i): the energy required to perform the authentication factor (in mJ).
- D_i (Authentication delay associated with factor f_i): the time required to execute the factor (in ms).

The security of authentication factors is expressed as a Security Level (SL) on a scale from 1 to 10. This SL is inversely related to the probability of a successful attack: for example, a factor with a security level of 9 corresponds to a success probability of approximately 1% for an attack, computed as $(10 - \text{security_level})/100$. Lower security levels naturally increase the attack success probability. Energy cost is consistently present in both approaches, while delay was explicitly introduced in the multi-objective optimization model.

3.3.2 Linear approach

In AAT-L, the selection of authentication factors is based on a linear scoring mechanism that balances security and energy cost. For each agent to be authenticated following the trust evaluation and consistency check phases, an appropriate security level is determined. This level is determined based on several key criteria,

Table 3.1: Overview of parameters and functions in the authentication decision process of AAT-L.

Parameter/Function	Description
Θ_{min}	Minimum trust threshold to justify an authentication.
Θ_{high}	High trust threshold indicating a trustworthy agent.
τ_{min}	Minimum criticality threshold to justify an authentication.
$Trust$	Trust value in the claimed identity of the sender.
$Crit$	Criticality level of shared information.
w_{SL}	Weight for Security Level (SL).
w_{EC}	Weight for Energy Cost (EC).
MAX_{SL}	Maximum allowable weight for w_{SL} .
a, b	Coefficients balancing the impact of $Trust$ and $Crit$.
$Score_{SL}$	Weighted average SL across utilized factors.
$Score_{EC}$	Sum of energy costs of all utilized factors.
$Score_{Global}$	Overall score for the selected set of factors.

including the level of trust associated with the claimed identity and the criticality of the shared information. We assume the availability of a diverse set of authentication factors, each offering specific trade-offs between energy cost and security level. All parameters and functions used in the AAT-L model are summarized in Table 3.1.

The objective here is then to select the optimal combination of factors according to these criteria. To achieve this, we define :

- *Trust*: Trust value in the claimed identity by the agent seeking authentication, $Trust : id \rightarrow [\Theta_{min}, 1]$, where Θ_{min} is the minimum trust threshold to justify authentication, and 1 represents the maximum trust level.
- *Crit*: The criticality level of the shared information, $Crit : info \rightarrow [\tau_{min}, 1]$, where τ_{min} is the minimum criticality threshold to justify authentication, and 1 is the maximum criticality level.
- w_{SL} and w_{EC} : Weights for SL and Energy Cost (EC), calculated as follows:

$$w_{SL} = \min(MAX_{SL}, a \times Trust + b \times Crit) \quad (3.1)$$

$$w_{EC} = 1 - w_{SL} \quad (3.2)$$

where, $a + b = 1$, MAX_{SL} denotes the maximum allowable weight for w_{SL} , while a and b serve as adjustable coefficients aimed at balancing the impact of *Trust* and *Crit* on both security and cost considerations.

- $Score_{SL}$: Calculated as the weighted average SL across all utilized factors.

$$Score_{SL} = \sum_{i=1}^n w_i \times (10 - SL_i)/100 \quad (3.3)$$

where $\sum_{i=1}^n w_i = 1$, SL_i is the SL of the i -th authentication factor in the combination, and w_i are the weights assigned to each factor, which may be equal or vary according to other criteria.

- $Score_{EC}$: Calculated as the sum of the energy costs of all utilized factors.

$$Score_{EC} = \sum_{i=1}^n cost_i \quad (3.4)$$

where $cost_i$ represents the energy cost of the i -th authentication factor in the combination. These scores must be normalized to ensure that all components contribute in a balanced way to the overall evaluation.

- $Score_{Global}$: A global score is then calculated for each combination of factors using the following formula:

$$Score_{Global} = w_{SL} \times Score_{SL} + w_{EC} \times Score_{EC} \quad (3.5)$$

The objective is to minimize the global score $Score_{Global}$. A low $Score_{Global}$ indicates an effective combination of high SL (low $(10 - SL)/100$) and low energy cost, demonstrating optimal performance of the authentication system in terms of security and energy efficiency. Finally, the combination with the smallest global score is selected and used to perform the authentication.

3.3.3 Multi-Objective Optimization Approach

While the linear weighted approach (AAT-L) provided a solution for trust-based authentication factor selection, it required predefined weights that might not capture the full complexity of trade-offs in dynamic A-IoT environments. To address this limitation, AAT-MOO generates a Pareto front of optimal solutions, allowing for more flexible and context-aware selection. This improvement introduces two

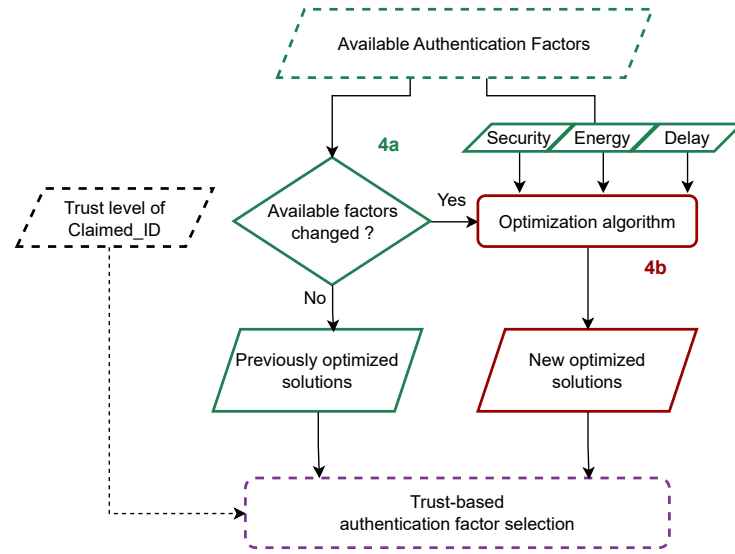


Figure 3.4: New sub-steps in AAT-MOO

additional sub-steps (4a and 4b) to be executed before step 4. This change is illustrated in Figure 3.4 and described as follow:

4a. Authentication Factor Availability Check: The truster checks if the set of available authentication factors has changed. Since the optimization model derives a set of optimal solutions based on these factors, any change requires re-optimization. In dynamic A-IoT environments, factors may become unavailable due to power constraints (e.g., a biometric sensor) or new factors may emerge following device upgrades. If the set of available factors is unchanged, previously computed optimal trade-offs remain valid, allowing the process to proceed directly to trust-based decision-making. Conversely, any modification, such as a factor becoming unavailable or a new factor being introduced, triggers re-execution of the optimization process to generate an updated set of trade-offs.

4b. Optimization Algorithm Execution: If the optimization algorithm must be executed, either during the initial run or due to changes in authentication factor availability, it takes as input the characteristics of the available authentication factors: (i) security level, (ii) energy consumption, and (iii) delay. The algorithm explores the trade-offs among these metrics, producing a Pareto front of optimal authentication solutions. Each solution represents a combination of two or three

factors from the available set, enabling adaptive selection that aligns with system constraints while optimizing security, energy consumption, and latency.

Problem Formulation

The selection of authentication factors is modeled as a multi-objective optimization problem aimed at balancing three competing objectives: maximizing security (equation 3.6), minimizing energy consumption (equation 3.7), and reducing authentication delay (equation 3.8). The optimization problem is formulated as follows: given a set of available authentication factors F_{avail} , we aim to select a subset $A \subseteq F_{\text{avail}}$ that forms an MFA scheme while optimizing the following objectives:

$$\text{Maximize } SL(A) = \sum_{f_i \in A} SL_i \quad (3.6)$$

$$\text{Minimize } E(A) = \sum_{f_i \in A} E_i \quad (3.7)$$

$$\text{Minimize } D(A) = \sum_{f_i \in A} D_i \quad (3.8)$$

Subject to the constraints:

- A must contain at least one authentication factor: $|A| \geq 1$.
- The number of selected factors is constrained by system resources: $1 \leq |A| \leq k_{\text{max}}$, where k_{max} is the upper limit on the number of factors an agent can handle simultaneously.
- Each selected factor must be available in the current context: $A \subseteq \mathcal{F}_{\text{avail}}$, where $\mathcal{F}_{\text{avail}}$ can change dynamically depending on environmental conditions and device capabilities.
- Each authentication factor can be used at most once in a single authentication solution: $\forall f_i, f_j \in A, i \neq j \Rightarrow f_i \neq f_j$.

In a MOO problem, it is generally impossible to find a single solution that optimizes all objectives simultaneously. Instead, the solution space consists of a set of Pareto-optimal solutions, where improving one objective often comes at the cost of another. To navigate these trade-offs, the final decision-making process will be guided by a trust-based approach, ensuring that the selected solution aligns with contextual priorities and dynamically balances the competing factors.

Multi-Objective Optimization with NSGA-II

As discussed in section 2.2.2, NSGA-II is well-suited for the authentication factor selection problem formulated in the previous subsection due to three key characteristics: (i) *fast non-dominated sorting*, which efficiently ranks solutions based on dominance relations, (ii) *elitism preservation*, ensuring high-quality solutions are retained across generations, and (iii) *crowding-distance-based diversity maintenance*, which prevents premature convergence by promoting diversity in the Pareto front. Figure 3.5 illustrates the NSGA-II procedure. The algorithm achieves diversity in solutions through the crowding distance estimation procedure while ensuring convergence toward the Pareto-optimal front using fast non-dominated sorting. The fitness of an individual in NSGA-II is determined by its non-dominance rank within the population of size N . First, fast non-dominated sorting ranks the population members by evaluating their dominance relations, classifying them into different Pareto fronts (denoted as $F_1, F_2, F_3, \dots$). Solutions that are not dominated by any others belong to the first front (F_1), while dominated solutions are assigned to subsequent fronts. Once the fronts are defined, the crowding distance sorting measures the density of solutions in the objective space. It is calculated as the average distance between neighboring solutions along each objective. This sorting ensures that solutions with higher crowding distances are favored to maintain diversity. After sorting and ranking, the new population (P_{t+1}) is generated by combining the parent population (P_t) and the new solutions population (Q_t), and the Pareto fronts are updated accordingly. The process continues, refining the population by selecting the best solutions (denoted as R_t) for the next generation.

In the context of authentication factor selection, an individual in NSGA-II represents a specific combination of factors from $\mathcal{F}_{\text{avail}}$. Each individual is encoded as a subset $A \subseteq F_{\text{avail}}$, where the size of A corresponds to the chosen authentication scheme (e.g., two-factor or three-factor authentication). The optimization process starts with a diverse population of such subsets, and each solution is evaluated based on the three objectives: security, energy consumption, and delay, as defined in equations (3.6), (3.7), and (3.8).

Through iterative selection, crossover, and mutation, NSGA-II refines these combinations, favoring non-dominated solutions while maintaining diversity using crowding distance. Constraints are incorporated by ensuring only feasible solutions are generated during initialization, crossover, and mutation. The process continues until a termination criterion is met, such as a fixed number of generations or convergence of the Pareto front. This enables the algorithm to efficiently explore the

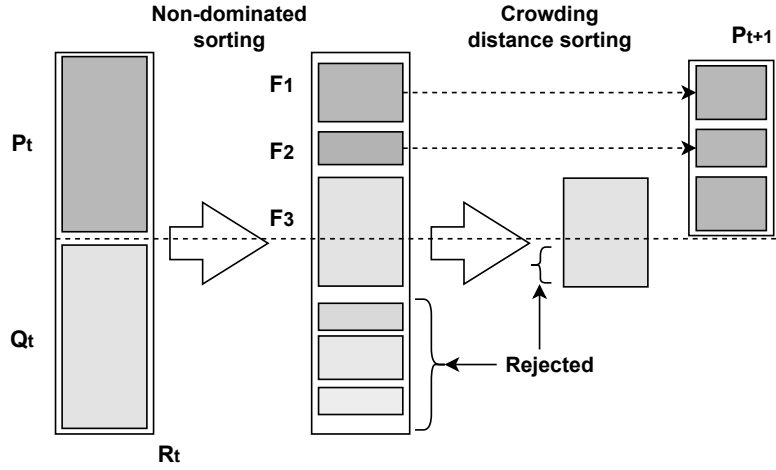


Figure 3.5: General structure of NSGA-II [73]

trade-offs between authentication factors' characteristics, ultimately generating a Pareto front of optimal authentication schemes. Once the Pareto front is obtained, trust values guide the final solution selection, as explained in the following subsection.

Trust-Based Decision Making on the Pareto Front

Given a set of non-dominated authentication solutions obtained through NSGA-II: $\mathcal{S} = \{S_1, S_2, \dots, S_n\}$, the decision process involves selecting the appropriate solution S^* for a given `claimed_ID`. Let $T(\text{ID})$ represent the trust score assigned to the claimed identity, where $T : \text{ID} \rightarrow [0, 1]$.

The selection function $\mathcal{F}_{sel}$ is defined as:

$$S^* = \mathcal{F}_{sel}(\mathcal{S}, T(\text{ID})) \quad (3.9)$$

where $\mathcal{F}_{sel}$ determines the most appropriate solution S^* based on trust and the available set of non-dominated solutions $\mathcal{S}$.

High-trust claimed identities trigger the use of stricter authentication factors to prevent potential exploitation of high-trust identities by malicious actors through identity spoofing, thereby protecting trust-dependent operations. Conversely, for low-trust identities, the truster may opt for less resource-intensive authentication methods or outright rejection of the interaction to minimize risk and resource

expenditure:

$$\mathcal{F}_{sel}(\mathcal{S}, T(\text{ID})) = \arg \max_{s \in \mathcal{S}} (\alpha \cdot \text{SL}(s) - \beta \cdot \text{E}(s) - \gamma \cdot \text{D}(s)) \quad (3.10)$$

where:

- $\text{SL}(s)$ is the security level of authentication solution s .
- $\text{E}(s)$ represents the energy consumption of s .
- $\text{D}(s)$ represents the delay introduced by s .
- $\text{SL}(s)$, $\text{E}(s)$, and $\text{D}(s)$ are normalized to ensure a balanced contribution to the overall evaluation.
- α, β, γ are weight coefficients that vary based on trust:

$$\alpha = \lambda \cdot \frac{1}{1 + e^{(-k \cdot (T(\text{ID}) - c))}} \quad (3.11)$$

$$\beta = (1 - \alpha) \cdot \phi \quad (3.12)$$

$$\gamma = (1 - \alpha) \cdot (1 - \phi) \quad (3.13)$$

where α is determined by a sigmoid function that depends on the trust value $T(\text{ID})$. This function ensures a smooth transition in the weight assigned to security as trust increases. k is a steepness parameter controlling this transition from low to high security priority as trust increases, c represents the tipping threshold; it sets the midpoint of the curve. For example, if $c = 0.5$, the sigmoid reaches 0.5 precisely when the input is 0.5, ϕ represents the importance of energy and delay ($\phi = 0.5$ for equal importance), the coefficients satisfy $\alpha + \beta + \gamma = 1$. Additionally, $\lambda \rightarrow [0, 1]$ controls the influence of trust on the trade-off, allowing for easier integration of other contextual factors in the selection process. For example $\lambda = f(T(\text{ID}), C_1, C_2, \dots)$ where $C_1, C_2, \dots$ represent other contextual factors (e.g., device type, criticality of the operation). For now, $\lambda = 1$ as trust is the primary factor. This allows for a smooth and dynamic adjustment of priorities based on the trust value of the claimed identity.

This ensures that for high-trust identities, security takes priority, while for low-trust identities, resource efficiency is favored. To summarize, the authentication process consists of the following steps:

1. An agent presents a `claimed_ID`.

2. The receiving agent (truster) retrieves $T(\text{claimed_ID})$ and the Pareto-optimal authentication solutions $\mathcal{S}$ from the available set of authentication factors.
3. The function $\mathcal{F}_{sel}$ selects the most suited solution S^* .
4. The selected authentication method is applied.

3.4 Conclusion

In this chapter, we establish a theoretical framework of a trust-based adaptive authentication for A-IoT environments. The concept of opportunistic sensor-based authentication factors is introduced, highlighting how existing IoT infrastructures can be repurposed to provide additional authentication factors without incurring significant hardware costs. It can adapt to different types of sensors and data sources, making it suitable for a wide range of applications. This flexibility introduces an element of unpredictability and diversity into the authentication process, making it more challenging for attackers to predict the authentication factors that will be used. The illustrative scenario of a parking access system show how such factors can be integrated into existing infrastructures, therefore reinforcing existing mechanisms.

Building on this, a trust-based adaptive authentication decision-making model is proposed that explicitly exploits the positive feedback loop between trust and authentication. Trust levels not only influence whether authentication is required but also determine the choice and rigor of authentication factors to be used. The goal of this framework is to address the inefficiencies of static authentication by aligning resource consumption with the assessed level of trustworthiness. Furthermore, we formalize the authentication decision-making workflow and introduce two distinct approaches for authentication factor selection: AAT-L, a linear strategy balancing security and energy consumption, and AAT-MOO a multi-objective optimization approach using NSGA-II to navigate the trade-offs among security, energy, and delay. By restricting the trust-based selection space to Pareto-optimal solutions, this method avoids exhaustive re-evaluation of all factors, except when the set of available authentication factors itself changes.

Impersonation attacks are addressed through a strategy that combines adaptive security, MFA, and trust-based resource allocation. Adaptive security levels enable dynamic adjustment of authentication rigor based on trust levels and contextual factors, ensuring that higher security measures are applied when necessary. This

approach recognizes that not all authentication attempts have equal risk and that security measures should be proportional to both the trust level of the claimed identity and the potential impact of a successful attack. MFA increases the difficulty of successful impersonation attempts by requiring attackers to compromise multiple authentication mechanisms simultaneously. The framework's approach to factor selection aims to choose a combination of factors that provides optimal security while minimizing resource consumption, making attacks both more difficult and more expensive for potential adversaries. Trust-based resource allocation ensures that authentication resources are applied where they provide the most value, focusing intensive authentication procedures on high-risk scenarios while allowing more lightweight authentication for lower-risk interactions.

The potential of these contributions also lies in their generality and application in several IoT environments. Opportunistic factors can be instantiated across various domains, such as smart homes, healthcare, and smart cities, where sensors are already present. By incorporating trust-based decision-making, the framework aims to ensure that authentication is proportionate, efficient, and resilient to impersonation attacks, thereby addressing key gaps between trust and authentication in the current literature. At the same time, important challenges remain, notably in defining systematic methodologies for selecting appropriate opportunistic factors, preserving privacy when repurposing contextual data, and evaluating trust dynamics and influences on authentication in adversarial settings.

Overall, this chapter provides a structured theoretical model for trust-based authentication decision-making in A-IoT. The next step is to evaluate this framework through simulation and experimental validation, in order to assess its effectiveness against impersonation attacks and its ability to maintain an optimal balance between efficiency and adaptability in dynamic A-IoT scenarios.

4

Experimental Evaluation of the Adaptive Authentication Framework

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This chapter presents the experimental evaluation of the proof-of-concept idea of opportunistic authentication factors, as well as AAT-L and AAT-MOO. The practical relevance of opportunistic authentication factors and the adaptive authentication framework can only be demonstrated through experimentation. This chapter therefore details the experiments conducted to assess the feasibility and performance of the proposed approaches under various conditions representative of A-IoT environments.

The experimental study is organized into two complementary parts. First, Section 4.1 provides illustrative experiments demonstrating the feasibility of opportunistic sensor-based authentication factors derived from existing IoT sensors. These preliminary evaluations serve as a proof of concept by showing how sensor-based data can be leveraged to construct additional authentication signals. The results validate the central intuition that opportunistic use of existing IoT infrastructure can reinforce traditional mechanisms and improve multi-factor authentication schemes.

The second part of this chapter focuses on the evaluation of the adaptive authentication factor selection, more specifically, the evaluation of trust-based authentication decision-making models (AAT-L and AAT-MOO). These models aim to dynamically adapt authentication requirements by integrating trust assessments with multi-criteria trade-offs involving security, energy efficiency, and latency. To assess their effectiveness, Section 4.2 presents a simulation setup of a multi-agent IoT navigation scenario in which autonomous agents interact, exchange information, and authenticate one another in uncertain environments populated with honest, malicious, and unpredictable behaviors. Through this simulation-based evaluation, Section 4.3 details the results of comparing the proposed models against baseline authentication decision strategies. Their performance is analyzed across different experimental phases, varying both the complexity of authentication schemes and the composition of agent populations. Particular attention is given to how trust levels influence authentication decisions, and to the balance achieved between security and resource consumption.

4.1 Illustrative Experiments for Opportunistic Authentication Factors

Considering conditions similar to the described illustrative scenario (presented in section 3.1.2), this section evaluates the capacity of these existing IoT systems (surveillance and acoustic monitoring systems) to reinforce the authentication process by creating signatures of the vehicles entering the parking. Specifically, we take advantage of a microphone from the acoustic monitoring system and a camera from the surveillance system to build new signatures of vehicles. These signatures can then serve as an additional authentication layer, complementing existing mechanisms.

4.1.1 Measuring the Similarity of Engine Sounds

To compare two engine sounds, a simple yet effective signal processing method is employed. The objective is to demonstrate the feasibility of the proposed mechanism, focusing on the capability of identifying vehicles based on their engine sound rather than proposing the optimal signal processing method. The spectral centroid feature (Equation 4.1) is used to determine if two engine sounds are similar. It provides the average frequency or the center of mass of a sound spectrum:

$$SpectralCentroid = \frac{\sum f(n)M(n)}{\sum M(n)} \quad (4.1)$$

where $M(n)$ represents the magnitude or weighted frequency value of bin number n , and $f(n)$ represents the center frequency at bin n .

With this feature, to find the similarity or dissimilarity between two signals, the spectral centroid of each signal is calculated separately, then the mean of each spectral centroid is calculated before computing the Euclidean distance between the two means (Equation 4.2):

$$distance = \sqrt{\sum (\overline{SC1} - \overline{SC2})^2} \quad (4.2)$$

where $\overline{SC1}$ and $\overline{SC2}$ are the means of the first and second spectral centroids, respectively. As the distance increases, the dissimilarity between the two signals grows, while a distance close to 0 indicates greater similarity.

4.1.2 Measuring the Similarity of Vehicle Pictures

In the experiments, a series of vehicle images were captured against a consistent background, and the background was then removed to isolate the vehicles. To illustrate the significance and potential of the features extracted from such information, the color histogram of the captured images is used as a feature for authentication. Although color alone may not be a uniquely distinguishing attribute, it can be crucial in identifying significant variations, particularly if a potentially malicious vehicle has a markedly different color. To compare vehicles pictures, color histograms are generated from these isolated vehicle images, representing the distribution of colors for each vehicle. This comparison helps determine if two vehicle images represent the same vehicle based on the distribution of color data. We calculate the Bhattacharyya distance [83] between two normalized color

histograms (Equations 4.3 and 4.4):

$$D_B(P, Q) = -\ln(BC(P, Q)) \quad (4.3)$$

$$BC(P, Q) = \sum_x \sqrt{P(x)Q(x)} \quad (4.4)$$

where BC represents the Bhattacharyya coefficient, and P and Q are the normalized color histograms of the two vehicle images. BC measures the similarity between the two histograms. A coefficient of 0 indicates that the histograms are identical, while higher values indicate greater dissimilarity.

4.1.3 Experimental Setup

The experiments are conducted on two different types of vehicles: cars and mobile robots, and in two different environments: an indoor environment for the mobile robots, where it is calm and there is not much noise when recording the engine sounds, and an outdoor environment for the cars, where there is much more noise. The engines sounds were recorded from several distances from the vehicles to obtain realistic data and to detect if small distances can make a significant difference in the comparison. Tests were repeated on several samples of different durations. For example, one test was performed on samples of 5 seconds, another on samples of 2 seconds, and another on 1 second samples. Regarding the second test, we captured multiple pictures of each vehicle, varying the camera angle slightly for each shot. The collected data from cameras and microphones are used for feature extraction. The extracted features would then be used to generate opportunistic authentication factors, to be integrated with the RFID data into the multi-factor authentication mechanism.

4.1.4 Results

The results of the illustrative experiments conducted are presented here to evaluate the effectiveness of the proposed opportunistic sensor-based authentication factors in the described scenario. The results are divided into two main categories: audio-based factors and visual-based factors.

Audio-Based Results|Engine Sounds

The effectiveness of using engine sounds for vehicle authentication is measured using the spectral centroid feature and the Euclidean distance between spectral

Table 4.1: Average distance between all the spectral centroids.

	Robot 1	Robot 2	Car 1	Car 2	Car 3	Car 4
Robot 1	43.6	467.8	951.3	1078.4	980.5	720.8
Robot 2	467.8	38.5	514.3	635.5	422.8	290.5
Car 1	951.3	514.3	17.2	131.1	76.6	223.9
Car 2	1078.4	635.5	131.1	16.8	110.7	343.4
Car 3	980.5	422.8	76.6	110.7	72.3	280.4
Car 4	720.8	290.5	223.9	343.4	280.4	17.9

Note: **bold** values represent average distances that are true positives, while **red** ones indicate false positives.

centroids of recorded engine sounds. The results are presented in Table 4.1 and Figures 4.1 and 4.2.

The conducted experiments resulted in a 100% accuracy between the robots (indoor) and an 89% accuracy between the cars (outdoor), utilizing a fixed similarity score threshold of 100. Noting that the outdoor environment contains a significant amount of noise, and Figures 4.1 and 4.2 represent the symmetric distances between spectral centroids of different samples of engine sounds for cars and robots, respectively. In these heatmaps, lighter cells represent more similar sounds, while darker cells indicate more different sounds. The distinct clusters of lighter squares diagonally in each heatmap correspond to different vehicles, demonstrating that the engine sounds are consistent within the same vehicle and distinct between different vehicles.

Table 4.1 represents a symmetric distance matrix showing the average distance between all spectral centroids for all the tested vehicles (robots and cars). The diagonal elements of the matrix represent the average distances between different spectral centroid samples of the same vehicle, while the off-diagonal elements represent the distances between different vehicles. From the table, we observe one false positive where the average distance between samples from Car 1 and Car 3 is below 100, as indicated in bold and red. However, it is evident that the average distance between spectral centroids of different vehicles is significantly greater than that between multiple spectral centroid samples from the same vehicle. All diagonal elements were under the fixed threshold, as indicated in bold and black, demonstrating the potential for using such information for authentication purposes.

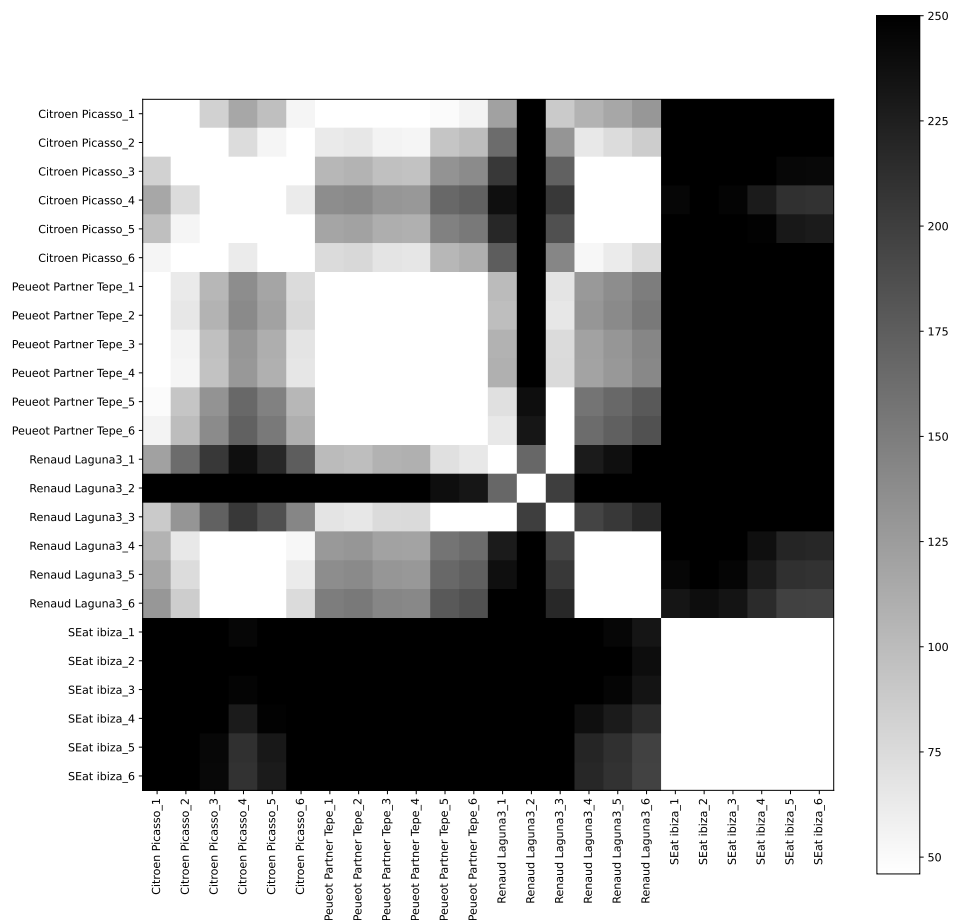


Figure 4.1: Outdoor: Distances between several samples of engine sounds of four different cars.

Visual-Based Results|Color Histograms

The effectiveness of using color histograms for vehicle authentication is measured by calculating the Bhattacharyya distance between the color histograms of captured vehicle images. The results are presented in Table 4.2.

The conducted experiments resulted in a 93.7% accuracy when comparing color histograms of several samples for each car, utilizing a fixed similarity score threshold of 0.2. For the robots, a 100% accuracy was achieved. Three of the four chosen cars for this experiment were selected to have similar color characteristics, minimizing potential bias in the results. Similar to the previous table, Table 4.2 presents

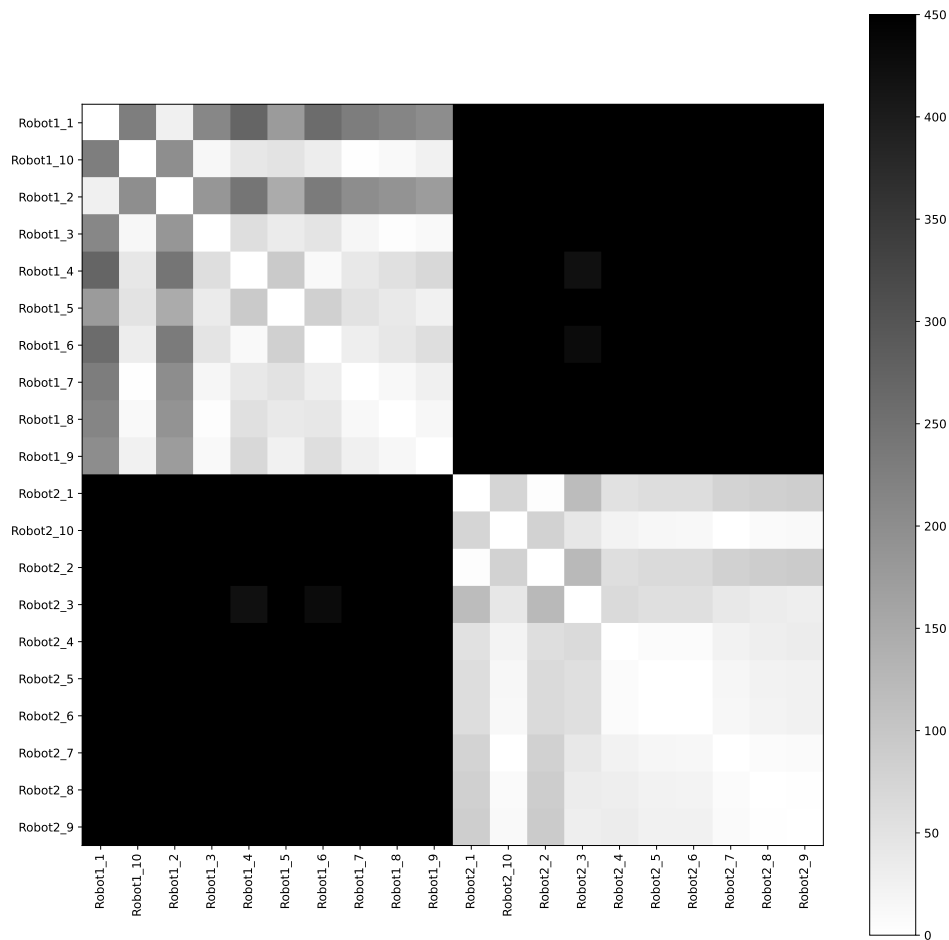


Figure 4.2: Indoor: Distances between several samples of engine sounds of two different robots.

a symmetric distance matrix illustrating the average distance between all color histograms for all the tested vehicles. The average distance between color histograms of different vehicles is significantly greater than the average distance between color histograms of different picture samples from the same vehicle, often exceeding twice the value. The only false negative result is observed in Car 3, where the average distance exceeds the fixed threshold, as indicated in bold and red. This discrepancy can be attributed to slight variations in camera angles during the shots.

The results from the illustrative experiments demonstrate the potential of opportunistically using sensors from neighboring systems to propose additional authen-

Table 4.2: Average distance between the color histograms

	Robot 1	Robot 2	Car 1	Car 2	Car 3	Car 4
Robot 1	0.038	0.31	-	-	-	-
Robot 2	0.31	0.042	-	-	-	-
Car 1	-	-	0.069	0.222	0.382	0.376
Car 2	-	-	0.222	0.145	0.342	0.348
Car 3	-	-	0.382	0.342	0.224	0.382
Car 4	-	-	0.376	0.348	0.382	0.165

Note: **bold** values represent average distances that are true positives, while **red** ones indicate false negatives.

tication factors as a proof of concept ¹. The experiments using color histograms and spectral centroid for vehicle authentication are illustrative, showing that even simple methods can be used to build new authentication factors based on patterns and characteristics. Both methods highlight the potential of leveraging existing sensor data to enhance security. While color histograms can be processed efficiently and implemented using existing camera systems, and the spectral centroid method can analyze engine sounds for authentication, both approaches may be sensitive to various environmental factors. Therefore, their use should be context-aware and ideally part of a multi-factor authentication scheme to ensure robustness and reliability.

In the following subsection, the implications of these results are discussed, as well as the potential benefits, and the challenges associated with implementing opportunistic authentication factors in IoT systems.

4.1.5 Discussion

The concept of opportunistically leveraging existing IoT sensors to create new authentication factors is a promising approach to enhancing the security of IoT systems ². The high accuracies achieved in our illustrative experiments with both audio-based and visual-based authentication demonstrate the feasibility and potential of this method. These results validate the integration of opportunistic factors

¹ The code used for the illustrative experiments is available on GitHub: https://github.com/Marc-SA/illustrative_exp_opportunistic_factors

² This proposition was developed in [84]

with existing authentication mechanisms to create a robust MFA system. There are several potential benefits to combining opportunistic factors:

- **Increased security:** By integrating existing authentication factors, as RFID-based identification in the scenario, with opportunistic factors, such as audio and visual data, the overall security of the authentication system can be significantly enhanced. This multi-factor approach ensures that even if one authentication factor is compromised, additional factors provide robust verification, making unauthorized access much more difficult.
- **Reduced false positives:** Utilizing multiple, diverse authentication factors helps reduce the likelihood of false positives. Each factor adds a layer of verification, increasing the overall reliability of the authentication process.
- **Enhanced robustness:** A multi-factor authentication system that leverages opportunistic data from existing sensors would be more resilient against potential attacks. Attackers would need to compromise multiple authentication factors simultaneously, which is considerably more challenging than targeting a single factor. Additionally, the flexibility of opportunistic factors introduces an element of unpredictability, complicating attackers' efforts as they would need to bypass multiple, distinct, and not predetermined authentication factors.

This method can be adapted to various types of IoT devices and environments by identifying useful data from different sensors and developing context-aware strategies for their use. By quantifying the energy cost, security efficiency, and latency of each sensor-based authentication factor, we can adapt the opportunistic method to other types of IoT sensors and applications. For example, temperature sensors could detect environmental changes, motion detectors could identify unusual activities, and in an industrial IoT environment, vibration sensors could monitor equipment conditions to detect anomalies that suggest unauthorized access or tampering. Designers of IoT authentication protocols play a crucial role in selecting appropriate sensors, integrating sensor data, and implementing flexible authentication processes that enhance security across diverse applications. However, our experiments with sensor data from two environments showed context-dependence, with indoor results outperforming outdoor ones. This outperformance is due to consistent lighting conditions, fewer weather-related disruptions, and less background noise in indoor environments compared to outdoor settings. This highlights the importance of considering the context in which the sensors operate when evaluating the

relevance of each factor.

Furthermore, the ability to build a large set of possible authentication factors provides the system with the flexibility to dynamically adapt to different contexts and threat scenarios. This adaptability allows the system to select the most relevant authentication factors for each specific situation, thus improving the overall security of the IoT environment. Additionally, thoughtful design and context-aware selection strategies are essential for effectively deploying this approach in diverse IoT environments, as these strategies help balance resource demands and security needs. In the following, we present an evaluation of AAT-L and AAT-MOO, focusing on their capability to dynamically select the most appropriate authentication factors based on the trust estimation of the claimed identity of the agent to be authenticated.

4.2 Simulation Setup for Trust-based Authentication Decision-Making

To evaluate the effectiveness and adaptability of AAT-L and AAT-MOO, we focus on assessing how agents make authentication decisions under varying levels of trust and available authentication factors with different characteristics. In particular, the objective is to observe how the different models balance security and energy consumption while leveraging trust assessments. To test and validate these models, a simulation of an IoT environment is implemented that represents a multi-agent navigation scenario. This scenario is suited for evaluating our mechanism due to the dynamic nature of the environment, where agents rely on information from other agents to navigate efficiently. The simulation was developed using the MESA agent-based framework [85]. AAT-L and AAT-MOO, which include a trust evaluation model and an adaptive authentication decision-making process, have been fully integrated into this simulation. Each IoT device is represented as an autonomous agent capable of dynamically evaluating trust levels and making authentication decisions based on the criteria in our models.

4.2.1 Multi-Agent Navigation Scenario

Environment

The navigation space is represented as a 2D grid that serves as a map, providing a spatial framework for the agents' movements. Obstacles are randomly placed on the

map, marking positions that agents cannot cross. The environment is characterized by dynamic events, including shifts in obstacle positions or the introduction of new obstacles, which contribute to its inherent uncertainty.

Agents

Two types of agent are distinguished in the simulation:

1. **Navigators.** A *navigator* agent is an autonomous agent randomly placed on the map and endowed with navigation capabilities. Its objective is to reach a specific, unknown destination while minimizing the navigation distance. It has limited knowledge of the map, being able to detect only its immediate surroundings, which includes all the cells adjacent to the one where it is located on the map.
2. **Guides.** A *guide* agent is an autonomous agent that has no physical presence on the map, but has global knowledge of the map, including the position of obstacles and the destinations of the *navigators*. The guides communicate with navigators, transmitting essential information such as the locations of obstacles and the designated destinations, helping them navigate the map more efficiently and safely.

Figure 4.3 illustrates the MESA simulation environment, represented as a 2D grid. The navigator agents are represented as small circles in red, while obstacles are shown as big circles in black.

Interaction and collaboration

The *guides* communicate the location of obstacles on the map and the destinations to be reached to each *navigator*. The *navigators*, relying on this interaction to obtain the necessary information for navigation, evaluate the information received based on their trust in the *guides* to make decisions about their route on the map. Consequently, information about the same object (map and destinations) is received from different *guides*.

Trust

Navigators use the Beta Reputation System (BRS) [33] to evaluate the trustworthiness of *guides* based on their past actions. Other trust management models can be used, as the choice of trust model is not the central contribution of this study. BRS uses the positive and negative results of previous interactions to calculate the

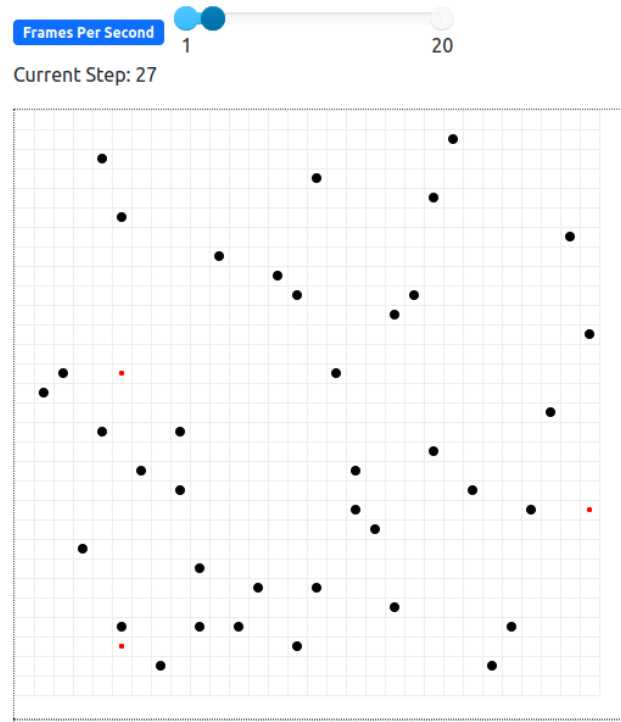


Figure 4.3: Example of the simulation environment with MESA

probability that an agent will act reliably in the future. Mathematically, the trust value of an agent in BRS is calculated using the following formula:

$$Trust = \frac{\alpha}{\alpha + \beta} \quad (4.5)$$

$$\alpha = r + 1 \text{ and } \beta = s + 1 \quad (4.6)$$

where r and s respectively represent the number of positive and negative interactions an agent (*truster*) has had with the trustee in question. This formula provides an estimate of the probability that the agent will behave honestly in a future interaction. A value close to 1 indicates high reliability, while a value close to 0 suggests low reliability.

In this multi-agent navigation scenario, the evaluation of a *guide*'s trustworthiness takes place continuously during navigation, particularly when obstacles are encountered. A negative interaction occurs when there is a mismatch between

the map provided by the *guide* and the navigator's real experience. For example, if the guide indicates an obstacle that does not exist, or omits an obstacle that is actually present, the navigator records this as a negative feedback, thereby reducing the guide's trust level. Similarly, if the final destination given by the guide turns out to be incorrect, this also constitutes a negative feedback. Such inaccuracies can significantly impact the *navigator's* ability to navigate effectively, leading to a negative evaluation of the *guide's* trustworthiness.

Conversely, a positive interaction occurs when the information provided by the *guide* accurately reflects reality. Each time the navigator encounters a real obstacle that matches the guide's map, the system records a positive feedback that generates a positive trust update. Likewise, if the final destination indicated by the guide is correct, it reinforces trust in that guide. Thus, the trust level of a guide is directly shaped by how well their information about obstacles and destinations corresponds to the actual environment.

Given that *navigators* can detect inaccuracies in obstacle positions more rapidly than errors in destination information, we introduce a weight w_β to the parameter β to more strongly penalize *guides* who provide incorrect destination information:

$$s_{t+1} = s_t + (1 \cdot w_\beta) \quad (4.7)$$

This weighting reflects the greater impact of destination inaccuracies on navigation effectiveness and trust. If $w_\beta > 1$, it increases the penalty for incorrect destination information. By applying this penalty, we aim to prevent *guides* from maintaining a high trust score if they provide accurate obstacle information but frequently share incorrect destination details.

Authentication

Navigator agents can use one of several authentication methods that we compare in our scenario, including our proposed AAT-L and AAT-MOO models, along with other baseline approaches, to authenticate *guide* agents. We define several artificial authentication factors for the simulation. The energy cost, delay, and security level values assigned to each authentication factor are synthetic but inspired by plausible characteristics observed in IoT authentication systems, drawn from typical energy consumption patterns, estimated processing and communication times, and multi-criteria security evaluations in the literature [86, 87, 88]. These values are intended to facilitate the exploration of trade-offs between energy efficiency, security, and

delay, rather than representing any specific authentication protocols. Specifically, we account for the inherent correlation between security and energy consumption, as stronger authentication mechanisms typically demand more computational resources. Future work may incorporate empirical data to refine these values. Each factor is characterized by three key metrics:

- Security level (SL) $\rightarrow [1, 10]$ where higher values indicate stronger security
- Energy consumption (EC) $\rightarrow [0.1, 2.5]$ represented in millijoules (mJ)
- Delay (D) $\rightarrow [0.1, 100]$ represented in milliseconds (ms)

To simulate attacks, the security level of each authentication factor, ranging from 1 to 10, is directly linked to its robustness against attacks. For example, a security level of 9 corresponds to an attack success probability of approximately 1%, calculated as $(10 - \text{security_level})/100$, with lower security levels increasing the likelihood that an attack succeeds. Here, an attack is considered successful if the attacker is able to correctly impersonate another agent, thereby bypassing the authentication mechanism.

Depending on the adopted strategy, an agent selects authentication factors in various ways from the available options. The five authentication strategies used in our simulation are:

- Random Authentication Method: A baseline method that randomly selects authentication factors from the available options.
- Static Authentication (SA) Method: A baseline method that uses the same factors for all agents. The selection of these factors follows a high-security strategy that relies exclusively on strong authentication factors, regardless of their energy consumption.
- Trust Rule-based Authentication Method: A baseline threshold-based approach that adjusts factor selection based on three predefined trust thresholds (e.g., $T(\text{ID}) < 0.3 \rightarrow$ low energy, $0.3 < T(\text{ID}) < 0.7 \rightarrow$ balances objectives, $T(\text{ID}) > 0.7 \rightarrow$ high security).
- AAT-L Method: This method (Section 3.3.2) utilizes trust levels to decide which identities to authenticate and then to select appropriate authentication factors for each authentication. The criticality level in this model was originally calculated based on the distance the *navigator* must travel to reach its

destination. However, for the sake of comparison with other methods, this parameter is omitted in the following simulations. This work was detailed in [89].

- AAT-MOO Method: This method (Section 3.3.3) formulates authentication decisions as a multi-objective optimization problem, balancing security requirements, energy consumption, then applies trust-based decision on the Pareto-front. This work was detailed in [90].

4.2.2 Experimental Setup

The experimental evaluation is conducted through a multi-phase simulation study to assess the performance of the different authentication methods across varying scenarios in the multi-agent navigation scenario. Each simulation runs over 1000 episodes, where each episode begins with the initialization of the environment and the placement of the *navigators* on the map, and ends when all the *navigators* have reached their destinations while receiving information from various guide agents. In each episode, the *guides* provide the *navigators* with information about the map. The latter authenticate the messages received and evaluate and update the trust values during navigation.

Table 4.3: Phase 1 - Experimental Setup

Element	Specification
Available authentication factors	8
Selected factors per authentication	2
Episodes per simulation	1000
Navigator agents	5 (each adopting one of the 5 authentication methods)
Guide agents	4 honest, 4 changing-behavior, 2 malicious

The objective is to evaluate the different authentication methods based on three key performance metrics: (i) security efficiency to assess how well each method resists impersonation attacks from agents attempting to deceive navigators, (ii) overall energy consumption during the authentication processes throughout navigation episodes, and (iii) execution time of each method under varying complexity scenarios. The analysis is conducted across the first three experimental phases described above, examining the trade-offs between these metrics as the authentication complexity increases.

Phase 1: Initial Comparison with 8 Available Authentication Factors

The experimental setup of this phase is summarized in Table 4.3. The evaluation begins with a baseline scenario using the 8 authentication factors defined in Table 4.4, where each *navigator* agent selects two authentication factors when authenticating *guide* agents (out of $\binom{8}{2} = 28$ possible combinations). Each *navigator* interacts with 10 *guides* each episode, updating trust values about them dynamically based on the information they send. *Guides* follow one of three behaviors:

- Honest agents, who share correct map information most of the time with some noise added randomly regarding the position of some obstacles.
- Malicious agents, who always share erroneous map information regarding obstacle positions and destinations.
- Changing-behavior agents, who alternate unpredictably between sharing correct and false information.

Table 4.4: 8 Artificial authentication factors

Factor	Security level	Energy cost (mJ)	Delay (ms)
Factor 1	2	0.1	5
Factor 2	2.5	0.25	10
Factor 3	5.5	0.5	30
Factor 4	6	0.65	20
Factor 5	7.5	1.0	50
Factor 6	7.5	1.4	35
Factor 7	8.5	1.6	75
Factor 8	9.5	2	90

Phase 2: Scalability Analysis with 15 Available Authentication Factors

To evaluate scalability in more complex authentication environments, we expand to 15 available authentication factors while maintaining two-factor combinations. Seven additional factors were added to the initial set of eight, as presented now in Table 4.5. The other elements of the experimental setup presented in Table 4.3 were kept unchanged.

Table 4.5: 15 Artificial authentication factors

Factor	Security level	Energy cost (mJ)	Delay (ms)
Factor 1	2	0.1	5
Factor 2	3	0.15	15
Factor 3	2.5	0.25	10
Factor 4	4	0.2	25
Factor 5	5.5	0.5	30
Factor 6	6	0.3	40
Factor 7	6	0.65	20
Factor 8	7	0.8	45
Factor 9	7.5	1.0	50
Factor 10	7.5	1.4	35
Factor 11	8	1.3	60
Factor 12	8	1.2	65
Factor 13	8.5	1.6	75
Factor 14	9.5	2	90
Factor 15	9	2.3	75

Phase 3: Higher Complexity with Three-Factor Authentication

In this phase, three-factor combinations are examined to assess the trust-based authentication decision-making performance under increased complexity. The other elements of the experimental setup are similar to phase 2.

Phase 4: Different Agent Configurations

In this phase, we evaluate the authentication methods under varying agent composition scenarios to understand how different trust distributions among agents affect performance. Instead of our baseline configuration (4 honest, 4 changing-behavior, 2 malicious), we test five different agent configurations that represent diverse multi-agent environments with varying behaviors. Table 4.6 presents the five agent configurations tested.

Table 4.6: Agent Configuration Scenarios for Phase 4 Analysis

Configuration	Description	Honest	Malicious	Changing
C1	Moderate trust env	4	3	3
C2	High trust env	8	1	1
C3	Low trust env	2	5	3
C4	Unpredictable env	4	0	6
C5	Hostile env	1	8	1

4.3 Results and Evaluation of AAT-L and AAT-MOO

4.3.1 Setup for AAT-MOO and Analysis of the Pareto Front

Before presenting the comparative results, we provide illustrations of how the AAT-MOO method operates. NSGA-II algorithm was implemented to optimize the selection of two-factor and three-factor authentication schemes. The key parameters used in NSGA-II process are as follows: crossover rate p_c of 0.9 and mutation rate p_m of 0.1. The fitness evaluation considers the three objectives defined previously. The number of possible solutions out of the 15 available authentication factors for two-factor schemes is $\binom{15}{2} = 105$, while for three-factor schemes, it increases to $\binom{15}{3} = 455$.

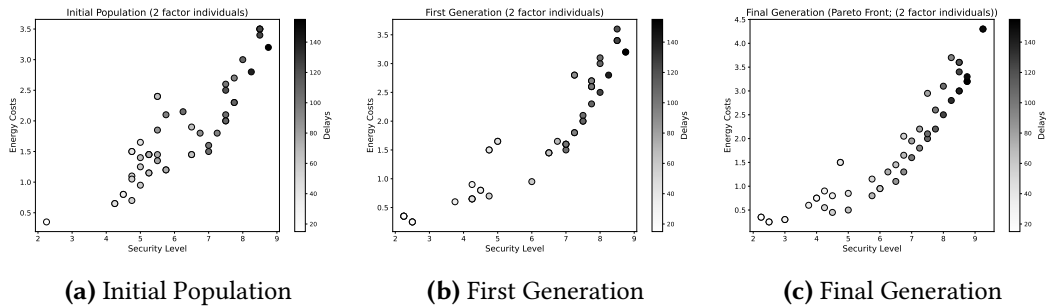
**Figure 4.4:** Results of two-factor authentication factor combinations

Figure 4.4 and Figure 4.5 illustrate the evolution of the optimization process using NSGA-II for two-factor and three-factor authentication schemes (combinations from Table 4.5), respectively. The 3D plots represent security on the x-axis, energy cost on the y-axis, and delay indicated by the color-bar. Each figure displays three key stages: the initial population shows a random distribution of authentication factor combinations across the objective space, the first generation demonstrates

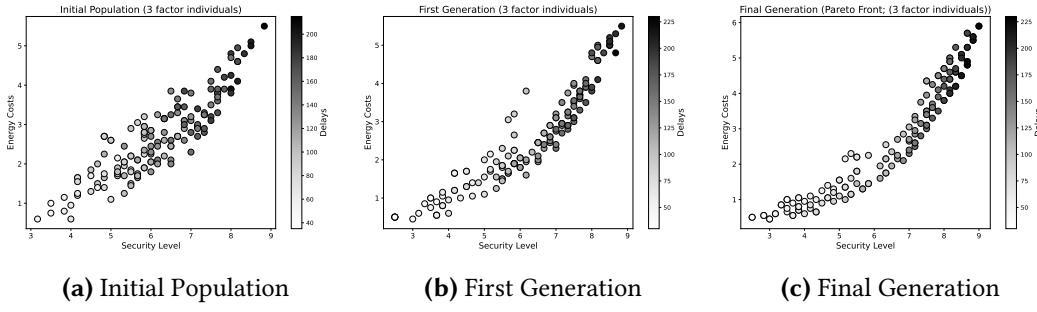


Figure 4.5: Results of three-factor authentication factor combinations

the beginning of the optimization process, and the final Pareto front represents the optimal trade-off solutions. In both authentication schemes, the Pareto front was successfully obtained after 3 to 5 generations, demonstrating the efficiency of the NSGA-II algorithm in converging to optimal solutions that balance security, energy consumption, and delay considerations.

As the optimization progresses, the algorithm gradually refines the population towards a set of non-dominated solutions forming the Pareto front. The final Pareto fronts for two-factor and three-factor schemes highlight the fundamental trade-offs inherent in authentication scheme selection: (i) Trade-off between security, energy, and delay: increasing security generally leads to higher energy consumption and increased delay. Solutions that minimize energy consumption tend to offer lower security levels. (ii) Impact of adding a third factor: the Pareto front for three-factor schemes extends beyond the two-factor front, reflecting the potential for achieving higher security. However, this comes at the cost of increased energy consumption and delay, emphasizing the growing complexity with additional factors.

The analysis of the Pareto front highlights that selecting an optimal authentication scheme requires navigating these trade-offs. The Pareto front analysis underscores the importance of balancing security, energy, and delay when selecting authentication schemes. The trust-based decision process presented in the next subsection dynamically guides this selection.

The trust-based selection mechanism (Equation 3.10) is then applied on the Pareto-optimal solutions obtained via NSGA-II. This mechanism adjusts the selection based on dynamic trust levels, with parameters α , β , and γ computed according to the trust value and steepness parameter k . Higher values of k make the selection more

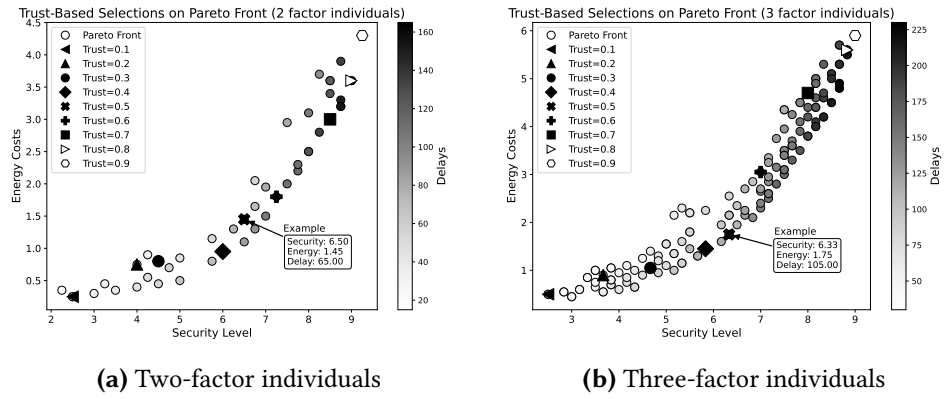


Figure 4.6: Visualization of trust-driven selection on the Pareto front

sensitive to changes in trust.

For illustration purposes, a series of tests are performed using a range of trust values $T(\text{ID}) \in [0, 1]$. For each trust value, the selection process was applied to the Pareto-optimal solutions generated by NSGA-II. The impact of varying trust values on the chosen authentication solution was observed across several scenarios. Figure 4.6 illustrates the relationship between the trust value and the selected authentication solution. As trust increases, the selection mechanism prioritizes solutions with higher security levels, while minimizing energy consumption and delay as secondary factors. Conversely, for low-trust values, the selection process favors solutions that reduce resource consumption, opting for less secure but more energy-efficient or lower-delay authentication methods.

4.3.2 Phase 1 Results: Baseline Evaluation

We evaluate all five authentication methods using 8 available authentication factors (Table 4.4) with 2-factor combinations. We first present results from a single simulation run over 1000 episodes to demonstrate the behavioral patterns of each method, followed by statistical analysis based on mean results from 100 independent runs to ensure robust validation of the findings.

Security vs. Energy Performance Analysis

Figures 4.7 and 4.8 present the security and energy performance of each authentication method, revealing distinct trade-offs between security effectiveness and

resource efficiency.

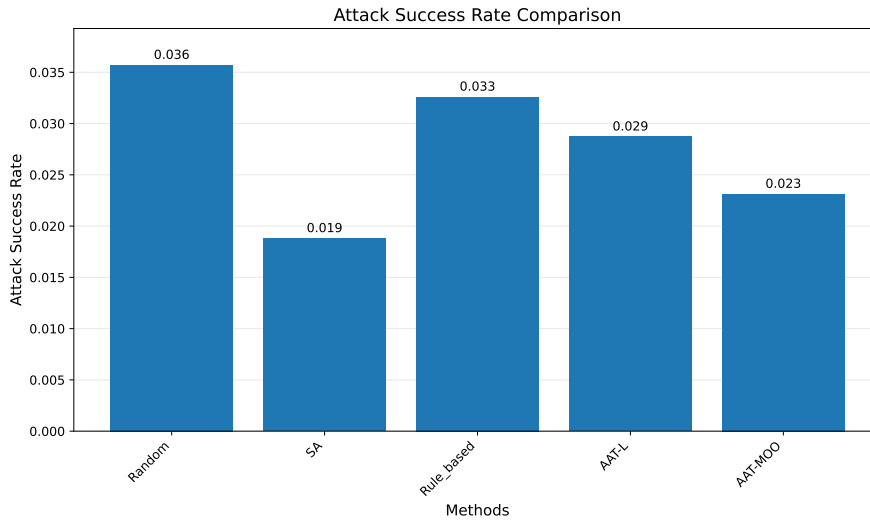


Figure 4.7: Attack success rate - Phase 1

The random approach, by selecting factors arbitrarily, leads to both higher attack rates and greater energy consumption. Static Authentication demonstrates the highest security resilience with the lowest attack success rate of 0.015. While it minimizes the likelihood of attacks, it incurs significant energy consumption, as shown by the green line. This superior performance comes from its consistent use of high-security authentication factors for every authentication decision, providing maximum protection against impersonation attempts but completely disregarding energy efficiency considerations. The trade-off represents a "security-first" approach that prioritizes protection over resource optimization. The rule-based approach, which relies on trust values assigned to three clusters, outperforms the random strategy in terms of security. However, it does not reach the same balance between security and energy as AAT-L and AAT-MOO. Despite their limitations, these three approaches, remain the quickest and simplest in terms of computational requirements, each taking on average less than $0.09ms$.

AAT-L achieves a comparable security performance to AAT-MOO with an attack success rate of 0.023. However, it consumes more energy than AAT-MOO because the selected factors are not necessarily drawn from the Pareto front, making them both more energy-demanding and less resilient to attacks. In addition, although

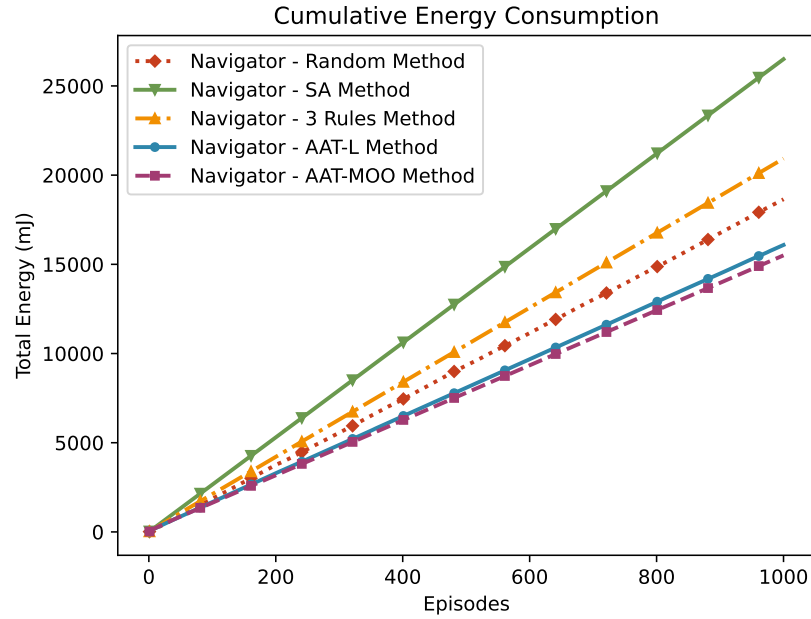


Figure 4.8: Cumulative energy consumption - Phase 1

the method achieves close security to AAT-MOO through its trust-based evaluation process, it incurs higher computational overhead in terms of execution time, as it must evaluate all possible combinations of factors for each authentication decision. Specifically, the selection process of 2 factors among 8 takes on average $0.394ms$.

AAT-MOO demonstrates a balanced approach between security and energy efficiency, achieving the second-best security performance with 0.021 attack success rate while maintaining the lowest energy consumption overall. The method effectively navigates across the Pareto front to select factor combinations that provide strong security when needed while conserving energy when the claimed identities are less trustworthy. This adaptive approach showcases the effectiveness of multi-objective optimization in achieving practical compromises between competing objectives. In this phase, the selection process of the best trade-off solution among the Pareto-optimal solutions takes on average $0.312ms$, while the NSGA-II algorithm takes on average $2.553ms$ to provide the Pareto front. However, as long as the set of available authentication factors does not change, NSGA-II needs to be executed only once to generate the Pareto front.

Generally, in both AAT-MOO and AAT-L, by avoiding authentication of agents with whom previous interactions have provided sufficient but inconclusive trust information, and by adapting authentication factors based on established trust and the criticality of exchanged information, we enhance energy efficiency. Authentication occurs only when the risk justifies the energy expenditure. This approach allows resources to be focused where they provide the most value, directly contributing to observed energy savings.

Impact of Attacks on Navigation Performance

Table 4.7 illustrates how the security-energy trade-offs ultimately affect navigation efficiency. The optimal navigation path for each navigator agent was 20 steps per episode under normal conditions with accurate information from trustworthy guide agents. Each successful attack attempt allows the attacker to transmit erroneous map information to the *navigators*, leading to misguided paths. Specifically, when a *navigator* accepts an incorrect map sent by an attacker, the number of steps required to reach the destination can increase to over 50. This impact may not always be apparent in the average steps per episode, due to the relatively small number of successful attacks compared to the total interactions.

The results reveal that methods achieving better security-energy balance also maintain navigation performance closer to optimal. Static Authentication's superior security keeps navigation steps closest to the optimal 20 steps, but at significant energy cost. AAT-MOO achieves nearly comparable navigation efficiency while consuming substantially less energy, demonstrating the practical value of balanced optimization. Methods with higher attack vulnerability show progressively worse navigation efficiency, as successful impersonation attacks force navigator agents onto longer, suboptimal paths.

Table 4.7: Attack success rate and average steps per episode over 1000 episodes

	Random	SA	Rule-based	AAT-L	AAT-MOO
Successful attack rate	3.6%	1.9%	3.3%	2.9%	2.3%
Avg steps per episode	21.33	20.21	21.26	20.75	20.55

Statistical Validation

The statistical performance analysis across 100 independent simulation runs is summarized in Figure 4.9 and Table 4.8. AAT-MOO achieves the lowest mean energy consumption over 1000 episodes (15,512.84 mJ) with moderate variability (std = 232), outperforming all other methods. Static Authentication provides the best security performance (1.66% attack success rate) with a small variability (std = 0.13) but at the cost of the highest energy consumption (26,500.00 mJ). The results demonstrate that AAT-MOO provides the optimal balance between energy efficiency and security effectiveness, consuming 41.5% less energy than Static Authentication while maintaining competitive security performance with only 0.69 percentage points difference in attack success rates.

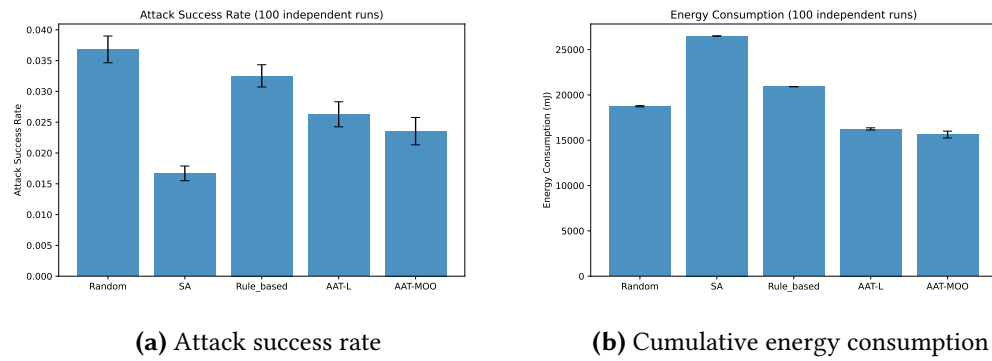


Figure 4.9: Comparison of attack rate and energy consumption over 100 independent runs

Table 4.8: Phase 1 - Statistical performance analysis over 100 independent runs

Method	Energy Consumption (mJ)			Attack Success Rate (%)		
	Mean	Min	Max	Mean $\pm$ Std	Min	Max
Random	18,737.14 $\pm$ 81	18,569	18,968	3.68 $\pm$ 0.17	3.33	4.04
SA	26,500.00 $\pm$ 0	26,500	26,500	1.66 $\pm$ 0.13	1.27	1.97
Rule-based	20,911.65 $\pm$ 23	20,809	20,980	3.23 $\pm$ 0.22	2.82	3.84
AAT-L	16,229.57 $\pm$ 146	15,737	16,537	2.61 $\pm$ 0.21	2.22	3.28
AAT-MOO	15,512.84 $\pm$ 232	13,655	15,756	2.35 $\pm$ 0.20	1.89	2.86

Bold values indicate best performance per metric

4.3.3 Phase 2 Results: Scalability Evaluation

We expand the authentication complexity by increasing the available factors from 8 to 15 (Table 4.5) while maintaining 2-factor combinations for authentication. The statistical validation of this phase is presented in Table 4.9. The overall performance rankings and behavioral characteristics remain consistent with Phase 1 results. However, this phase reveals insights into the computational scalability of different authentication methods as the solution space expands from $\binom{8}{2} = 28$ to $\binom{15}{2} = 105$.

The results reveal significant differences in scalability performance. The AAT-L method execution time increases from $0.39ms$ to $1.558ms$ when the number of factors increases from 8 to 15. This growth comes from AAT-L's requirement to exhaustively evaluate all possible factor combinations for each authentication decision, making it increasingly impractical as the solution space expands. In contrast, AAT-MOO demonstrates excellent scalability characteristics. The factor selection time increases minimally from $0.312ms$ to $0.475ms$, demonstrating that AAT-MOO, by navigating the Pareto-front thanks to the multi-objective optimization approach, scales efficiently with problem complexity. The NSGA-II algorithm shows moderate increase from $3.553ms$ to $9.689ms$. The results demonstrate that AAT-MOO not only achieves optimal security-energy trade-offs but also maintains practical feasibility as system complexity increases. Additionally, AAT-L considers only security and energy consumption, whereas AAT-MOO also optimizes for delay.

Table 4.9: Phase 2 - Statistical performance analysis over 100 independent runs

Method	Energy Consumption (mJ)			Attack Success Rate (%)		
	Mean	Min	Max	Mean $\pm$ Std	Min	Max
Random	$19,297.78 \pm 71$	18,209	18,457	3.53 ± 0.24	3.29	4.18
SA	$28,000.00 \pm 0$	28,000	28,000	1.14 ± 0.10	0.97	1.32
Rule-based	$19,800.79 \pm 42$	19,689	19,821	2.82 ± 0.13	2.44	2.89
AAT-L	$16,650.23 \pm 80$	16,592	16,751	2.41 ± 0.16	2.22	2.78
AAT-MOO	$16,187 \pm 412$	15,748	16,721	2.06 ± 0.23	1.76	2.33

Bold values indicate best performance per metric

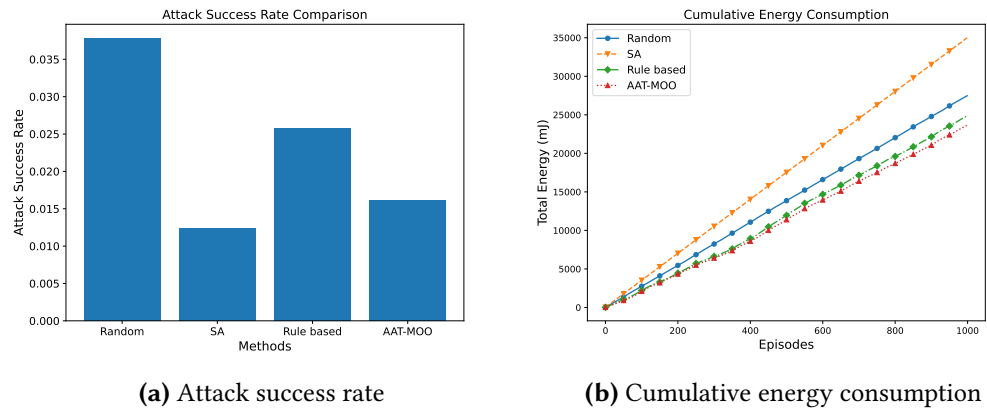
4.3.4 Phase 3 Results: Higher Complexity Evaluation

We increase the authentication complexity by transitioning from 2-factor to 3-factor combinations for authentication while maintaining the 15 available authentication factors. This expansion significantly increases the solution space from $\binom{15}{2} = 105$ to

Table 4.10: Execution Time Scalability Analysis: AAT-L vs AAT-MOO Across All Phases

Configuration	Combinations	Execution Time (ms)		
		AAT-L	AAT-MOO	NSGA-II
8 factors, 2-factor selection	28	0.39	0.31	2.55
15 factors, 2-factor selection	105	1.55	0.47	9.69
15 factors, 3-factor selection	455	4.85	0.64	30.83

$\binom{15}{3} = 455$ possible combinations, representing a 4.3 times increase in complexity. Table 4.10 presents the execution time comparison across all three experimental phases, demonstrating the critical scalability differences between authentication methods. AAT-L increases significantly from 1.55ms to 4.85ms, making it impractical in this configuration. AAT-MOO maintains excellent scalability with execution time increasing minimally from 0.47ms to 0.64ms across all phases. The NSGA-II algorithm for generating Pareto fronts shows increases from 2.55ms to 9.69ms to 30.83ms as complexity grows. AAT-MOO becomes computationally critical only when the set of available authentication factors changes frequently, requiring NSGA-II to regenerate new Pareto fronts. In stable environments where the factor set does not change very often, the optimization overhead becomes minimal across all authentication operations, making AAT-MOO highly efficient for practical deployment.

**Figure 4.10:** Comparison of attack rate and energy consumption on three-factor individuals

The results, shown in Figure 4.10, present the security and energy performance of the authentication methods excluding AAT-L over 1000 episodes with 3-factor combinations. The results confirm the performance patterns observed in previous

phases, while the overall attack success rate decreases as a result of using three factors per authentication. Static Authentication maintains the highest security resilience with the lowest attack success rate but consumes the most energy due to its consistent use of high-security factors. AAT-MOO achieves again the optimal trade-off, providing competitive security performance while maintaining superior energy efficiency.

4.3.5 Phase 4 Results: Different Agent Configurations

The results in this phase, presented in Figure 4.11, demonstrate AAT-MOO's adaptive behavior across different trust environments (Table 4.6), revealing both its strengths and limitations based on agent composition.

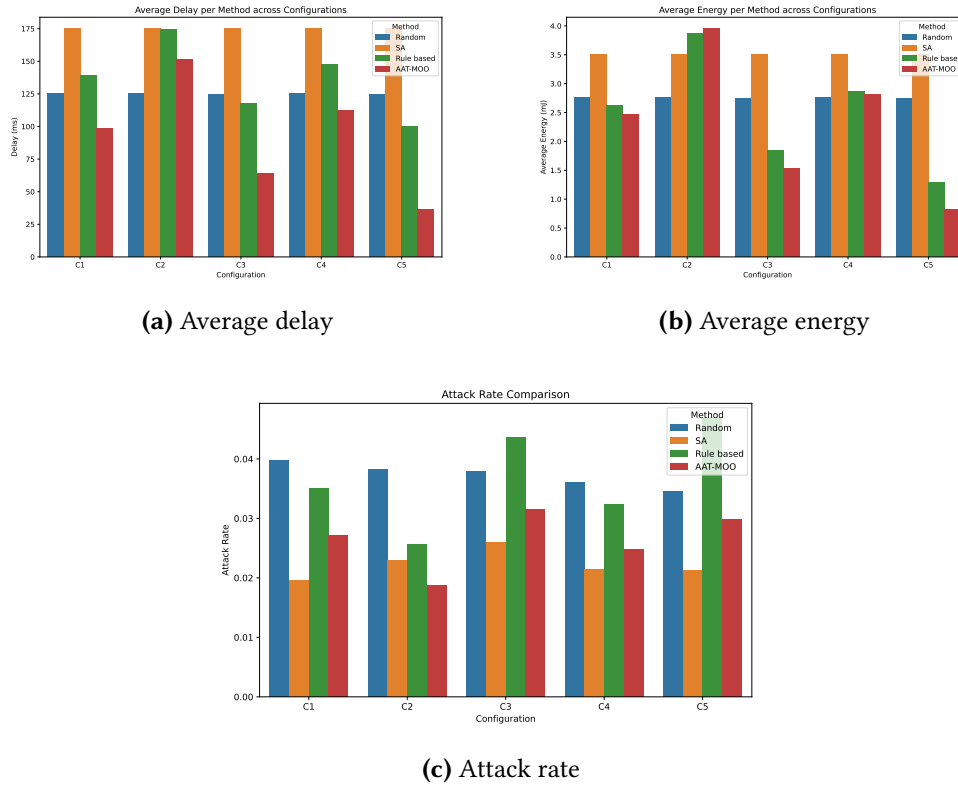


Figure 4.11: Comparison of results in different configurations

In configuration C2, where 80% of agents are honest, AAT-MOO achieves exceptional security performance, demonstrating even higher resistance to attacks than

static authentication method. AAT-MOO selects highly secure authentication factors when interacting with the dominating trustworthy agent population, resulting in superior security performance. However, this comes at increased energy consumption as the method prioritizes security when trust levels justify the additional resource expenditure.

In balanced configurations, such as C1 and C4, AAT-MOO maintains its characteristic trade-off optimization, adapting factor selection based on the mixed trust distribution, similar to the initial configuration tested in the previous phases. However, in low trust environments, such as C3 and C5, with only 20% and 10% honest agents respectively, AAT-MOO shows significant vulnerability. The method's trust-based optimization leads to the selection of less secure authentication factors overall, resulting in decreased resistance to attacks compared to other configurations.

This analysis reveals that the trust-based authentication decision-making effectiveness is inherently linked to the trust distribution within the multi-agent environment. In high-trust scenarios, the method's adaptive nature allows it to outperform even Static Authentication by applying strong security measures where trust levels warrant the investment. Conversely, in low-trust environments, the method's energy-conscious optimization may compromise security, making it less suitable for hostile multi-agent scenarios. While extreme scenarios, like C5 (with 90% malicious agents), represent challenging cases that are relatively rare in typical IoT deployments, they still require careful consideration during system design.

4.4 Conclusion

This chapter presents a comprehensive experimental evaluation of the trust-based adaptive authentication framework. The results collectively demonstrate both the feasibility of opportunistically leveraging IoT existing IoT sensors as additional authentication factors, and the effectiveness of trust-based decision-making in dynamically selecting authentication factors under diverse conditions.

The first illustrative experiments confirm that analyzing sensor-based data, even with simple processing methods, such as a spectral centroid analysis of engine sounds or histogram-based comparison of vehicle images, can generate reliable opportunistic patterns for authentication. While these factors alone are not sufficient for robust identity verification, they provide valuable complementary evidence that can be integrated into multi-factor authentication processes. This proof of

concept validates the central premise that existing IoT infrastructures can be opportunistically exploited to enhance security without requiring significant additional hardware deployment. By creating a set of possible authentication factors, this approach allows for dynamic selection based on current context and constraints. This selection of factors depends on several key criteria, including but not limited to resource availability, threat level (criticality of the interaction or the exchanged information), trust values, and environmental conditions.

The second part of the evaluation focused on the trust-based adaptive authentication models, AAT-L and AAT-MOO, within a simulated multi-agent navigation scenario. Results show that AAT-MOO achieves the most favorable balance between security, energy efficiency, and computational scalability. By employing multi-objective optimization, AAT-MOO first generates a Pareto front of non-dominated authentication solutions, capturing optimal trade-offs between the objectives. A sigmoid-based trust-weighting mechanism then dynamically prioritizes solutions from this Pareto front, favoring high-security configurations for highly trusted identities to prevent trust exploitation by identity spoofing, and energy-efficient configurations for low-trust claimed identities. Overall, AAT-MOO outperforms AAT-L, as well as static and rule-based baselines in most scenarios. Nevertheless, the analysis also highlights several limitations. The effectiveness of AAT-MOO decreases in highly adversarial or trust-scarce environments, where malicious agents dominate the population, revealing the dependency of trust-based decision-making on the surrounding trust distribution. Moreover, AAT-MOO's computational cost rises when authentication factors evolve frequently, requiring new Pareto fronts to be regenerated. In such conditions, AAT-L becomes a preferable alternative: its simpler linear formulation provides faster adaptation and lower overhead, which may be more suitable for embedded or constrained IoT devices.

Future work should extend these evaluations toward more realistic and adversarial scenarios, including coordinated multi-agent attacks and dynamic trust manipulation strategies. Testing the framework on real-world testbeds, for example, using embedded IoT platforms would allow verification of its robustness and scalability under real constraints. Finally, validating the framework with real authentication factors rather than synthetic ones remains an essential step toward demonstrating its practical relevance.

5 From Passive to Active Trustees: A Bidirectional Trust Model as an Asset

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This chapter introduces a distinct theoretical contribution to trust management, differing from the contributions presented in the two previous chapters. It shifts the focus toward resource-tolerant IoT environments, where higher computational capabilities enable the integration of more advanced technologies such as blockchain-based mechanisms.

Traditional computational trust and reputation models have largely been built upon a unidirectional and passive paradigm. A truster agent actively evaluates passive trustee agents by aggregating historical data, often in the form of direct experiences or third-party opinions, to calculate a final trust or reputation score. The trustee is a subject of evaluation, unaware of its own standing or the criteria by which it is judged. Therefore, trustees are evaluated by others through feedback or observation, but they do not actively participate in the construction of their own trust profile. Their trust standing is effectively external and may remain opaque to them until a decision is made elsewhere. In addition, the one-way decision process optimizes the truster's choice of trustee and implicitly assume that a truster's selection is the final step before an interaction. The trustee is assumed to accept service provision whenever selected, ignoring the fact that trustees also incur costs and risks and must also find the truster worthy of its service. Furthermore, a trustee may wish to avoid trusters with a history of providing false information or giving unfair feedback. If trustees cannot leverage their good behavior as a portable, verifiable asset under their ownership, the long-term incentive to behave honestly is reduced.

In the context of open and large-scale IoT environments, these limitations become even more critical (Challenges 1 and 3). Each party, whether truster or trustee, faces potential risks regarding data integrity, reliability, and security. In such settings, trust decisions cannot rely solely on passive evaluation, they must instead support mutual verification, enabling both sides to evaluate one another before establishing cooperation.

This chapter argues that the traditional paradigm fails to capture the reciprocal, participatory, and strategic nature of trust. This chapter therefore presents theoretical propositions toward a bidirectional trust model in which trustees are active participants in the trust relationship. In this view, trustees can obtain and present verifiable proofs of past reliability; trust decisions become bidirectional, with both parties evaluating and conditioning their behavior on cryptographically verifiable evidence of past interactions; and these proofs function as incentive assets that convert honest behavior into future selection advantages. The proposed model removes the purely passive role traditionally assigned to trustees, reconceptualizing trust as a portable, contextual, and verifiable asset.

To make this vision possible, this chapter considers IoT infrastructures that are capable of ensuring the immutability, traceability, and verifiability of such proofs of trustworthiness. Technologies such as blockchain can play a crucial role in this context by enabling decentralized storage and validation of trust records without

relying on a central authority. This chapter therefore focuses on blockchain-enabled IoT environments that are more tolerant to resource-constraints challenges, where devices can generate and exchange verifiable trust assets through lightweight, token-based mechanisms.

Building on this motivation, this chapter pursues the following theoretical objectives:

1. Define proof of trustworthiness as an asset.
2. Formalize bidirectional trust dynamics.
3. Establish incentive mechanism.
4. Ensure decentralization and verifiability.

This chapter first discusses the need to re-think multi-agent trust management systems, then introduces the proposed theoretical model. It presents the underlying concepts, the use of Non-Fungible Tokens (NFTs) as trust assets, and a high-level architecture of the model. An interaction protocol is then outlined to illustrate the bidirectional trust dynamics, followed by a description of the mechanisms for issuing and verifying tokens.

5.1 Why re-think trust management?

This section presents ideas and discussions that aim to shift the way multi-agent trust management models are built. It begins by identifying the asymmetry problem in existing approaches, which largely consider trust as a one-way evaluation process. Then, it discusses the concept of *trust as relational capital* to provide a theoretical foundation for viewing trust as an asset that agents can actively build and manage. Following this, the section discusses how such a perspective can shift trust management from a passive evaluation process toward active trust engagement, where agents take ownership of their trust credentials. The discussion then turns to the incentive challenges that arise in ensuring trustworthy behavior and how the trust as an asset approach helps align self-interest with cooperation within blockchain-enabled IoT environments.

5.1.1 The Asymmetry Problem in Traditional Trust Models

Trust management in MAS has historically been conceived as a unidirectional evaluation process: trusters assess trustees, make selection decisions, and the cycle repeats. This paradigm, while computationally tractable and widely implemented, has a fundamental asymmetry that misrepresents the nature of trust in collaborative environments. Trustees are cast as passive objects of evaluation (measured, scored, and selected) without agency in the relationship formation process and often without knowledge of how they are perceived by others.

This asymmetry creates several critical problems. First, it ignores the reality that trustees also face risks when engaging with unreliable or malicious trusters. A trustee that provides high-quality services may find its resources exploited by trusters who provide false feedback, fail to fulfill their obligations, or misuse the provided services. Second, the unidirectional model provides no mechanism for trustees to accumulate and demonstrate their trustworthiness proactively. Instead, they must wait passively to be evaluated, with no control over their reputation or ability to leverage past good behavior. Third, the model fails to create proper incentives for trustworthy behavior, particularly in environments where repeated interactions are not guaranteed.

5.1.2 Trust as Relational Capital: A Theoretical Foundation

[10] highlight the gap in this asymmetric paradigm through their theory of trust as relational capital. They argue that being trusted is itself a strategic resource: it increases the probability of being selected as a partner, raises an agent's negotiation power, and functions as an individual form of social capital that can be accumulated, managed, and even strategically invested. This perspective transforms trust from a mere evaluation metric into an asset that agents can deliberately cultivate and strategically deploy.

The relational capital view introduces several crucial insights that traditional trust models fail to capture. Being trusted is not simply a passive state but an active achievement that requires effort, consistency, and strategic awareness. Agents who are trusted gain tangible advantages: they are selected more frequently for collaborations, can command better terms in negotiations, and enjoy greater flexibility in their strategic choices. Conversely, agents who lack trust capital find themselves excluded from valuable opportunities, regardless of their actual competence or

good intentions.

This conceptualization reveals a gap in existing computational trust models: they measure and compute trust values, but they provide no mechanism for agents to actively build, prove, or leverage their trustworthiness as capital. The trust scores generated by traditional reputation systems are externally imposed labels rather than owned assets. Agents cannot present proof of their trustworthiness to new partners, cannot selectively disclose their trust credentials in different contexts, and cannot make strategic decisions about how to invest effort in building trust capital.

5.1.3 From Passive Evaluation to Active Trust Engagement

The recognition of trust as relational capital necessitates a re-conceptualization of trust management systems. If trustworthiness is a form of capital that agents can accumulate and deploy strategically, then trust management systems must provide mechanisms that enable agents to:

Own and Control Their Trust Credentials: Rather than having trust scores computed and stored by external authorities, agents should possess verifiable proof of their trustworthy behavior that they control and are aware of.

Invest in Trust Building: Agents should be able to make strategic decisions to invest effort in building trust credentials, with the expectation that these investments will yield future benefits in future interactions.

Leverage Past Trustworthiness: Good behavior in past interactions should create tangible assets that empower the agent holding them, and can be demonstrated to new partners.

5.1.4 The Incentive Challenges

An often overlooked aspect of trust management systems in multi-agent environments is the question of incentives. While these systems aim to evaluate the trustworthiness of agents, they rarely address the underlying motivation: why should an agent choose to behave cooperatively or provide accurate information in the first place? Trust scores or reputation mechanisms may reflect past behaviors, but without tangible benefits linked to maintaining a positive reputation, there is little to deter agents from malicious actions. In other words, the system may be

able to detect and record misbehavior, but if no real cost is imposed or no reward is granted for acting correctly, agents may not have sufficient reason to conform to expected norms. More specifically, the benefits of trustworthy behavior accrue primarily to trusters (who receive good service) rather than to trustees themselves.

The trust as capital perspective addresses this challenge by reconceptualizing trust not merely as a score, but as an asset that can be accumulated, owned, and leveraged by agents. In this view, trustworthy behavior generates "trust capital" which provides tangible future benefits. This transforms trust into a resource that agents actively seek to build, aligning individual self interest with collective system reliability. Additionally, recognizing the bidirectional nature of trust implies incentive symmetry. If both parties benefit from being trusted, both have symmetric incentives to behave trustworthy regardless of their current role.

5.1.5 Toward Verifiable Trust Proofs

The recognition that trust can function as relational capital suggests that trust proofs should possess properties similar to other forms of capital: they should be owned by the agent who earned them, verifiable by others, context-dependent, and resistant to forgery or manipulation. This requires moving beyond simple numerical scores toward rich, verifiable proofs that capture specific instances of trustworthy behavior in context.

In this perspective, we envision IoT environments where devices are capable of producing and exchanging cryptographically verifiable proofs of trustworthy behavior, without relying on a central authority. This assumption implies the presence of a distributed infrastructure ensuring immutability and decentralized verification. These conditions can be met in blockchain-enabled IoT systems. Blockchain provides the necessary foundation for registering, validating, and auditing trust evidence, while allowing autonomous IoT agents to manage their own trust-related data securely.

Although several studies have explored the integration of blockchain and IoT to improve transparency and decentralized trust management [91, 92], existing approaches generally remain transaction or reputation-oriented. They focus on recording interactions or computing global reputation scores on-chain, keeping the evaluation process truster-centric and the trustee passive. In contrast, treating trust itself as a capital or asset that an agent owns and manages proactively goes beyond the simple registration of interactions. The model proposed in this chapter

introduces a new paradigm in which agents can (i) earn cryptographically verifiable proofs for trustworthy behavior, (ii) present these proofs to potential partners for verification, (iii) manage their own trust portfolio while independently verifying the authenticity and history of others' proofs.

Building on all the ideas discussed in this section, this chapter presents perspectives and preliminary theoretical foundations for building a bidirectional trust management model that presents trust as an owned, verifiable, and deployed form of asset. While previous chapters focused on resource-constrained IoT settings that demand lightweight and adaptive authentication mechanisms, this chapter explores perspectives for IoT environments where blockchain integration is feasible.

5.2 The BINT Model

This section presents the core architecture and fundamental principles of the Bidirectional NFT-based Trust (BINT) model. We detail the foundational concepts, formally define the Trust NFT (TNFT), outline the model architecture and bidirectional agent interactions.

5.2.1 Concepts and Definitions

This section introduces the fundamental concepts and defines the key terminology that will be used throughout this chapter.

MAS-related concepts

- **Service ($\mathcal{S}$):** It represents the primary motivation for interactions between agents. A service is defined as a capability offered by one agent to another, such as information provision, task execution, or decision support, enabling agents to fulfill individual or collective goals.
- **Interaction ($\mathcal{I}$):** A discrete exchange or sequence of exchanges between two agents (a Truster and a Trustee), typically involving the request and provision of a service. Each interaction has a defined outcome.
- **Outcome ($\mathcal{O}$):** The result of an interaction, which can be positive (e.g., service successfully delivered, information accurately provided) or negative (e.g., service failure, malicious behavior, non-compliance).

- **Agents ($\mathcal{A}$):** Agents can dynamically assume one of two primary roles based on the context of an interaction:
 - **Truster (Consumer) ($\mathcal{AC}$):** An agent that initiates a service request, seeking information or a capability from another agent (producer). The truster generally depends on the requested service, however, it may also receive and consume a service (e.g., information) without explicitly requesting it.
 - **Trustee (Producer) ($\mathcal{AP}$):** An agent that offers and produces a service to another consumer agent, either in response to a request or proactively.
- **Bidirectional trust evaluation:** This enables reciprocal trust assessment. It refers to the process where both the truster evaluates the Trustee (as is common) and the Trustee evaluates the truster. This mutual assessment ensures that both parties consider the reliability and intentions of the other before committing to an interaction. This reciprocity promotes a more balanced and equitable environment, where agents are encouraged to maintain good behavior regardless of their current role. It also enables trustees to decline interactions with potentially unreliable Trusters, thereby protecting their resources.

Blockchain-related concepts

In addition to the following, the essential blockchain concepts that will be used in this chapter are presented in Table 5.1.

- **Non-Fungible Token (NFT):** An NFT is a unique digital asset that represents ownership or proof of authenticity of a specific item, with its ownership and metadata immutably recorded on a blockchain. This creates a transparent and tamper-proof history of the asset's provenance and associated actions. The NFT is not limited to a single type of blockchain; blockchains such as Ethereum and Solana are examples, which makes the proposed model independent of any specific blockchain and allows it to evolve alongside the technology itself.
- **Trust Non-Fungible Token (TNFT):** A TNFT is a granular, contextual, and cryptographically verifiable digital asset that attests to a specific instance of trustworthy behavior. Each TNFT is issued upon the successful completion of a verified interaction (i.e., a positive outcome) where the agent acted as a Trustee. Each TNFT is uniquely identifiable and linked to the Trustee who

Table 5.1: Blockchain Terminology

Term	Explanation
Wallet	A digital account that stores private keys and allows users to send, receive, and manage blockchain assets.
Transaction	A record of a transfer of assets or data between blockchain addresses, verified and stored on the ledger.
Smart Contract	A self-executing program on the blockchain that enforces rules and automates agreements without intermediaries.
Balance	The amount of cryptocurrency, tokens or NFT held by a wallet at a specific point in time.
NFT	A unique digital asset represented on the blockchain, usually associated with art, collectibles, or unique items.
Address	A unique identifier on the blockchain used to send or receive assets, typically derived from a wallet's public key.
Hash	A fixed-length string generated from data using a cryptographic function, ensuring integrity and uniqueness of transactions or blocks.

earned it, it serves as verifiable and immutable digital proofs of an agent's trustworthiness during future exchanges, and it encodes details about the interaction that led to its minting (e.g., service type, date, a reference to the Trustee's feedback).

- **Verifiable Trust Portfolio (VTP):** The collection of blockchain addresses, each pointing to a specific TNFT owned by an agent. This collection constitutes an agent's verifiable portfolio of past trustworthy behaviors, serving as cryptographic proof of their reliability across different contexts and interactions. When an agent presents its VTP, it is providing tangible evidence that allows other agents to quickly and objectively assess its accumulated trustworthiness without relying solely on subjective feedback or limited direct experience. The verification of metadata stored within these TNFTs is performed by querying the blockchain via smart contracts.
- **Smart Contract (SC):** Self-executing code deployed on the blockchain, serving as the immutable support for trust mechanisms. In BINT, they are responsible for: i) verifying TNFTs as a proof, ii) creating (minting) new TNFTs upon verified outcomes, iii) applying penalties for detected misbehavior (e.g.,

through burning TNFTs), and iv) managing the lifecycle of agents' trust-related data on the network.

- **Blockchain Nodes:** Servers connected to the same blockchain network that store and exchange records, perform data queries, and participate in the consensus process, ensuring the integrity and immutability of the stored information. They host smart contracts and enable communication with other solution modules via APIs, facilitating the decentralized operation of the BINT Model.

These definitions establish the concepts and terminology used in BINT, providing a basis for the subsequent discussion of its architecture and interaction protocols.

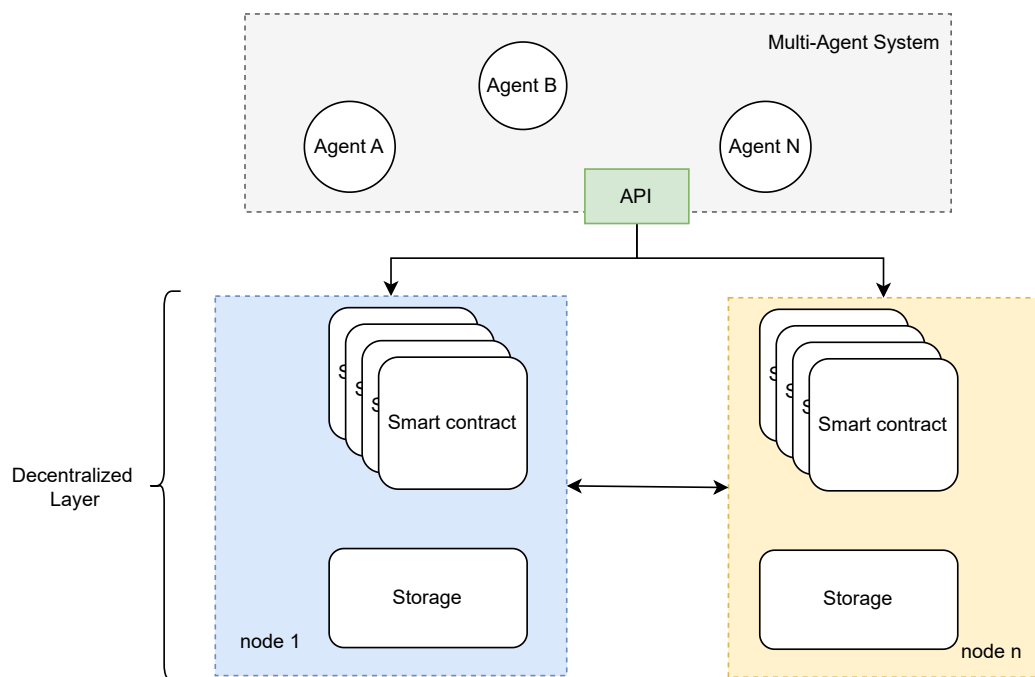


Figure 5.1: BINT Architecture

5.2.2 Model Architecture

The BINT model is conceptualized as a multi-layered architecture. Its structure ensures a separation of concerns, facilitating the interaction of agents, the immutable recording of trust-related data, and the autonomous management of TNFTs. The

architecture is composed of three main layers: the MAS layer, the API layer, and the Blockchain layer as shown in Figure 5.1.

MAS Layer

This layer hosts the agents, who perform their primary functions within the system. It represents the environment in which agents interact, exchange services, and evaluate each other's trustworthiness. Agents dynamically participate in interactions, both providing and consuming services, while continuously assessing the trustworthiness of their counterparts. Trust evaluations are based on verifiable information provided by each agent. This involves processing the VTP (proof of trustworthiness), examining the quantity, type, and recency of issued tokens. This objective data can optionally be combined with the agent's personal interaction history, producing a composite trust estimate to inform decision-making.

API Layer

The API layer serves as the intermediary between agents and the blockchain infrastructure, as shown in Figure 5.2. It provides standardized interfaces for agents to query, submit, and verify trust-related data, such as TNFTs. Through this interface, agents can perform blockchain-dependent actions, including querying TNFT metadata and ownership records, submitting validated interaction outcomes to mint new TNFTs, triggering penalties such as TNFT burning, and interacting with smart contracts more generally.

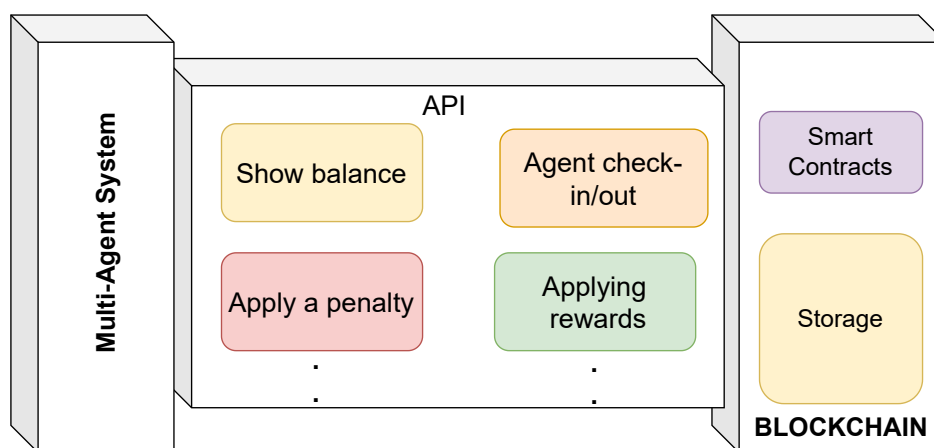


Figure 5.2: BINT API connecting agents with the Blockchain

Blockchain Layer

The blockchain layer constitutes the decentralized, immutable, and transparent infrastructure of the BINT model. It serves as the core repository for all trust-related data and provides a secure execution environment for smart contracts. The blockchain network maintains a shared, tamper-proof ledger of all transactions, including TNFT transfers, agent interaction outcomes, and other trust-related activities. Distributed nodes participate in the consensus process to validate and immutably record this information, ensuring transparency, fault tolerance, and auditability.

Smart contracts deployed on the blockchain automate the core trust mechanisms and critical system functionalities. The TNFT smart contract manages the full lifecycle of TNFTs, including minting, verification, and burning, and enforces that TNFTs are issued only upon verified positive outcomes of interactions. The architecture is extensible, allowing additional specialized smart contracts to be deployed for specific functionalities as needed.

Together, the blockchain layer ensures that all trust-related data is secure, verifiable, and decentralized, providing a robust foundation upon which agents can build and verify trust in a fully transparent manner.

5.2.3 Formalization of core entities

In the following, we formally define the structure of interactions, outcomes, the nature of TNFTs, and how an agent's accumulated trustworthiness (VTP) is represented and queried.

Interactions

Each Interaction $\mathcal{I}$ is uniquely identifiable and represents a specific instance of service exchange between two agents. It includes the participants, the type of the service, and other metadata regarding the interaction. Formally, an interaction $\mathcal{I}$ is defined as a tuple:

$$\mathcal{I} = \langle interactionID, \mathcal{AC}, \mathcal{AP}, serviceType, metadata_{\mathcal{I}} \rangle \quad (5.1)$$

where:

- *interactionID*: A unique identifier assigned to each interaction directly by the blockchain.

- $\mathcal{AC}$: The Trustee (consumer) agent initiating the interaction (the address of the agent's wallet).
- $\mathcal{AP}$: The Trustee (producer) agent providing the service (the address of the agent's wallet).
- $serviceType$: The type of service being requested or provided.
- $metadata_I$: A set of optional, context-specific information relevant to the interaction. This may include attributes such as start time, end time, among others.

Outcomes

Each Outcome O is a definitive result of an Interaction I . An outcome O is formally defined as a tuple:

$$O = (interactionID, status, metadata_O) \quad (5.2)$$

where:

- $interactionID_I$: The unique identifier of the interaction to which this outcome pertains, ensuring a direct link to the specific I .
- $status \in \langle \text{success, failure, disputed} \rangle$.
- $metadata_O$: Additional details or feedback regarding the outcome, which may include a cryptographic hash of detailed feedback.

Trust NFT

A TNFT is minted on the blockchain upon successful completion of a verified interaction, serving as a reusable credential for future trust decisions. Each TNFT represents a specific, positive instance of an agent's reliable behavior. It is formally defined by the following immutable attributes:

$$TNFT = \langle tokenID, owner, issuer, interactionID, metadata_{TNFT} \rangle \quad (5.3)$$

where:

- $tokenID$: The token address on the blockchain.
- $owner$: The address of the agent who owned the TNFT.

- *issuer*: The address of the agent who triggered the minting of the TNFT.
- *interactionID*: Reference to the interaction from which the TNFT was generated.
- *metadata_{TNFT}*: Additional details regarding the issued TNFT.

Verifiable Trust Portfolio (VTP)

The VTP of an agent a is the set:

$$VTP_a = \{TNFT_i \mid TNFT_i.owner = a\} \quad (5.4)$$

where a is the blockchain address uniquely identifying the agent. The VTP serves as the agent's complete proof of trustworthiness, providing a comprehensive and objectively verifiable record of their past reliable behaviors.

To enable efficient verification, the BINT smart contract exposes a query function:

$$getVTP(a) \mapsto VTP_a \quad (5.5)$$

This function returns the structured information associated with all TNFTs owned by a , allowing other agents to objectively evaluate a 's accumulated trustworthiness by consulting the blockchain.

For illustration, the implementation of the $getVTP(a)$ query, detailed in Algorithm 1, can be integrated into the TNFT Manager smart contract as part of the TNFT lifecycle management module, alongside essential functions such as minting a new TNFT (Algorithm 2), burning an existing one (Algorithm 3), and other design-related functions. This integration ensures that trust-related operations (creation, revocation, and verification) are handled within a unified, verifiable, and decentralized framework.

5.3 Bidirectional Interactions and Incentive Mechanisms

This section presents the detailed interaction protocol that governs how agents engage in bidirectional trust evaluation, mutual selection, and service provision within the BINT model. The process follows the bidirectional philosophy of the proposed model, where both truster and trustee act as active decision-makers, revealing their verifiable trust information. We begin by describing the protocol,

Algorithm 1 TNFT Manager - getVTP(a)

Require: a : Wallet address AC **Ensure:** VTP_a

- 1: **Check** if AC is a valid wallet in the registered blocks
 - 2: **if** AC is invalid **then**
 - 3: **return** "Invalid wallet"
 - 4: **else**
 - 5: Count the number of TNFTs associated with AC
 - 6: **if** $\text{count}(\text{TNFT}) > 0$ **then**
 - 7: **for** each TNFT **do**
 - 8: Group TNFTs by serviceType
 - 9: Sort TNFTs of each serviceType in descending order of creation
 - 10: **end for**
 - 11: **return** list of TNFTs organized by serviceType
 - 12: **else**
 - 13: **return** $VTP_a = \phi$
 - 14: **end if**
 - 15: **end if**
-

Algorithm 2 TNFT Manager - Create TNFT

Require: Valid wallet address AC , issuer AP , interactionID, metadata**Ensure:** New $TNFT$ registered on blockchain

- 1: **Function** createTNFT(AC , AP , interactionID, metadata):
 - 2: **Check** if AC is a valid wallet
 - 3: **if** AC is invalid **then**
 - 4: **return** "Invalid wallet address"
 - 5: **end if**
 - 6: **Check** if AP is a valid issuer
 - 7: **if** AP is invalid **then**
 - 8: **return** "Invalid issuer"
 - 9: **end if**
 - 10: Generate a new unique $tokenId$
 - 11: Create new $TNFT = \{tokenId, owner = AC, issuer = AP, interactionID, metadata\}$
 - 12: Register $TNFT$ on the blockchain ledger
 - 13: **return** Confirmation of creation with $TNFT$ details
-

Algorithm 3 TNFT Manager - Burning TNFT (Penalty)

Require: AP, serviceType_i**Ensure:** TNFT with serviceType_i removed (burned) if exists

```

1: Function burnTNFT(AP, serviceType_i):
2:   Retrieve list of TNFTs owned by AP
3:   if AP has no TNFTs then
4:     return "No TNFTs found for this AP"
5:   end if
6:   for each TNFT in AP's list do
7:     if TNFT.serviceType == serviceType_i then
8:       Remove (burn) this TNFT from the blockchain
9:       return "TNFT with serviceType_i has been burned"
10:    end if
11:  end for
12: return "No TNFT with serviceType_i found"

```

then elaborate on the incentive mechanisms that drive trustworthy behavior, and finally discuss potential enhancements for more sophisticated trust management scenarios.

5.3.1 Bidirectional Trust Protocol

The BINT model implements a structured interaction protocol that ensures both parties can evaluate each other before committing to collaboration, while maintaining verifiable records of all trust-relevant actions on the blockchain. Figure 5.3 illustrates the interaction sequence between a Truster, Trustee, and Smart Contract. This version of the protocol demonstrates bidirectional evaluation, proactive provision of trust proofs, and interactions with the smart contract. Several extensions to this version are discussed in subsequent sections.

Initial Request and Mutual Verification

The interaction begins when an agent acting as Truster identifies a need for a service and selects potential Trustees to approach. BINT forces Trusters to proactively include their VTP address in the initial request. This VTP address serves as a pointer to the Truster's blockchain-based collection of TNFTs, providing immediate access to verifiable trust credentials. The initial request thus contains both the service specification (including required capabilities, context, etc) and the Truster's

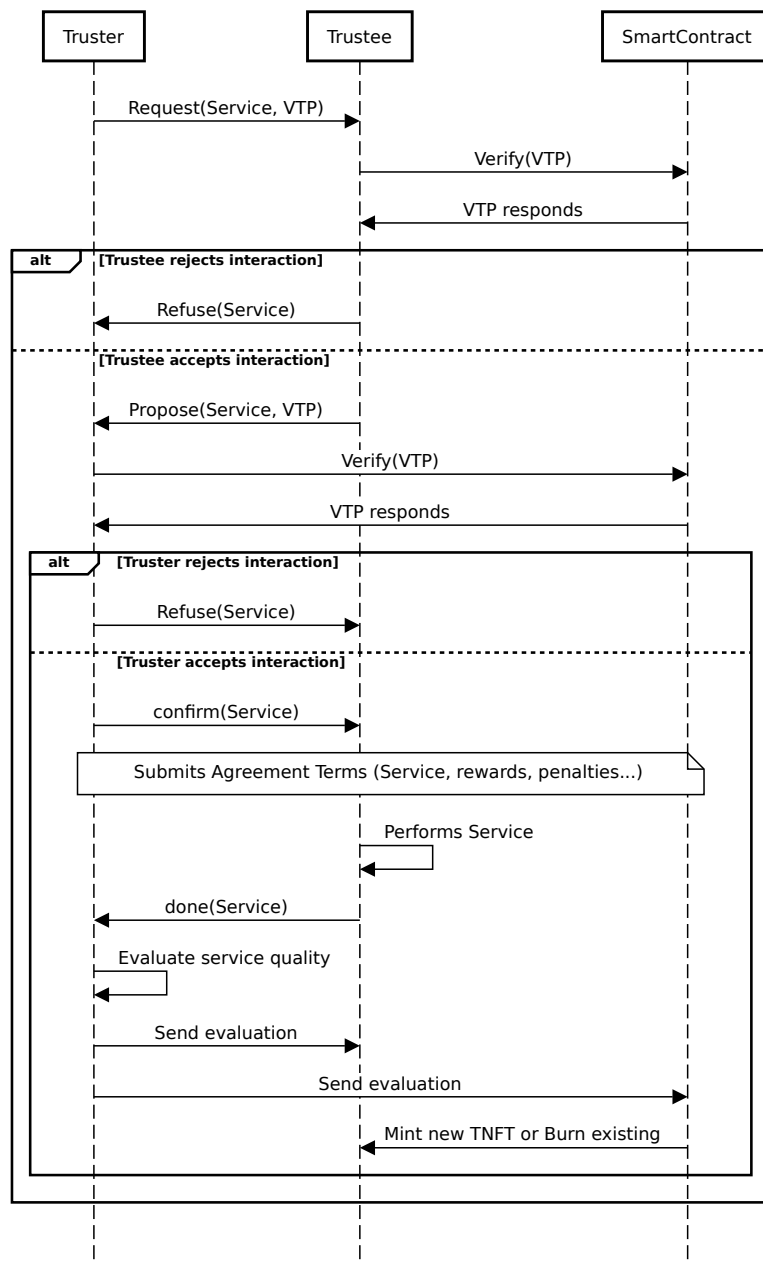


Figure 5.3: Interaction sequence of BINT agents

VTP address. Rather than waiting to be evaluated, the Truster immediately demonstrates its own trustworthiness, signaling good faith and providing the Trustee

with information needed for its own risk assessment.

Upon receiving this request, the Trustee queries the blockchain to verify the Truster's credentials. This verification is performed by sending the VTP address to the smart contract, which returns detailed information about the TNFTs contained within that portfolio. The smart contract response includes the list of TNFT identifiers owned by that address, contextual information for each TNFT such as service type, date, and interaction details, and metadata that enables the Trustee to assess the relevance and recency of each credential. The Trustee can leverage smart contract functions to compute aggregate trust metrics rather than performing these calculations locally. The smart contract might compute total TNFTs in contexts relevant to the current request, recency-weighted trust scores that give more importance to recent good behavior, or behavioral consistency indicators that reveal patterns over time.

Based on verification results, the Trustee makes its first selection decision. If the Truster's VTP fails to meet the Trustee's trust threshold, the Trustee can refuse the interaction, protecting its resources from potentially unreliable partners. If proceeding, the Trustee responds with a proposal including its own VTP address. The Truster now performs identical verification of the Trustee's credentials, assessing historical reliability in similar contexts and overall trustworthiness. Based on this verification and the proposed terms, the Truster makes its final selection decision. Only when both parties have independently evaluated each other and explicitly decided to proceed does the interaction advance. This bilateral selection ensures both agents enter collaboration with confidence in their partner's reliability.

Service Agreement, Execution, and Evaluation

Following mutual acceptance, terms of the agreement are submitted to the blockchain. The agreement captures service specifications, timelines, rewards including potential TNFT issuance, and penalties for breach or poor performance. The smart contract records this agreement and returns a unique hash serving as an immutable identifier. Recording the agreement on-chain provides non-repudiation, creates an objective reference for evaluation, enables automated enforcement, and establishes an evidence basis for TNFT issuance decisions.

The Trustee then performs the service. Upon completion, the Trustee signals delivery, triggering the evaluation phase. In this presented protocol, only the Truster is responsible of evaluating the service quality and submit the assessment

to the smart contract. Based on evaluations, the smart contract manages TNFT issuance. Only the Trustee receives TNFTs for successful service delivery. If the Truster's evaluation indicates a positive feedback, the smart contract mints a new TNFT assigned to the Trustee's VTP, encoding contextual information about the interaction. Conversely, negative feedback triggers penalty mechanisms, potentially burning existing TNFTs. This asymmetric reward structure where only Trustees earn TNFTs is intentional and foundational to the incentive mechanism.

5.3.2 Incentive Mechanism

The proposed protocol, which is more focused on Trustees, enforces an asymmetric reward rule: only trustees receive TNFTs for cooperative behavior. This design, illustrated in Figure 5.4, creates a positive feedback loop: agents who behave well as trustees become more likely to be accepted as trusters (and trustees) later, since their accumulated TNFTs increase their perceived reliability. Consequently, when the same agent later acts as Truster, it can present its accumulated VTP to potential Trustees, demonstrating reliability through TNFTs earned in previous Trustee roles. A rich VTP signals successful collaboration history, increasing the probability that high-quality Trustees will provide services. Trusters with strong VTPs can then experience higher acceptance rates, better service terms, priority treatment, and access to Trustees serving only high-reputation agents. This creates a cycle: good Trustee behavior accumulates TNFTs, improving Truster treatment, enabling better collaborations, creating more opportunities to earn additional TNFTs. Trustees compete to build strong VTPs through excellent service, while Trusters with weak VTPs struggle accessing quality Trustees. This competitive dynamic drives continuous behavior improvement, as agents must build trust capital to access collaboration opportunities. The mechanism also requires agents to adopt long-term perspectives, investing in reputation through consistent performance for future returns.

However, this asymmetry raises a valid theoretical question: if trusters do not earn TNFTs, what prevents them from acting maliciously, and what incentivizes them to participate fairly in the system? Trusters could provide false negative feedback to deny TNFTs to Trustees who performed adequately. However, the bidirectional verification phase enables Trustees to assess Truster reliability before committing to service provision. A Trustee examining a potential Truster's VTP can observe not only the TNFTs the Truster has earned as a Trustee, but also patterns in their evaluation behavior, since such metadata is recorded on the blockchain and can be accessed via smart contracts. If a Truster has a history of consistently providing negative evaluations while other Trusters who interacted with the same Trustees

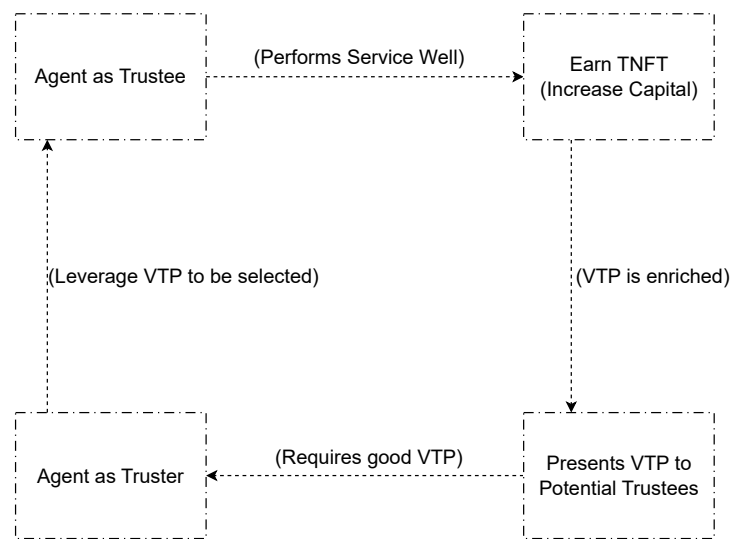


Figure 5.4: BINT Trust Loop

provided positive evaluations, this pattern signals potential malicious behavior. Trustees can use this information to refuse service to Trusters with suspicious evaluation patterns, creating indirect consequences for dishonest feedback.

This version of the model remains however vulnerable to malicious Truster behavior, particularly in scenarios where Trusters can afford to sacrifice their own future Truster prospects. This limitation motivates the enhanced mechanisms discussed in the following section, particularly staking mechanisms, bilateral evaluation consensus and third-party witnesses.

5.4 Discussion: Extensions and Future Enhancements

The protocol presented previously provides a foundation for bidirectional trust management as an asset, but several extensions could enhance fairness and sophistication. We present these as directions for future development, acknowledging that each introduces complexity and trade-offs requiring careful evaluation.

5.4.1 Bilateral Staking and Negotiable Engagement

Beyond simply displaying VTPs and minting or burning TNFTs after service performance, agents can proactively determine and stake a portion of their TNFT portfolios before engaging. In this staking-based negotiation phase, both the truster and the trustee declare how much of their TNFTs they are willing to risk to enter the agreement. This converts trust evidence into tangible commitment, making cooperation decisions carry real economic and reputational consequences. The baseline protocol carries no direct cost beyond opportunity costs or reputation damage. By contrast, bilateral TNFT staking introduces real, quantifiable stakes. For example, both parties submit collateral through the smart contract proportional to their perceived risk or service value. Upon successful completion of the service, both agents recover their stakes, and the trustee additionally earns new TNFTs. In case of misbehavior or performance breach, the smart contract automatically burns or locks the corresponding staked TNFTs. This mechanism creates symmetric accountability, raises misbehavior costs, enables graduated penalties, and allows automatic compensation. Agents can also negotiate stake size dynamically, with higher stakes signaling stronger confidence and commitment to fair behavior.

5.4.2 Bilateral Evaluation Consensus

The presented protocol relies primarily on Truster evaluation for TNFT issuance, creating vulnerability to malicious Trusters. Bilateral consensus mechanisms could require agreement from both parties or weight both perspectives in decisions. A mutual agreement requirement would issue TNFTs only when both submit positive evaluations, with conflicts entering dispute resolution. This protects Trustees from unfair denial and encourages honest assessment but may make earning TNFTs harder and create new gaming opportunities requiring careful balance between fairness and incentive sufficiency. Also, independent third-party witnesses could observe and verify outcomes, providing objectivity. Implementation could involve designated witnesses agreed upon during contracting. Witnesses provide objective perspectives, reduce gaming through external oversight, and create specialized verification business models. However, they introduce privacy concerns, require careful incentive design, add complexity and delays, and may themselves be biased or collude. A hybrid approach might selectively invoke witnesses only for high-stakes interactions or disputed cases, balancing objectivity benefits against efficiency costs. The blockchain layer could maintain a witness registry with performance records, enabling agents to evaluate witness reliability through the same VTP mechanisms used for service providers.

5.4.3 The Cold Start Problem

A critical challenge facing any reputation-based system is the cold start problem: how do new agents without established trust histories gain initial opportunities to build their VTPs? In traditional unidirectional trust systems, this manifests as new agents struggling to attract trusters despite potentially being highly reliable. BINT's bidirectional nature potentially exacerbates this challenge, as both Trusters and Trustees may refuse interactions with VTP-deficient partners. To address this, some mechanisms can facilitate trustworthy onboarding while preserving system integrity:

- Provisional trust tokens: newly registered agents could receive temporary, specific TNFTs serving as "trust loans". Misbehavior incurs amplified penalties, ensuring strong incentives for compliant conduct. Over time, provisional tokens are replaced by earned TNFTs.
- Sponsorship: established agents may sponsor newcomers, especially those that belong to the same organization, by staking parts of their own VTPs, incurring penalties for poor sponsorship outcomes. This mechanism transfers social accountability to the blockchain.
- Graduated low-risk service access: depending on the context, agents with minimal VTPs could access low-risk services only, building credentials that unlock progressively higher-value interactions.

5.5 Conclusion

This chapter proposes a conceptual shift in computational trust management, moving from a passive, unidirectional perspective to an active, bidirectional paradigm in which trust is reconceptualized as a verifiable and owned asset in blockchain-enabled IoT environments. Through the BINT model, a theoretical framework is introduced in which agents can not only be evaluated but also actively manage, present, and leverage their own trustworthiness. By formalizing the notions of TNFTs and VTPs, the model enables trust evidence to be cryptographically verifiable and owned by agents. This approach addresses limitations in traditional trust and reputation systems, including the asymmetry between truster and trustee roles, the passive role of trustees, and the absence of incentive mechanisms. By treating trustworthy behavior as a source of digital capital that can be accumulated, staked, and verified, BINT aligns individual incentives with collective reliability. This reflects a fundamentally different philosophy about the nature of trust in MAS.

Traditional approaches treat trust as something done to agent, while the BINT perspective reconceptualizes trust as something agents do, an activity of building, presenting, and leveraging verifiable evidence of reliability.

The proposed bidirectional trust evaluation protocol illustrates how both parties can evaluate one another through verifiable proofs before collaboration, establishing mutual accountability. The integration of blockchain technologies provides immutability, transparency, and decentralized verification. As discussed in Section 5.4, several areas require further development before large-scale deployment. The proposed extensions represent directions for enhancing the basic model's fairness and robustness. However, experimental evaluation is necessary to validate the theoretical foundations and assess the practical feasibility of TNFTs in real-world IoT environments.

This thesis addresses fundamental challenges in securing and enabling reliable interactions within A-IoT environments. These environments are inherently dynamic and uncertain: agents frequently depend on others for information or services without prior knowledge of their reliability or intentions. Such dependency situations give rise to a fundamental challenge; how to make secure and rational decisions when other entities may not behave as expected. By developing trust-based authentication mechanisms and novel approaches to trust management, this research contributes to the intersection of MAS and IoT. It proposes solutions to enhance trust management in blockchain-enabled IoT environments, while balancing security requirements with the inherent resource constraints and dynamic characteristics of distributed, energy-constrained IoT environments.

6.1 Summary of Contributions

The primary contributions of this thesis can be organized around three interconnected research themes: opportunistic sensor-based authentication factors, trust-based adaptive authentication decision-making, and bidirectional model of trust as an asset.

The first contribution introduces a proof of concept demonstrating how authentication factors can be conceived in an opportunistic way within IoT environments. Rather than always viewing authentication as requiring dedicated hardware or protocols, this research demonstrates that existing sensor infrastructure deployed for non-security purposes can be repurposed to generate additional authentication factors. The concept of opportunistic sensor-based authentication factors addresses [RQ1](#) by showing how IoT's inherent sensor richness becomes an advantage rather than only a management challenge. The illustrative experiments using microphones from acoustic monitoring systems and surveillance cameras demonstrated this concept, showing that even simple techniques can extract discriminative features suitable for enhancing authentication. This approach not only avoids additional hardware costs but also introduces unpredictability and diversity into authentication processes, making them more resilient against attackers who

cannot anticipate which factors will be deployed in a given context.

The second contribution establishes a theoretical and practical framework for integrating trust evaluation with authentication factor selection, directly addressing [RQ2](#) and [RQ3](#). It was argued that traditional approaches treat authentication as independent from trust assessment, resulting in static security policies that either over-provision security (wasting resources) or under-provision it (creating vulnerabilities). This contribution demonstrates that trust and authentication form a positive feedback loop: authentication validates identities that enable meaningful trust assessment, while trust levels inform appropriate authentication rigor. Two distinct approaches were developed for authentication factor selection. The AAT-L method provides a straightforward weighted approach that balances security and energy consumption based on trust levels and contextual criticality. The AAT-MOO method enhances this by formulating authentication decisions as a multi-objective optimization problem. By employing NSGA-II to generate Pareto-optimal solutions that balance security, energy consumption, and delay, AAT-MOO enables trust-guided selection from non-dominated solutions rather than exhaustive evaluation of all possible factor combinations.

The experimental evaluation through multi-agent navigation simulations demonstrated that AAT-MOO and AAT-L achieve superior balance between security and resource efficiency across varying complexity scenarios. Compared to static authentication baselines, AAT-MOO reduced energy consumption by approximately 41.5% while maintaining competitive security performance, with only 0.69 percentage points difference in attack success rates. The framework's scalability was validated across increasing authentication complexity, from 8 to 15 available factors and from 2-factor to 3-factor combinations. AAT-MOO maintained practical execution times as long as the set of available authentication factors did not change frequently, whereas AAT-L incurred higher computational costs but proved more suitable in scenarios where the set of authentication factors changes frequently.

The third contribution addresses [RQ4](#) by reconceptualizing trust relationships in MAS. Traditional trust models position trustees as passive subjects of evaluation, creating an asymmetry that fails to capture the reciprocal nature of trust and provides insufficient incentives for trustworthy behavior. The proposed BINT model introduces trust as verifiable asset that agents can accumulate, own, and potentially strategically leverage. By implementing Trust NFTs (TNFTs) on blockchain infrastructure, BINT enables trustees, and trusters, to proactively demonstrate their trustworthiness through cryptographically verifiable credentials. Each TNFT repre-

sents a specific instance of reliable behavior, creating a verifiable trust portfolio (VTP) that agents control and present to potential interaction partners. This transformation from passive evaluation to active trust agency addresses the incentive problem from several aspects. Trustworthy behavior generates tangible assets that provide future selection advantages, acting as an incentive mechanism to maintain trustworthy behavior. The bidirectional evaluation mechanism ensures that both trusters and trustees assess each other before interactions, creating symmetric incentives for maintaining trustworthy behavior regardless of role.

This research contributes to both theoretical understanding and practical implementation of trust and security in MAS and IoT. Theoretically, it challenges the unidirectional paradigm that has dominated computational trust research, demonstrating how trust can function as an asset rather than only an evaluation metric. Additionally, the integration of trust with authentication decision-making provides a framework for dynamic security adaptation. Practically, the opportunistic authentication concept offers IoT system designers a path to enhanced security without requiring additional hardware investments. The adaptive authentication frameworks provide concrete mechanisms for implementing context-aware security policies that respond to both trust levels and resource constraints. The blockchain-based trust management model offers a decentralized infrastructure for trust verification as an asset without relying on centralized authorities, particularly relevant for large-scale resource-tolerant IoT deployments. Together, these contributions form a coherent research trajectory that redefines the relationship between trust and authentication, from mutual dependence to adaptive co-evolution, and finally to mutual accountability.

6.2 Future Directions

Despite the contributions presented, several limitations warrant acknowledgment and suggest directions for future research. The opportunistic authentication factors, while demonstrated feasible through illustrative experiments, require more extensive validation across diverse IoT contexts with different types of sensor data. Privacy considerations represent another important direction for future research. The repurposing of contextual sensor data for authentication raises privacy concerns that require careful analysis. Future work should develop privacy-preserving mechanisms that enable opportunistic authentication while protecting sensitive information about behaviors and contexts. The evaluation of AAT-L and AAT-MOO was conducted in a controlled simulation that, while dynamic, abstracts

certain real-world factors such as communication noise, uncertainty and limited knowledge of the environment. Further validation on real IoT testbeds or emulated environments with real sensors would strengthen the conclusions regarding scalability and performance. Additionally, this trust-based authentication framework showed performance dependencies on the trust distribution within agent populations. While AAT-L and AAT-MOO show better results in balanced or high-trust environments, their security effectiveness degrades in highly hostile scenarios dominated by malicious agents. This suggests that adaptive authentication strategies should incorporate environmental trust assessments to adjust their optimization objectives dynamically. Future research might explore hybrid approaches that transition between trust-based adaptation and static high-security modes when environmental conditions warrant maximum protection.

Current experimental evaluation relied on synthetic authentication factor characteristics and simulated attack scenarios. Future work should incorporate empirically measured security levels, energy consumption profiles, and latency characteristics from actual IoT authentication protocols to strengthen practical relevance. To further enhance the effectiveness and applicability of the trust-based authentication framework, future experiments could focus on testing its performance across diverse contexts and applications. For instance, evaluating the system in varying IoT environments, such as industrial, healthcare, and smart home settings, could provide deeper insights into its adaptability and reliability. Another critical avenue for improvement involves developing and simulating strategic attack methods, such as coordinated multi-agent impersonation or evolving adversarial strategies, to test the resilience of the framework.

The BINT model, while theoretically developed in Chapter 5, requires comprehensive experimental validation. Implementation challenges including blockchain scalability, transaction costs, and the computational overhead of TNFT verification need systematic investigation in resource-tolerant and resource-constrained IoT devices. The integration of blockchain adds transparency but also introduces latency and energy concerns that need careful tuning depending on the deployment context. The model must also be tested against different types of trust attacks to prevent trust manipulation and the abusive use of TNFTs between malicious agents. Additionally, the cold start problem raises fundamental questions about the nature of trust bootstrapping in decentralized systems. Traditional approaches often rely on centralized authorities providing initial certification or guarantee. BINT's decentralized architecture must balance two competing demands: maintaining high trust standards to protect system integrity, and providing accessible

entry points to ensure system growth and inclusivity. The solution space involves careful calibration of entry barriers, provisional mechanisms, and graduated accesses that collectively enable new agent integration without compromising the verifiable trust foundation that gives TNFTs their value. Furthermore, staking mechanisms introduce several design challenges. First, determining appropriate stake sizes requires careful calibration to avoid either excessive risk aversion (stakes too high, inhibiting collaboration) or insufficient deterrence (stakes too low, failing to prevent misbehavior). Second, the mechanism must account for asymmetries in VTP richness; agents with extensive portfolios can afford higher stakes, potentially creating barriers for newer or less established agents. Third, stake resolution in disputed interactions requires robust arbitration mechanisms to prevent strategic manipulation.

Many of the challenges and future enhancements identified above, from adapting to hostile environments to calibrating stake mechanisms and trust thresholds, share a common thread: they require dynamic decision-making in response to evolving conditions. Reinforcement learning (RL) offers a promising direction to address this adaptability gap, enabling agents to continuously improve their authentication and trust strategies through experience. RL-based agents could learn how to adapt authentication intensity and trust thresholds in response to changing environments, attacks, or behavioral patterns, thereby reducing dependency on pre-defined rules or optimization models. Another direction involves the deployment and evaluation of the proposed frameworks in real or large-scale IoT testbeds, involving heterogeneous devices and communication standards. This would provide deeper insight into the scalability, interoperability, and robustness of the proposed mechanisms under realistic conditions. Another promising avenue lies in the fusion of trust and risk assessment. Incorporating explicit notions of perceived risk and mission criticality into authentication decisions could make the models even more context-sensitive and efficient. This extension would allow agents to reason not only about “who to trust,” but also about “how much to risk” in each interaction.

The intersection of IoT and MAS creates both opportunities and challenges for securing distributed autonomous systems. This thesis demonstrates that by rethinking fundamental assumptions about authentication and trust management, we can develop more adaptive, efficient, and resilient security mechanisms suited to A-IoT environments. The shift from static to adaptive authentication, from passive to active trust agency, and from isolated security mechanisms to integrated trust-authentication frameworks represents meaningful progress toward securing

the increasingly autonomous and distributed systems that characterize modern IoT deployments.

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